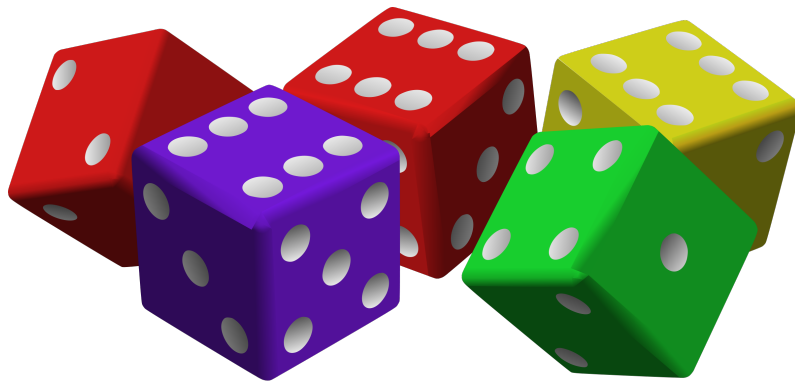




STA3045F

Order out of Chaos

II



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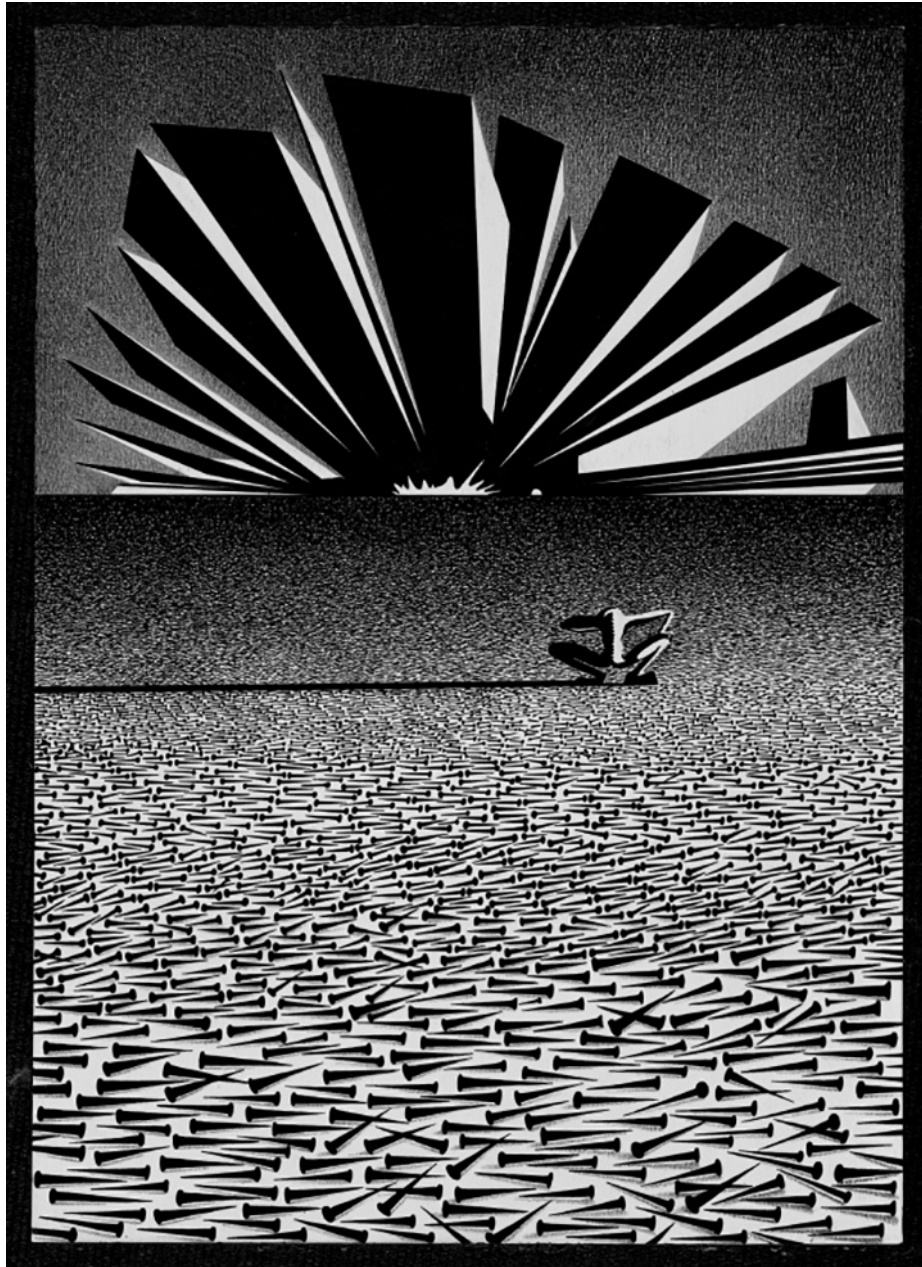


Figure 1: “Order out of chaos” by A. Fomenko.

“The subject matter of probability theory is the mathematical analysis of random events, i.e., of those empirical phenomena which—under certain circumstance—can be described by saying that:

They do not have *deterministic regularity* (observations of them do not yield the same outcome);
whereas at the same time

They possess some *statistical regularity* (indicated by the statistical stability of their frequency).”

— A.N. Shiryaev.

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Part I

Probability and Measure Theory

Chapter 1

The Story in a Nutshell

The study of elementary probability begins with what is called a *random experiment*: an experiment whose outcome cannot be predetermined with certainty. An example of such a random experiment is the tossing of three fair coins, with the (uncertain) outcome being the face shown by each of them. Associated with a random experiment is the set of all possible outcomes Ω , called the *sample space*. Each possible outcome $\omega \in \Omega$ is also called a *sample point*, and we define an *event* to be a subset of Ω . We say that an event A *occurs* if the outcome of the experiment ω belongs to A , i.e., if $\omega \in A$.

In the example of tossing three coins, we have

$$\Omega = \{\text{HHH}, \text{HHT}, \dots, \text{TTT}\} = \{\omega = \omega_1\omega_2\omega_3 : \omega_i = \text{H or T for } i = 1, 2, 3\},$$

where ‘H’ means “heads” and ‘T’ means “tails”. For instance, the event A that exactly two heads come up corresponds to the set

$$A = \{\text{HHT}, \text{HTH}, \text{THH}\}.$$

If the outcome is $\omega = \text{HHT}$, then A occurs since $\omega \in A$. If A does not occur, then its complement, A^c , occurs.

Let $\mathcal{F}$ be the set of all events (also called *measurable sets*). We will assume, for now, that every subset of Ω is an event, i.e., $\mathcal{F} = 2^\Omega$ = the set of all subsets of Ω (power set). Our aim is to assign to each event $A \in \mathcal{F}$, a real number $\mathbb{P}(A)$, called the *probability* of the event A occurring. We wish to make this assignment — which is just a function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ — in such a way that

ℙ1. $\mathbb{P}(\emptyset) = 0$;

ℙ2. $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$;

ℙ3. If $\{A_n : n = 1, 2, 3, \dots\}$ is a countable collection of pairwise disjoint events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n);$$

This is called *σ -additivity*.

ℙ4. $\mathbb{P}(\Omega) = 1$.

In our example, for instance, we can define $\mathbb{P}$ (i.e., assign probabilities to events in $\mathcal{F}$) as follows. First, for singletons we define

$$\mathbb{P}(\{\omega\}) := \frac{1}{8} \quad \text{for every } \omega \in \Omega.$$

Since $\mathcal{F}$ has 2^8 elements, we will complete the definition of $\mathbb{P}$ to the rest of $\mathcal{F}$ by

$$\mathbb{P}(A) := \sum_{\omega \in A} \mathbb{P}(\{\omega\}), \quad \emptyset \neq A \in \mathcal{F} \text{ and } \mathbb{P}(\emptyset) = 0.$$

It can easily be shown that $\mathbb{P}$ satisfies properties $\mathbb{P}1 - \mathbb{P}4$.

We define a *random variable* as a function $X : \Omega \rightarrow \mathbb{R}$. That is, X maps each outcome ω of the random experiment to the real number $X(\omega)$. Continuing with the previous example of three coin tosses, an example of a random variable X is one that counts the number of heads in each outcome. That is, the function $X : \Omega \rightarrow \mathbb{R}$ defined by

$$X(\omega) := \text{number of H in } \omega, \quad \omega \in \Omega.$$

For example, $X(\text{HHT}) = X(\text{HTH}) = 2$, while $X(\text{TTT}) = 0$. With this definition, we can now calculate probability statements about X as follows. If $B \subseteq \mathbb{R}$ is a subset of interest (e.g. an interval or a singleton), then we can calculate the probability that the value of X (after the experiment is conducted) lies in the set B as

$$\mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}) = \mathbb{P}(X^{-1}(B)).$$

That is, we first find the set $X^{-1}(B)$ of all the outcomes ω for which $X(\omega)$ lies in B , and then find the probability of $X^{-1}(B)$ as a subset of Ω . We should keep in mind though that $\mathbb{P}(X^{-1}(B))$ will only make sense if $X^{-1}(B) \in \mathcal{F}$ for all such sets B , since the domain of $\mathbb{P}$ is $\mathcal{F}$. This is not a problem yet since we chose $\mathcal{F}$ to be the power set of Ω .

In our example, we can (for instance) calculate the probability that $X = 0$ or 3 (i.e., $X \in B := \{0, 3\}$) as

$$\mathbb{P}(\omega \in \Omega : X(\omega) \in \{0, 3\}) = \mathbb{P}(\{\text{HHH}, \text{TTT}\}) = \frac{2}{8}.$$

Similarly, the probability that the value of X lies between -2 and 2 (both inclusive) is (with $B = [-2, 2]$)

$$\mathbb{P}(X^{-1}([-2, 2])) = \mathbb{P}(\{\text{HHH}\}^c) = \frac{7}{8}.$$

Thus, associated with each random variable X is another function $\mathbb{P}_X$ that assigns probabilities to subsets of $\mathbb{R}$ by the formula

$$\mathbb{P}_X(B) := \mathbb{P}(X^{-1}(B)), \quad B \in \mathcal{B}(\mathbb{R}),$$

where $\mathcal{B}(\mathbb{R})$ is the collection of all subsets of $\mathbb{R}$ that are of interest to us (intervals, singletons, etc. — see *Borel sets* in the next chapter). One can show that $\mathbb{P}_X : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ also satisfies $\mathbb{P}1 - \mathbb{P}4$, with $\mathbb{R}$ taking the place of Ω , provided $\mathcal{B}(\mathbb{R})$ is “well-behaved” under set theoretic operations. The function $\mathbb{P}_X$ is called the *probability distribution* or *Law* of X .

Failure of Naive Probability

Unfortunately though, it is not always possible to assign probabilities to all subsets of a given sample space Ω that satisfy $\mathbb{P}1 - \mathbb{P}4$. We present two examples of when this is not possible or not appropriate.

Example 1: The Vitali Set

Consider the random experiment of picking a real number uniformly on the interval $[-1, 2]$. We model this experiment on the sample space $\Omega = [-1, 2]$ and (for now) take the set of events $\mathcal{F}$ to be the power set of Ω . Since the numbers are picked uniformly, we can define $\mathbb{P}$ on intervals as

$$\mathbb{P}((a, b)) = \mathbb{P}((a, b]) = \mathbb{P}([a, b)) = \mathbb{P}([a, b]) = \frac{b - a}{3}, \quad \text{for } -1 \leq a \leq b \leq 2.$$

It is clear how to extend $\mathbb{P}$ to sets A that are countable unions of disjoint intervals. What is not clear is how we can extend $\mathbb{P}$ to all of $\mathcal{F} = 2^\Omega$. We will take this for granted and assume that such an extension exists. Another useful property that $\mathbb{P}$ satisfies is *translation invariance*:

$$\mathbb{P}(x + A) = \mathbb{P}(A) \quad \text{for } x \in \mathbb{R} \text{ and } A \in \mathcal{F} \text{ such that } x + A \in \mathcal{F},$$

where $x + A := \{x + a : a \in A\}$. We will now show that such a construction is not possible, by constructing a set $V \subseteq \Omega$ such that $\mathbb{P}(V)$ does not exist.

Define the following equivalence relation on $\mathbb{R}$:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

The equivalence class of each $x \in \mathbb{R}$ is $[x] = x + \mathbb{Q} = \{x + q : q \in \mathbb{Q}\}$. Using the axiom of choice, we form a collection V , where V consists of exactly one representative v from each equivalence class, with $0 \leq v \leq 1$. Now let $\{q_1, q_2, \dots\} := \mathbb{Q} \cap [-1, 1]$ an enumeration of the rational numbers in $[-1, 1]$. For each $k \geq 1$, define $V_k := q_k + V$. Then the sets $\{V_k : k = 1, 2, \dots\}$ are pairwise disjoint. Indeed, let $n \neq m$ and assume there exists $z \in V_n \cap V_m$. Then $z = q_n + r = q_m + s$ where $r, s \in V$. This implies that $r - s = q_m - q_n \in \mathbb{Q} \implies r \sim s \implies [r] = [s]$, which is a contradiction since V contains only one representative from each class. Hence there is no such z .

Next we claim that

$$[0, 1] \subseteq \bigcup_{k \geq 1} V_k \subseteq [-1, 2]. \quad (1.1)$$

To see this, first let $x \in [0, 1]$. Then $x \sim v$ for some $v \in V$. Hence $x - v = q \in \mathbb{Q} \implies x = v + q$. Since $x, v \in [0, 1]$, it follows that $q \in [-1, 1]$ and $q = q_k$ for some $k \geq 1$, which implies that $x = v + q_k \in V_k$ for some $k \geq 1$. Hence

$$[0, 1] \subseteq \bigcup_{k \geq 1} V_k.$$

For the other inclusion, note that if $x \in \bigcup_{k \geq 1} V_k$, then $x = v + q_k$ for some $k \geq 1$ and $v \in V \subseteq [0, 1]$. So, since $q_k \in [-1, 1]$ and $v \in [0, 1]$, it follows that $x = v + q_k \in [-1, 2]$.

Since $\mathbb{P}$ is increasing, it follows from (1.1) that

$$\frac{1}{3} = \mathbb{P}([0, 1]) \leq \mathbb{P}\left(\bigcup_{k \geq 1} V_k\right) \leq \mathbb{P}([-1, 2]) = 1,$$

and by σ -additivity,

$$\frac{1}{3} \leq \sum_{k \geq 1} \mathbb{P}(V_k) \leq 1.$$

Since $\mathbb{P}$ is translation-invariant, $\mathbb{P}(V_k) = \mathbb{P}(q_k + V) = \mathbb{P}(V)$. Hence,

$$\frac{1}{3} \leq \sum_{k \geq 1} \mathbb{P}(V) \leq 1,$$

which is impossible, since the sum $\sum_{k \geq 1} \mathbb{P}(V)$ is either 0 (when $\mathbb{P}(V) = 0$) or ∞ (when $\mathbb{P}(V) > 0$). Thus, $\mathbb{P}(V)$ is undefined (i.e., V is non-measurable).

Another famous example of a non-measurable set is given by the *Banach-Tarski paradox*.

Example 2: (Lack of) Information

The second reason why we cannot always pick $\mathcal{F} = 2^\Omega$ has to do with the information available after the random experiment is conducted.

In general, after the random experiment is conducted, we are not always given the true outcome ω , but information that allows us to decide whether or not a collection of events have occurred. To see this, consider the coin toss experiment. Suppose that we only have information about the outcome of the first toss, rather than the full outcome of the experiment. Then the events that can be decided after the experiment are

$$A := \{HHH, HHT, HTH, HTT\}, \quad B := \{THH, THT, TTH, TTT\}, \quad \Omega, \quad \text{and} \quad \emptyset.$$

Indeed, the event A occurs precisely if the outcome of the first toss is H, while B occurs if the outcome of the first toss is T. The event Ω always occurs, while $\emptyset$ never occurs. Furthermore, no other events can be decided. For instance, the event $C := \{THT, THH\}$ cannot be decided since the information available does not distinguish between elements of B . Informally, the information can't see inside A and B — it simply treats the elements of each event as one block. If we knew that C occurred, that would automatically exclude $\{TTH, TTT\}$ from occurring. Such additional information is not available. It makes sense then to pick $\mathcal{F} = \{A, B, \emptyset, \Omega\}$ since these will be the events of interest.

In summary, if the information available after the random experiment is conducted allows us to decide the occurrence or non-occurrence of a collection of events $\mathcal{A}$, we will pick $\mathcal{F} = \mathcal{A}$. The special case when the true outcome ω is known corresponds to knowing whether or not every event $A \in 2^\Omega$ has occurred.

Now What?

These examples illustrate the fact that, in general, we cannot take the set of events $\mathcal{F}$ to be the power set of Ω ; we have to impose some kind of “access control” on the set of events $\mathcal{F}$.

This then begs the question: how do we decide which subsets are admissible events? The second example gave us a decision rule based on the information available after the experiment is conducted. However, the first example shows that for some experiments, there is a theoretical limit on which events are admissible. The Vitali set, for instance, can never be admissible in that experiment.

The fact that $\mathcal{F}$ is the domain of $\mathbb{P}$ gives us more information about the structure of $\mathcal{F}$. Indeed, if we know $\mathbb{P}(A)$ (i.e., if $A \in \mathcal{F}$) then we also know $\mathbb{P}(A^c) (= 1 - \mathbb{P}(A))$; thus, $A \in \mathcal{F} \implies A^c \in \mathcal{F}$. The same holds for unions and intersections of events. So, $\mathcal{F}$ must “closed” with respect to the common set theoretic operations. From $\mathbb{P}1$ and $\mathbb{P}4$, we also have that $\emptyset, \Omega \in \mathcal{F}$. It is easy to show that all these requirements are met if $\mathcal{F}$ satisfies the following conditions:

A1. $\emptyset \in \mathcal{F}$;

A2. For every $A \subseteq \Omega$, $A \in \mathcal{F} \implies A^c \in \mathcal{F}$;

$\mathcal{A}3$. For every $A, B \subseteq \Omega$, $A, B \in \mathcal{F} \implies A \cup B \in \mathcal{F}$;

$\sigma\mathcal{A}4$. If $\{A_n : n = 1, 2, 3, \dots\}$ is a collection of subsets of Ω with $A_n \in \mathcal{F}$ for each $n \geq 1$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

A collection of subsets of Ω that satisfies $\mathcal{A}1$, $\mathcal{A}2$ and $\mathcal{A}3$ is called an *algebra* on Ω , and an algebra that also satisfies $\sigma\mathcal{A}4$ is called a σ -*algebra*¹. Thus, the collection of admissible events $\mathcal{F}$ should necessarily be a σ -algebra. The members of $\mathcal{F}$ are also called *measurable sets*.

Sadly, by reducing $\mathcal{F}$, we have inadvertently tempered with our random variable X . Consider the three coin toss experiment again, now with information only on the outcome of the first toss. We therefore choose

$$\mathcal{F} = \{A, A^c, \emptyset, \Omega\},$$

where $A := \{\text{HHH}, \text{HHT}, \text{HTH}, \text{HTT}\}$. The probabilities are easily calculated as $\mathbb{P}(A) = \mathbb{P}(A^c) = 0.5$, $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$. Define X to be the number of heads again. We now cannot, for instance, calculate the probability that X takes the value 2, since the set

$$X^{-1}(\{2\}) = \{\omega \in \Omega : X(\omega) = 2\} = \{\text{HHT}, \text{HTH}, \text{THH}\}$$

is no-longer in $\mathcal{F}$ (thus $\mathbb{P}(X^{-1}(\{2\}))$ does not make sense since $\mathbb{P}$ is defined only for elements of $\mathcal{F}$). Another interpretation of this is that X contains more information than our sigma algebra $\mathcal{F}$. Knowing the value of X actually gives us more information than what is provided by $\mathcal{F}$. For instance, if $X = 2$, then we know that the event $\{\text{HHT}, \text{HTH}, \text{THH}\}$ has occurred, a conclusion we cannot make based on $\mathcal{F}$ alone. On the other hand, if we know that A has occurred, the value of X cannot be determined, since it could be any of 1, 2 or 3. In other words, the values of X *cannot be determined from the outcome of the random experiment*; we say that X is *not $\mathcal{F}$ -measurable*.

So we also need access control for random variables. To this end, we define $X : \Omega \rightarrow \mathbb{R}$ to be a random variable if and only if

$$\text{for any } B \in \mathcal{B}(\mathbb{R}), X^{-1}(B) \in \mathcal{F}.$$

This definition will solve our problems since $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$ will be well-defined for every ‘appropriate’ set $B \in \mathcal{B}(\mathbb{R})$.

Measure Theory

Finally, we observe that there are several other set functions that behave more or less like a probability. Here are some examples of set functions that satisfy conditions $\mathbb{P}1 - \mathbb{P}3$.

- Length of subsets of the real line. This is the function λ_1 that assigns to certain subsets of $\mathbb{R}$ the “length” of that subset. If $A = (a, b)$ is a bounded interval, then $\lambda_1(A) = b - a$. However, λ_1 is defined on a larger collection than intervals (the intervals by themselves do not form a σ -algebra). This *measure*, as we will refer to it, satisfies conditions $\mathbb{P}1 - \mathbb{P}3$, but not $\mathbb{P}4$. Also, $\lambda_1(\Omega) = \lambda_1(\mathbb{R}) = \infty$, hence λ_1 takes values on the *extended real numbers* $\bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$. This measure is called the *Lebesgue measure*, named after the French mathematician Henri Lebesgue, who was instrumental in laying the foundations of the field of *Measure Theory*.

¹note that for a σ -algebra, condition $\mathcal{A}3$ can be removed since it is implied by condition $\sigma\mathcal{A}4$.

- The above example can be extended to arbitrary dimensions to construct the N -dimensional Lebesgue measure λ_N on subsets of $\mathbb{R}^N$, where N is a positive integer. For $N = 2$ this corresponds to area in the plane, while λ_3 measures volume in $\mathbb{R}^3$.
- Mass for quantities of matter.
- Cardinality of sets also satisfies the same properties as well. We call this measure the *counting measure* and denote it by $\#$.

Given an arbitrary set Ω and a sigma algebra $\mathcal{F}$ of subsets of Ω , we say that a set function $\mu : \mathcal{F} \rightarrow \mathbb{R}$ is a *positive measure* (or *measure*) on $(\Omega, \mathcal{F})$ if it satisfies $\mathbb{P}1, \mathbb{P}2$ and $\mathbb{P}3$ (with μ replacing $\mathbb{P}$ of course). If, in addition, the measure μ also satisfies $\mathbb{P}4$, then μ is called a *probability measure*. This means that the study of measure theory contains probability theory as a special case! The pair $(\Omega, \mathcal{F})$ is called a *measurable space* and the triple $(\Omega, \mathcal{F}, \mu)$ is called a *measure space*. If μ is a probability measure, then $(\Omega, \mathcal{F}, \mu)$ is called a *probability space*. Thus, in our road to understanding probability theory we are going to take a detour through abstract measure theory. We will prove many important results that hold in general measure spaces and later apply them to probability spaces.

The Plan

Here's how we're going to do it:

1. Measure and Probability: Foundations. We will first lay the foundations, definitions and terminology used in measure theory, with an emphasis on probability theory. Here, we will give examples of measures in different measurable spaces, and prove several foundational results in the field. We also will define the *Borel σ -algebra* on $\mathbb{R}$ ($\mathcal{B}(\mathbb{R})$), which is the smallest σ -algebra containing the intervals.
2. Measurable functions and random variables. If $(\Omega, \mathcal{F})$ is a measurable space, we will define what it means for a function $f : \Omega \rightarrow \mathbb{R}$ to be measurable with respect to $\mathcal{F}$. For the special case when $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, we call $X : \Omega \rightarrow \mathbb{R}$ a random variable. On $\mathcal{B}(\mathbb{R})$, we define the *Law of X* as a probability measure $\mathbb{P}_X : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ such that $\mathbb{P}_X(B) := \mathbb{P}(X^{-1}(B))$ for each $B \in \mathcal{B}(\mathbb{R})$.
3. Integration and expectation. We are going to define the integral of a measurable function $f : \Omega \rightarrow \mathbb{R}$ with respect to a measure μ :

$$\int_{\Omega} f d\mu.$$

Again, when $f = X$ is a random variable and $\mu = \mathbb{P}$ is a probability measure, this integral defines the expected value of X :

$$\mathbb{E}(X) := \int_{\Omega} X d\mathbb{P}.$$

The main intuition behind the integral is that it is an abstraction of many forms of aggregation. That is, $\int_{\Omega} f d\mu$ represents a kind of “weighted sum” of the values of f , weighted by the measure of each value. This intuition is clear when $f = X$ is a discrete random variable. In other special cases, the integral specializes to the following:

- (a) When $\Omega = [a, b] \subseteq \mathbb{R}$ and $\mu = \lambda_1 =$ Lebesgue measure,

$$\int_{[a,b]} f d\lambda_1 = \int_a^b f(x) dx \quad (\text{Riemann integral})$$

if the Riemann integral exists.

(b) When $\Omega = \mathbb{N}$ and $\mu = \text{counting measure} = \#$, then

$$\int_{\mathbb{N}} f d\# = \sum_{n=0}^{\infty} f(n),$$

provided the series is absolutely convergent. So, summation is a type of integration.

These special cases (and many others) allow us to treat all the forms of aggregation familiar to us — Riemann integration, summation, expectation, etc — under one heading: the integral. We can then prove one theorem that can be applied to each of the special cases. For example, the following well-known results:

$$\sum_{n=0}^{\infty} (a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n; \quad \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

and

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y),$$

are all just one theorem:

$$\boxed{\int_{\Omega} (f + g) d\mu = \int_{\Omega} f d\mu + \int_{\Omega} g d\mu},$$

and they are obtained by choosing appropriate μ , f and Ω .

4. Products and independence. We then consider two separate measure spaces $(\Omega_1, \mathcal{F}_1, \mu_1)$ and $(\Omega_2, \mathcal{F}_2, \mu_2)$. When $\mu_i = \mathbb{P}_i$ ($i = 1, 2$) are probability measures, this corresponds to two independent random experiments. How do we combine them? We consider the product space $\Omega := \Omega_1 \times \Omega_2$ and the so-called *product σ -algebra* $\mathcal{F} := \mathcal{F}_1 \otimes \mathcal{F}_2$. On this measurable space $(\Omega, \mathcal{F})$, we define a measure $\mu := \mu_1 \otimes \mu_2$, called the *product measure* of μ_1 and μ_2 , that is compatible with μ_1 and μ_2 in a way that $\mu_1 \otimes \mu_2(A \times B) = \mu_1(A)\mu_2(B)$ if $A \in \mathcal{F}_1$ and $B \in \mathcal{F}_2$. We then prove the powerful Fubini-Tonelli theorems:

$$\int_{\Omega_1 \times \Omega_2} f d(\mu_1 \otimes \mu_2) = \int_{\Omega_1} \int_{\Omega_2} f d\mu_2 d\mu_1 = \int_{\Omega_2} \int_{\Omega_1} f d\mu_1 d\mu_2.$$

5. Finally, we generalize the idea of conditional expectation. We define the general conditional expectation of a random variable X given a σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}(X|\mathcal{G})$, as the ‘best’ $\mathcal{G}$ -measurable approximation of the random variable X .

Let’s have fun!

Chapter 2

Measure and Probability: Foundations

We begin with a broad look at the field of measure theory, with a strong emphasis on results that are useful in probability theory.

2.1 Basic Measure Theory

The aim of this section is to give a mathematical definition of the abstract notion of a measure on subsets of a given set Ω . As we have mentioned in the previous chapter, such a measure cannot in general be well-defined for all subsets of Ω if it is to satisfy some desired conditions. Thus, the domain of a measure will be a collection of ‘measurable’ subsets $\mathcal{F} \subseteq 2^\Omega$. We also convinced ourselves that $\mathcal{F}$ should be closed under some basic set-theoretic operations. The next definition makes this precise.

Definition 2.1.1. Let Ω be a (non-empty) set. A collection $\mathcal{A} \subseteq 2^\Omega$ of subsets of Ω is called an algebra on Ω if

$\mathcal{A}1.$ $\emptyset \in \mathcal{A}$;

$\mathcal{A}2.$ For every $A \subseteq \Omega$, if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$;

$\mathcal{A}3.$ For every $A, B \subseteq \Omega$, if $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

An algebra $\mathcal{F}$ is called a σ -algebra or σ -field if it also satisfies the following additional condition

$\sigma\mathcal{A}4.$ For every countable collection $\{A_n : n = 1, 2, 3, \dots\}$ of subsets of Ω with $A_n \in \mathcal{F}$ for each $n \geq 1$,

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Remark 2.1.1. Note that if $\mathcal{A}$ is an algebra on Ω , then by $\mathcal{A}1$ and $\mathcal{A}2$, we have $\emptyset^c = \Omega \in \mathcal{A}$ ■

It is easy to see (by induction) that an algebra $\mathcal{A}$ is closed under finite unions; i.e., if $A_1, \dots, A_n$ are elements of $\mathcal{A}$, then their union $A_1 \cup \dots \cup A_n$ is also in $\mathcal{A}$. A σ -algebra extends this condition to countably infinite unions. Also, an application of DeMorgan’s Laws shows that an algebra is also closed under finite intersections; the same applies to a σ -algebra and countable intersections. Thus, an algebra is a collection of subsets of Ω that is closed under finitely many basic set-theoretic operations, while a σ -algebra extends this closure property to countably many such operations.

Example 2.1.1. Here are some examples of algebras and σ -algebras. Check that these are indeed algebras or σ -algebras.

1. The “smallest” σ -algebra on any set Ω is $\mathcal{A} := \{\emptyset, \Omega\}$. Any other algebra (or σ -algebra) must necessarily contain it.

2. On the other hand, the “largest” σ -algebra on a set Ω is the power set of Ω , 2^Ω . Every algebra (or σ -algebra) is contained in 2^Ω .
3. Let $A \subseteq \Omega$, then $\mathcal{F}_A := \{A, A^c, \emptyset, \Omega\}$ is a σ -algebra (and therefore, an algebra) on Ω . We call this the (σ) -algebra *generated* by $\{A\}$.
4. For a concrete example, let $\Omega = \{1, 2, 3, 4\}$ and

$$\mathcal{F} = \{\{1\}, \{2, 3\}, \{4\}, \{2, 3, 4\}, \{1, 2, 3\}, \{1, 4\}, \emptyset, \Omega\}.$$

One can easily check that $\mathcal{F}$ is a σ -algebra on Ω .

Notice that in all the examples above, the number of elements in each algebra is a power of 2. We will see later that this is not a coincidence. Also, for finite collections, the notions of an algebra and a σ -algebra coincide. ■

Example 2.1.2. Now we consider more complicated examples.

1. Let $\Omega = \mathbb{R}$ and choose $\mathcal{F} = \{A \subseteq \mathbb{R} : \text{for every } x \in \mathbb{R}, x \in A \iff -x \in A\}$ to be the collection of all symmetric subsets of $\mathbb{R}$. This is a σ -algebra on $\mathbb{R}$.
2. Let $\Omega = \mathbb{R}$ again and define

$$\mathcal{A} = \{A \subseteq \mathbb{R} : A = \cup_{k=1}^n I_k \text{ for some intervals } I_k \subseteq \mathbb{R}, k = 1, 2, \dots, n, n \in \mathbb{N}^+\}$$

to be subsets of $\mathbb{R}$ that are finite unions of (open, closed, bounded, empty, etc) intervals in $\mathbb{R}$. Then $\mathcal{A}$ is an algebra but not a σ -algebra, since (for instance) $A_n := \{n\} = [n, n] \in \mathcal{A}$ for each $n \in \mathbb{N}$, yet $\cup_{n \in \mathbb{N}} A_n = \mathbb{N} \notin \mathcal{A}$.

3. Let Ω be an infinite set. Define

$$\mathcal{A} := \{A \subseteq \Omega : A \text{ is finite or } A^c \text{ is finite}\}.$$

Then $\mathcal{A}$ is an algebra but not a sigma algebra.

4. Let Ω be an infinite set. Define

$$\mathcal{F} := \{A \subseteq \Omega : A \text{ is countable or } A^c \text{ is countable}\}.$$

Then $\mathcal{F}$ is a sigma algebra. ■

If Ω is a set and $\mathcal{F}$ is a sigma algebra on Ω , we will call the pair $(\Omega, \mathcal{F})$ a *measurable space* and the elements of $\mathcal{F}$ will be called *measurable sets*.

Definition 2.1.2. Let $(\Omega, \mathcal{F})$ be a measurable space and $\mu : \mathcal{F} \rightarrow \bar{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$ be a set function. We say that μ is a *positive measure* or *measure* on $(\Omega, \mathcal{F})$ if

$$\mathcal{M1.} \quad \mu(\emptyset) = 0;$$

$$\mathcal{M2.} \quad \mu(A) \geq 0 \text{ for every } A \in \mathcal{F};$$

$\mathcal{M3.}$ If $\{A_n : n = 1, 2, 3, \dots\}$ is a countable collection of pairwise disjoint elements of $\mathcal{F}$, then

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

This is called σ -additivity.

The triple $(\Omega, \mathcal{F}, \mu)$ is then called a *measure space*. If the measure μ also satisfies

$$\mathbb{P}4. \mu(\Omega) = 1,$$

then we call μ a *probability measure*, and the triple $(\Omega, \mathcal{F}, \mu)$ is called a *probability space*. We will use $\mathbb{P}$ when μ is a probability measure; thus $(\Omega, \mathcal{F}, \mathbb{P})$ will always denote a probability space.

As alluded to in the previous chapter, a measure generalises the notions of length, area, volume, probability, cardinality, and mass. We are going to study all these notions of ‘measure’ under one heading: Measure Theory.

Example 2.1.3. Here are some examples of measures.

1. The Dirac Measure. Let Ω be any non-empty set and pick $a \in \Omega$. If we let $\mathcal{F} = 2^\Omega$ and define $\delta_a : \mathcal{F} \rightarrow \mathbb{R}$ by

$$\delta_a(A) := \begin{cases} 1 & \text{if } a \in A \\ 0 & \text{otherwise,} \end{cases}$$

then δ_a is a probability measure on $(\Omega, \mathcal{F})$, called the *Dirac measure*.

2. The Counting Measure. Let Ω be a set and define $\# : 2^\Omega \rightarrow \bar{\mathbb{R}}$ by

$$\#(A) := \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{otherwise,} \end{cases}$$

where $|A|$ represents the cardinality (number of elements) of A . Then this is a measure on $(\Omega, 2^\Omega)$, called the *counting measure*.

3. Lebesgue Measure. Later in the chapter we are going to define the so-called *Lebesgue Measure* on $\mathbb{R}$. This measure λ_1 is defined on a sigma algebra that includes all the intervals, and it extends the notion of length of an interval to such sets. If $I = (a, b)$ is an interval, then

$$\lambda_1(I) = b - a.$$

The N -dimensional analogue of Lebesgue measure is a measure λ_N on some subsets of $\mathbb{R}^N$ that extends area for $N = 2$ and volume for $N = 3$.

■

Definition 2.1.3. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. The measure μ is said to be a *finite measure* if $\mu(\Omega) < \infty$. We say that μ is *σ -finite* if there exists a countable collection $\{A_n : n = 1, 2, 3, \dots\}$ such that

$$\text{for every } n \geq 1, A_n \in \mathcal{F}, \quad \mu(A_n) < \infty,$$

and

$$\Omega = \bigcup_{n=1}^{\infty} A_n.$$

All probability measures are finite measures. The counting measure defined above is finite if and only if Ω is a finite set, and is σ -finite if and only if Ω is countable (check this!). We collect some basic properties of measures.

Proposition 2.1.1. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and $A, B, A_n \in \mathcal{F}$ for each $n \geq 1$.

1. (*Additivity*) If $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$. More generally, if $A_1, \dots, A_n$ are pairwise disjoint, then

$$\mu\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i).$$

2. (Increasing) If $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
3. If $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
4. If $\mu(A \cap B) < \infty$, then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$.
5. (σ -sub-additivity)

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

Proof. 1. Let $F_1 = A$, $F_2 = B$ and $F_n = \emptyset$ for $n \geq 3$. The (F_n) are pairwise disjoint. Then by σ -additivity we have

$$\mu(A \cup B) = \mu \left(\bigcup_{n=1}^{\infty} F_n \right) = \sum_{n=1}^{\infty} \mu(F_n) = \mu(F_1) + \mu(F_2) + 0 + \dots = \mu(A) + \mu(B).$$

The second part follows by induction (or by a similar construction).

2. By additivity, $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$ since $\mu(B \setminus A) \geq 0$.
3. This follows from 2 by subtraction.
4. Note that $A \cup B = (A \setminus (A \cap B)) \cup (B \setminus (A \cap B)) \cup (A \cap B)$ and the three sets are disjoint. Apply 1 and 3.
5. Let $F_1 := A_1$, $F_2 = A_2 \setminus A_1$, $F_3 = A_3 \setminus (A_1 \cup A_2)$, $\dots$, $F_n = A_n \setminus (\cup_{i=1}^{n-1} A_i)$, $\dots$. Clearly (F_n) are pairwise disjoint with

$$\bigcup_{n=1}^{\infty} F_n = \bigcup_{n=1}^{\infty} A_n \text{ and } \mu(F_n) \leq \mu(A_n) \quad n \geq 1.$$

Hence

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \mu \left(\bigcup_{n=1}^{\infty} F_n \right) = \sum_{n=1}^{\infty} \mu(F_n) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

■

Remark 2.1.2. Properties 3 and 4 do not hold in general without the assumption of finite measures of the sets concerned. ■

Exercise 2.1.1. Let $(\Omega, \mathcal{F})$ be a measurable space and $\mu : \mathcal{F} \rightarrow \bar{\mathbb{R}}$ be a function such that $\mu(A) \geq 0$ and $\mu(\emptyset) = 0$. Show that the following are equivalent:

1. μ is σ -additive
2. μ is additive and σ -sub-additive.

■

Proposition 2.1.2. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, and $A_n \in \mathcal{F}$ for each $n \geq 1$.

1. (Continuity from below) If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

2. (Continuity from above) If $A_1 \supseteq A_2 \supseteq \dots$ and $\mu(A_{n_0}) < \infty$ for some $n_0 \geq 1$, then

$$\mu \left(\bigcap_{n=1}^{\infty} A_n \right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Proof. We prove 1 and leave 2 as an exercise. Let $F_1 = A_1$, $F_2 = A_2 \setminus A_1, \dots, F_n = A_n \setminus A_{n-1}, \dots$. Then (F_n) are pairwise disjoint with

$$A_n = \bigcup_{i=1}^n A_i = \bigcup_{i=1}^n F_i, \quad \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} F_n.$$

Hence

$$\begin{aligned} \mu \left(\bigcup_{n=1}^{\infty} A_n \right) &= \mu \left(\bigcup_{n=1}^{\infty} F_n \right) = \sum_{n=1}^{\infty} \mu(F_n) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(F_i) = \lim_{n \rightarrow \infty} \mu \left(\bigcup_{i=1}^n F_i \right) \\ &= \lim_{n \rightarrow \infty} \mu \left(\bigcup_{i=1}^n A_i \right) = \lim_{n \rightarrow \infty} \mu(A_n). \end{aligned}$$

■

Definition 2.1.4. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $N \subseteq \Omega$. We call N a μ -null set if there exists $N' \in \mathcal{F}$ such that $N \subseteq N'$ and $\mu(N') = 0$. The measure space $(\Omega, \mathcal{F}, \mu)$ is complete if $\mathcal{F}$ contains all the μ -null sets.

As the definition suggests, not all null sets are measurable, even though their ‘measure’ ought to be zero. If a measure space is not complete, the next result gives a way of ‘completing’ it by carefully appending null sets to every element of $\mathcal{F}$. First we need the following Lemma.

Lemma 2.1.1. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

1. If $\{N_n : n = 1, 2, 3, \dots\}$ is a countable collection of μ -null sets, then

$$N = \bigcup_{n=1}^{\infty} N_n$$

is also a μ -null set.

2. If μ is a probability measure and $\{A_n : n = 1, 2, 3, \dots\}$ is a countable collection of events in $\mathcal{F}$ with $\mu(A_n) = 1$ for each $n \geq 1$, then

$$\mu \left(\bigcap_{n=1}^{\infty} A_n \right) = 1.$$

Proof. We prove 1 and leave the second one as an exercise. Since each N_n is a null set, there exists (for each $n \in \mathbb{N}^+$) $N'_n \in \mathcal{F}$ such that $N_n \subseteq N'_n$ and $\mu(N'_n) = 0$. Using the fact that μ is σ -sub-additive, we get

$$\mu \left(\bigcup_{n=1}^{\infty} N'_n \right) \leq \sum_{n=1}^{\infty} \mu(N'_n) = 0.$$

Since $N \subseteq \bigcup_{n=1}^{\infty} N'_n$, it follows that N is a null set.

■

Proposition 2.1.3. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $\mathcal{N}_\mu$ be the set of all μ -null sets. Define $\bar{\mathcal{F}} \subseteq 2^\Omega$ and $\bar{\mu} : \bar{\mathcal{F}} \rightarrow \mathbb{R}$ by

$$\bar{\mathcal{F}} := \{A \cup N : A \in \mathcal{F}, N \in \mathcal{N}_\mu\} \text{ and } \bar{\mu}(A \cup N) := \mu(A) \text{ for } A \in \mathcal{F}, N \in \mathcal{N}_\mu.$$

Then

1. $\bar{\mathcal{F}}$ is a σ -algebra on Ω that contains $\mathcal{F}$;
2. $\bar{\mu}$ is a measure on $\bar{\mathcal{F}}$;
3. $\bar{\mu} = \mu$ on $\mathcal{F}$;
4. $(\Omega, \bar{\mathcal{F}}, \bar{\mu})$ is complete.

We call $(\Omega, \bar{\mathcal{F}}, \bar{\mu})$ the completion of $(\Omega, \mathcal{F}, \mu)$.

Proof. 1. Clearly $\emptyset = \emptyset \cup \emptyset \in \bar{\mathcal{F}}$. Now let $A \cup N \in \bar{\mathcal{F}}$, where $A \in \mathcal{F}$ and $N \in \mathcal{N}_\mu$. Choose $N' \in \mathcal{F}$ such that $N \subseteq N'$ and $\mu(N') = 0$. Then $(A \cup N)^c = (A \cup N')^c \cup (N' \cap N^c) \in \bar{\mathcal{F}}$ since $(A \cup N')^c \in \bar{\mathcal{F}}$ and $N' \cap N^c \subseteq N' \in \mathcal{N}_\mu$. Finally, if $(A_n \cup N_n)_{n=1}^\infty$ is a sequence in $\bar{\mathcal{F}}$, then

$$\bigcup_{n=1}^\infty (A_n \cup N_n) = \left(\bigcup_{n=1}^\infty A_n \right) \cup \left(\bigcup_{n=1}^\infty N_n \right) \in \bar{\mathcal{F}}.$$

Since $\emptyset$ is a null set, it follows that $A = A \cup \emptyset \in \bar{\mathcal{F}}$ for every $A \in \mathcal{F}$. Hence $\mathcal{F} \subseteq \bar{\mathcal{F}}$.

2. We have $\bar{\mu}(\emptyset) = 0$ and $\bar{\mu}(A \cup N) = \mu(A) \geq 0$. If $(A_n \cup N_n)$ is a sequence of pairwise disjoint members of $\bar{\mathcal{F}}$, then

$$\bar{\mu} \left(\bigcup_{n=1}^\infty (A_n \cup N_n) \right) = \bar{\mu} \left(\bigcup_{n=1}^\infty A_n \cup \bigcup_{n=1}^\infty N_n \right) = \mu \left(\bigcup_{n=1}^\infty A_n \right) = \sum_{n=1}^\infty \mu(A_n) = \sum_{n=1}^\infty \bar{\mu}(A_n \cup N_n).$$

3. If $A \in \mathcal{F}$, then

$$\bar{\mu}(A) = \bar{\mu}(A \cup \emptyset) = \mu(A).$$

4. Let $B \subseteq \Omega$ be a null set of $\bar{\mu}$. This means that there exists a set $B' = C \cup N \in \bar{\mathcal{F}}$ ($C \in \mathcal{F}$, $N \in \mathcal{N}_\mu$) such that $B \subseteq B'$ and $\bar{\mu}(B') = 0$. But this implies that C is a null set of μ , hence $B \in \mathcal{N}_\mu \implies B = \emptyset \cup B \in \bar{\mathcal{F}}$. So the space is complete. ■

Example 2.1.4. Let $\Omega = \{1, 2, 3, 4, 5\}$, $\mathcal{F} = \{\{1, 2\}, \{3, 4, 5\}, \Omega, \emptyset\}$ and define the measure $\mu : \mathcal{F} \rightarrow \mathbb{R}$ by

$$\mu(\{1, 2\}) = \mu(\emptyset) = 0 \text{ and } \mu(\{3, 4, 5\}) = \mu(\Omega) = 1.$$

It is easy to see that μ is a probability measure. The null sets of μ are

$$\mathcal{N}_\mu = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Since $\{1\} \notin \mathcal{F}$, $(\Omega, \mathcal{F}, \mu)$ is not complete. Its completion is

$$\bar{\mathcal{F}} = \{\{1, 2\}, \{3, 4, 5\}, \Omega, \emptyset, \{1\}, \{2\}, \{1, 3, 4, 5\}, \{2, 3, 4, 5\}\}$$

and

$$\begin{aligned} \bar{\mu}(\{1, 2\}) &= \bar{\mu}(\{2\}) = \bar{\mu}(\{1\}) = \bar{\mu}(\{\emptyset\}) = 0 \\ \bar{\mu}(\{3, 4, 5\}) &= \bar{\mu}(\{1, 3, 4, 5\}) = \bar{\mu}(\{2, 3, 4, 5\}) = \bar{\mu}(\Omega) = 1. \end{aligned}$$
■

Definition 2.1.5. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $P = P(\omega)$ be a property that may or may not hold for each $\omega \in \Omega$. We say that P holds μ -almost everywhere in Ω if there exists a μ -null set $N \subseteq \Omega$ such that $P(\omega)$ holds for all $\omega \in N^c$. When $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, we will say that P holds $\mathbb{P}$ -almost surely.

In the above example, the function $f : \Omega \rightarrow \mathbb{R}$ defined by $f(x) = (x-3)(x-4)(x-5)(x-1)$ is equal to zero μ -almost everywhere $x \in \Omega$. If there is no possible confusion, we will simply use “almost everywhere” instead.

We say that $\mu : \mathcal{F} \rightarrow \bar{\mathbb{R}}$ is *concentrated* on a set $A \subseteq \Omega$ if A^c is μ -null. In the previous example, the measure μ is concentrated on $\{3, 4, 5\}$ (or on any superset of it).

If μ_1 and μ_2 are positive measures on $(\Omega, \mathcal{F})$ and a, b are non-negative real numbers, then the set function $\mu := a\mu_1 + b\mu_2$ defined by

$$\mu(A) := a\mu_1(A) + b\mu_2(A), \quad A \in \mathcal{F},$$

is also a measure on $(\Omega, \mathcal{F})$ (show this as an exercise). The next result extends this to infinite sums. First we need the following result.

Exercise 2.1.2 (Tonelli’s Theorem). Let $(a_{n,m})_{n,m=1}^\infty$ be a double sequence with non-negative entries. Show that

$$\sum_{(m,n) \in \mathbb{N}^+ \times \mathbb{N}^+} a_{n,m} = \sum_{n=1}^\infty \sum_{m=1}^\infty a_{n,m} = \sum_{m=1}^\infty \sum_{n=1}^\infty a_{n,m},$$

where

$$\sum_{(m,n) \in \mathbb{N}^+ \times \mathbb{N}^+} a_{n,m} := \sup_F \sum_{(m,n) \in F} a_{n,m}$$

and the supremum is taken over finite sets $F \subseteq \mathbb{N}^+ \times \mathbb{N}^+$. ■

Proposition 2.1.4. Let $(a_k)_{k=1}^\infty$ be a sequence of non-negative real numbers and $(\mu_k)_{k=1}^\infty$ be a sequence of positive measures on $(\Omega, \mathcal{F})$. Then $\mu := \sum_{k=1}^\infty a_k \mu_k$ defined by

$$\mu(A) := \sum_{k=1}^\infty a_k \mu_k(A), \quad A \in \mathcal{F}$$

is also a positive measure on $(\Omega, \mathcal{F})$.

Proof. First

$$\mu(\emptyset) = \sum_{k=1}^\infty a_k \mu_k(\emptyset) = \sum_{k=1}^\infty 0 = 0.$$

It is also clear that $\mu(A) \geq 0$ for each $A \in \mathcal{F}$, so we need to show σ -additivity. To this end, let $(A_n)_{n=1}^\infty$ be a sequence of pairwise disjoint members of $\mathcal{F}$. Then

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{k=1}^\infty a_k \mu_k\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{k=1}^\infty \sum_{n=1}^\infty a_k \mu_k(A_n) = \sum_{n=1}^\infty \sum_{k=1}^\infty a_k \mu_k(A_n) = \sum_{n=1}^\infty \mu(A_n).$$
■

2.2 More about σ -algebras

We now take a closer look at how σ -algebras are constructed. Fix a set Ω and let $\mathcal{A}$ be a collection of subsets of Ω . Out of all the sigma-algebras that contain $\mathcal{A}$, we want to find one that is smallest with respect to set inclusion. We will call this the *sigma algebra generated* by $\mathcal{A}$ and denote it by $\sigma(\mathcal{A})$. The idea behind finding $\sigma(\mathcal{A})$ is that we can create a sigma algebra by prescribing which elements $\mathcal{A}$ must be in the sigma algebra. Since the collection $\mathcal{A}$ might not be a sigma algebra on its own, $\sigma(\mathcal{A})$ extends $\mathcal{A}$ into a sigma algebra by adding the fewest sets possible (i.e., only the necessary sets, nothing more).

Proposition 2.2.1. *Let Ω be a set and $\mathcal{A} \subseteq 2^\Omega$ be a collection of subsets of Ω . Then $\sigma(\mathcal{A})$ exists and is equal to the intersection of all sigma algebras containing $\mathcal{A}$.*

Proof. Let Γ be the collection of all sigma algebras that contain $\mathcal{A}$. The set Γ is non-empty, since $2^\Omega \in \Gamma$. The intersection of all members of Γ clearly satisfies the required conditions (check as an exercise). ■

We now define the most important sigma algebra in the course.

Definition 2.2.1. *Let $\mathcal{A} := \{(a, b) : a, b \in \mathbb{R}, a < b\}$ be the set of all open intervals in $\mathbb{R}$. We define the Borel sigma algebra of $\mathbb{R}$, $\mathcal{B}(\mathbb{R})$, as the sigma algebra generated by $\mathcal{A}$. That is*

$$\mathcal{B}(\mathbb{R}) := \sigma(\mathcal{A}).$$

The elements of $\mathcal{B}(\mathbb{R})$ are called Borel sets.

Theorem 2.2.1. *The following collections of sets are all Borel sets. Furthermore, $\mathcal{B}(\mathbb{R})$ is generated by any of them.*

1. All closed intervals $\{[a, b] : a < b\}$
2. All half-open intervals $\{(a, b] : a < b\}$ or $\{[a, b) : a < b\}$
3. All the unbounded intervals $\{(-\infty, b) : b \in \mathbb{R}\}$ or $\{(-\infty, b] : b \in \mathbb{R}\}$
4. All the unbounded intervals $\{(a, \infty) : a \in \mathbb{R}\}$ or $\{[a, \infty) : a \in \mathbb{R}\}$.

Proof. We show the first one and leave the rest as an exercise. If $a, b \in \mathbb{R}$ with $a < b$, then

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n} \right) \in \mathcal{B}(\mathbb{R}).$$

Conversely,

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \frac{1}{n}, b - \frac{1}{n} \right].$$

■

Example 2.2.1. The following sets are Borel sets.

1. Singletons $\{x\}$ for each $x \in \mathbb{R}$. To see this, note that $\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x + \frac{1}{n} \right)$.
2. All countable sets. Indeed, if $A \subseteq \mathbb{R}$ is countable, then

$$A = \bigcup_{x \in A} \{x\}.$$

So, in particular, $\mathbb{N}, \mathbb{Z}$ and $\mathbb{Q}$ are Borel sets.

3. All compliments of countable sets, i.e., *co-countable* sets. So, in particular, the irrationals $\mathbb{R} \setminus \mathbb{Q}$ is a Borel set.
4. All intervals or countable unions of intervals, by the previous theorem.
5. All open¹ subsets of $\mathbb{R}$. All closed subsets of $\mathbb{R}$.

¹A set $U \subseteq \mathbb{R}$ is open if for every $x \in U$ there exists an open interval $I_x \subseteq \mathbb{R}$ such that $x \in I_x \subseteq U$. A set $F \subseteq \mathbb{R}$ is closed if $\mathbb{R} \setminus F$ is open.

Remark 2.2.1. By the density of $\mathbb{Q}$ in $\mathbb{R}$, $\mathcal{B}(\mathbb{R})$ is also generated by

$$\{(a, b) : a, b \in \mathbb{Q}\},$$

which is a countable collection of sets. A sigma algebra that is generated by countably many sets is said to be *countably generated*. So, $\mathcal{B}(\mathbb{R})$ is countably generated. ■

The above example might suggest that all subsets of $\mathbb{R}$ are Borel sets (i.e., $\mathcal{B}(\mathbb{R}) = 2^{\mathbb{R}}$). This is not true! There are subsets of $\mathbb{R}$ that are not Borel sets, but their construction is beyond the scope of the course. In fact, $|\mathcal{B}(\mathbb{R})| = |\mathbb{R}| = \mathfrak{c} < |2^{\mathbb{R}}|$.

If $\Omega \subseteq \mathbb{R}$, we define the Borel σ -algebra on Ω , $\mathcal{B}(\Omega)$, as $\mathcal{B}(\mathbb{R})$ restricted to Ω . That is,

$$\mathcal{B}(\Omega) := \{A \cap \Omega : A \in \mathcal{B}(\mathbb{R})\}.$$

Check that this is indeed a σ -algebra on Ω .

Finally we define the Borel σ -algebra of $\mathbb{R}^N$, $\mathcal{B}(\mathbb{R}^N)$. First define an *open rectangle* $R \subseteq \mathbb{R}^N$ as a set of the form

$$R = \prod_{i=1}^N (a_i, b_i) = \{x = (x_1, \dots, x_N) : a_i < x_i < b_i, i = 1, 2, \dots, N\}, \quad a_i < b_i.$$

We define

$$\mathcal{B}(\mathbb{R}^N) := \sigma(\{R : R \text{ is an open rectangle}\}).$$

The Borel σ -algebra of a subset $\Omega \subseteq \mathbb{R}^N$ is defined similarly.

Definition 2.2.2. Let Ω be a set and $\mathcal{A} \subseteq 2^{\Omega}$ be a collection of subsets of Ω .

1. We say that $\mathcal{A}$ is a π -system if it is closed under finite intersections; i.e., if for every $A, B \subseteq \Omega$, $[A, B \in \mathcal{A} \implies A \cap B \in \mathcal{A}]$.
2. We say that $\mathcal{A}$ is a λ -system if

$\lambda 1.$ $\Omega \in \mathcal{A}$;

$\lambda 2.$ If $A, B \in \mathcal{A}$ and $A \subseteq B$, then $B \setminus A \in \mathcal{A}$;

$\lambda 3.$ If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$ is an increasing sequence of sets with $A_n \in \mathcal{A}$ for each $n \geq 1$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}.$$

We will call $\pi(\mathcal{A})$ (resp. $\lambda(\mathcal{A})$) the π -system (resp. λ -system) generated by $\mathcal{A}$ (i.e., the smallest π -system (λ -system) containing $\mathcal{A}$).

It is clear that every sigma algebra is both a π and a λ -system. The converse is also true.

Lemma 2.2.1. A collection of subsets $\mathcal{E} \subseteq 2^{\Omega}$ is a sigma algebra on Ω if and only if it is both a π -system and a λ -system.

Proof. ($\implies$): It is clear that a sigma algebra is both a π and a λ -system.

($\impliedby$): Now assume that $\mathcal{E}$ is both a π and a λ system. By $\lambda 1$ and $\lambda 2$ we get $\emptyset = \Omega \setminus \Omega \in \mathcal{E}$. If $A \in \mathcal{E}$, then $A^c = \Omega \setminus A \in \mathcal{E}$ by $\lambda 1$ and $\lambda 2$. Now let $(A_n)_{n=1}^\infty$ be elements of $\mathcal{E}$. Define the sequence (F_n) by $F_1 = A_1$, $F_2 = A_1 \cup A_2, \dots, F_n = A_1 \cup \dots \cup A_n, \dots$. Then (F_n) is an increasing sequence with $F_n = (A_1^c \cap A_2^c \cap \dots \cap A_n^c)^c \in \mathcal{E}$ for each $n \geq 1$ since $\mathcal{E}$ is a π -system that is now closed under taking complements. Thus,

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} F_n \in \mathcal{E}$$

by property $\lambda 3$. Hence $\mathcal{E}$ is a σ -algebra. ■

The next two results are fundamental.

Theorem 2.2.2 (Monotone Class Theorem). *Let $\mathcal{A}$ be a π -system. Then $\sigma(\mathcal{A}) = \lambda(\mathcal{A})$.*

Proof. We give a sketch of the proof. Since $\sigma(\mathcal{A})$ is also a λ -system containing $\mathcal{A}$, it follows that $\lambda(\mathcal{A}) \subseteq \sigma(\mathcal{A})$. To show the reverse inclusion, we need to show that $\lambda(\mathcal{A})$ is a σ -algebra containing $\mathcal{A}$. Since $\lambda(\mathcal{A})$ is a λ -system, it is enough to show that it is also a π -system by the previous result. Pick $A \in \mathcal{A}$ and define

$$\mathcal{D}_A := \{B \in \lambda(\mathcal{A}) : A \cap B \in \lambda(\mathcal{A})\}.$$

Show that $\mathcal{D}_A$ is a λ -system that contains $\mathcal{A}$, and conclude that $\mathcal{D}_A = \lambda(\mathcal{A})$. Finally, define $\mathcal{D}_A$ for any $A \in \lambda(\mathcal{A})$ and show again that $\mathcal{D}_A$ is a λ -system that contains $\mathcal{A}$. ■

Theorem 2.2.3 (Dynkin's $\pi - \lambda$ Theorem). *Let $\mathcal{A}$ be a π -system and $\mathcal{D}$ be a λ -system such that $\mathcal{A} \subseteq \mathcal{D}$. Then $\sigma(\mathcal{A}) \subseteq \mathcal{D}$.*

Proof. Since $\mathcal{A}$ is a π system, the previous result implies that $\sigma(\mathcal{A}) = \lambda(\mathcal{A}) \subseteq \mathcal{D}$, since $\lambda(\mathcal{A})$ is the smallest λ -system containing $\mathcal{A}$. ■

These results imply the following useful uniqueness result for measures.

Theorem 2.2.4. *Let Ω be a set, $\mathcal{A} \subseteq 2^\Omega$ be a π -system on Ω and $\mathcal{F} := \sigma(\mathcal{A})$ be the sigma algebra generated by $\mathcal{A}$. Let $\mu_1, \mu_2 : \mathcal{F} \rightarrow \bar{\mathbb{R}}$ be two measures on $\mathcal{F}$ with $\mu_1(\Omega) = \mu_2(\Omega) < \infty$. If $\mu_1(A) = \mu_2(A)$ for every $A \in \mathcal{A}$, then $\mu_1(A) = \mu_2(A)$ for every $A \in \mathcal{F}$. That is, if μ_1 and μ_2 are finite measures that agree on a π -system generating $\mathcal{F}$, then they agree everywhere on $\mathcal{F}$.*

Proof. Again we give a sketch. We use the “principle of appropriate sets”. Define

$$\mathcal{D} := \{A \in \mathcal{F} : \mu_1(A) = \mu_2(A)\}$$

and show that $\mathcal{D}$ is a λ -system containing $\mathcal{A}$ as use the $\pi - \lambda$ theorem to conclude that $\mathcal{D} = \mathcal{F}$. ■

Let us now take a closer look at the semantics of sigma algebras, which will be useful in studying probability spaces. This is the point of view that sigma algebras contain information. Consider the random experiment of tossing a fair six-sided die. The sample space of this experiment is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Suppose the actual outcome ω of the experiment is not observed entirely, but the only information available after the experiment is conducted is whether or not ω is even and whether or not ω is prime. This means that at the end of the experiment, we can decide whether or not the following events have occurred:

$$A := \{2, 4, 6\}, \quad A^c = \{1, 3, 5\}, \quad B := \{2, 3, 5\} \quad \text{and} \quad B^c = \{1, 4, 6\}.$$

In addition, we can also decide whether or not certain combinations of these events have occurred. For instance, we can also decide whether or not the outcome is $\{2\}$, since this occurs if and only if the outcome ω is even ($\omega \in A$) and prime ($\omega \in B$) ($\{2\} = A \cap B$). If we consider all the

information that can be extracted from the available outcomes, we see that the set of events which can be decided after the experiment is

$$\begin{aligned} \mathcal{F} &= \{A, B, A \cap B, A \cup B, A^c, B^c, (A \cup B)^c, (A \cap B)^c, A^c \cap B, A^c \cup B\} \\ &\cup \{A \cap B^c, A \cup B^c, (A \cap B) \cup (A \cup B)^c, (A \cap B^c) \cup (A^c \cap B), \emptyset, \Omega\} \\ &= \{\{1\}, \{2\}, \{3, 5\}, \{4, 6\}, \{1, 2\}, \{1, 3, 5\}, \{1, 4, 6\}, \{2, 3, 5\}, \{2, 4, 6\}, \{3, 4, 5, 6\}\} \cup \\ &\quad \{\{1, 2, 3, 5\}, \{1, 2, 4, 6\}, \{1, 3, 4, 5, 6\}, \{2, 3, 4, 5, 6\}, \Omega, \emptyset\}, \end{aligned}$$

which is exactly the sigma algebra generated by $\{A, B\}$. Thus, the given two sources of information generate a sigma algebra of events that can be decided based on that information. Convince yourself that all the events listed above can indeed be decided based only on knowledge of whether or not each of A or B occurred.

Now if we take a closer look at the sigma algebra $\mathcal{F}$ above, we observe the following:

- Every member of $\mathcal{F}$ is a union of some of $\{1\}, \{2\}, \{3, 5\}$ and $\{4, 6\}$ ($\emptyset$ corresponds to the empty union). These sets also form a partition of Ω (i.e., they are non-empty, disjoint and their union is all of Ω).
- The elements 3 and 5 always appear together in any set. That is, the information available cannot distinguish between the outcomes 3 and 5. Thus, as far as the information available goes, the set $\{3, 5\}$ can be thought of as one element. We will call such a set an *atom* or *block*. The same is true for 4 and 6. The blocks of $\mathcal{F}$ are $\{1\}, \{2\}, \{3, 5\}$ and $\{4, 6\}$.
- The sigma algebra $\mathcal{F}$ is also generated by the blocks. That is, $\mathcal{F} = \sigma(\{\{1\}, \{2\}, \{3, 5\}, \{4, 6\}\})$. Furthermore, $|\mathcal{F}| = 2^4 = 2^{\text{number of blocks}}$.

Define the equivalence relation R on Ω as follows

$$\text{for every } \omega_1, \omega_2 \in \Omega, \omega_1 \sim \omega_2 \iff \forall A \in \mathcal{F} [\omega_1 \in A \iff \omega_2 \in A].$$

That is, two elements are related if they always ‘appear’ together in $\mathcal{F}$ (i.e., they cannot be distinguished by $\mathcal{F}$). Then the equivalence classes of R are exactly the blocks that generate $\mathcal{F}$.

However, this characterisation of blocks is not very useful if we want to find the blocks that generate $\mathcal{F}$ in order to find $\mathcal{F}$, because it requires knowledge of $\mathcal{F}$. We will solve this problem by modifying the definition of R as follows. If $\mathcal{A}$ is a given collection of sets that generate $\mathcal{F}$, then we will redefine R as follows:

$$\text{for every } \omega_1, \omega_2 \in \Omega, \omega_1 \sim \omega_2 \iff \forall A \in \mathcal{A} [\omega_1 \in A \iff \omega_2 \in A].$$

In our example we have $\mathcal{A} = \{A, B\}$, and one can check that $([x])$ is the equivalence class of x

$$[1] = \{1\}, [2] = \{2\}, [3] = [5] = \{3, 5\} \text{ and } [4] = [6] = \{4, 6\}.$$

We collect all these ideas into the next two theorems that also generalise to countably infinite collections.

Theorem 2.2.5. *Let $\mathcal{A} \neq \emptyset$ be a countable collection of subsets of Ω such that $\sigma(\mathcal{A}) = \mathcal{F}$. Define the relation R on Ω by*

$$\text{for every } \omega_1, \omega_2 \in \Omega, \omega_1 \sim \omega_2 \iff \forall A \in \mathcal{A} [\omega_1 \in A \iff \omega_2 \in A].$$

Then R is an equivalence relation on Ω and each block (equivalence class) belongs to $\mathcal{F}$; i.e., $[\omega] \in \mathcal{F}$ for each $\omega \in \Omega$.

Proof. The fact that R is an equivalence relation is clear. Let us show that $[\omega] \in \mathcal{F}$ for each $\omega \in \Omega$. Fix $\omega \in \Omega$ and let

$$\mathcal{A}_\omega := \begin{cases} \{A \in \mathcal{A} : \omega \in A\} & \text{if } \omega \in \bigcup_{B \in \mathcal{A}} B \\ \{\Omega \setminus \bigcup_{B \in \mathcal{A}} B\} & \text{otherwise.} \end{cases}$$

We claim that $[\omega]$ is the intersection of all members of $\mathcal{A}_\omega$, i.e.,

$$[\omega] = \bigcap_{A \in \mathcal{A}_\omega} A.$$

Indeed,

$$\omega' \in [\omega] \iff \omega' \sim \omega \iff \omega' \in A \quad \forall A \in \mathcal{A}_\omega \iff \omega' \in \bigcap_{A \in \mathcal{A}_\omega} A,$$

hence $[\omega]$ is an intersection of countable members of $\mathcal{F}$; thus, it is also in $\mathcal{F}$. ■

Theorem 2.2.6. *Let $\mathcal{A}$ be a countable collection of subsets of Ω such that $\sigma(\mathcal{A}) = \mathcal{F}$. Define the equivalence relation R on Ω by*

$$\text{for every } \omega_1, \omega_2 \in \Omega, \omega_1 \sim \omega_2 \iff \forall A \in \mathcal{A} [\omega_1 \in A \iff \omega_2 \in A].$$

Assume that the set of blocks (equivalence classes) $\mathcal{B} := \{[\omega] : \omega \in \Omega\}$ is countable; i.e., $\mathcal{B} = \{B_\alpha : \alpha \in I\}$ where I is countable. Then

1. $\mathcal{F} = \sigma(\mathcal{B})$, and for every $A \in \mathcal{F}$ there exists $J \subseteq I$ such that

$$A = \bigcup_{\alpha \in J} B_\alpha.$$

That is,

$$\mathcal{F} = \left\{ \bigcup_{\alpha \in J} B_\alpha : J \subseteq I \right\}.$$

2. If $\mathcal{A}$ is finite, then $\mathcal{B}$ and $\mathcal{F}$ are also finite, and $|\mathcal{F}| = 2^{|\mathcal{B}|}$.

Proof. Exercise. ■

Exercise 2.2.1. Show that if Ω is countable, then every σ -algebra on Ω is generated by countably many blocks. ■

2.3 Lebesgue Measure

In this section we construct a measure λ_1 on $\mathbb{R}$ that will extend the notion of length of intervals to a collection of subsets of $\mathbb{R}$ that is bigger than the intervals (and therefore, bigger than the Borel sets), since the intervals by themselves do not form a σ -algebra. We proceed as follows:

1. We will first define the length $\ell(I)$ of an interval $I \subseteq \mathbb{R}$. This definition easily extends to finite unions of pairwise disjoint intervals.
2. We then extend this definition of “length” to *all subsets* of $\mathbb{R}$, and call this extension the *Lebesgue outer measure* λ_1^* . Unfortunately, λ_1^* is not a measure on all subsets of $\mathbb{R}$ (it is not σ -additive).

3. Finally, we carefully choose a collection $\mathcal{L}(\mathbb{R})$ of subsets of $\mathbb{R}$ for which the restriction of λ_1^* to $\mathcal{L}(\mathbb{R})$ is a measure. We show that $\mathcal{L}(\mathbb{R})$ is a σ -algebra and $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}(\mathbb{R})$. We will then call the restriction of λ_1^* to $\mathcal{L}(\mathbb{R})$ the *Lebesgue measure* λ_1 .

The method of constructing a measure via the above steps is used in the construction of several other measures and is credited to the Greek mathematician Constantin Caratheodory.

Definition 2.3.1. Let $I \subseteq \mathbb{R}$ be an interval in $\mathbb{R}$ (possibly empty, e.g., $I = (1, 1)$). We define the length of I as

$$\ell(I) := \begin{cases} \sup I - \inf I & \text{if } I \neq \emptyset \\ 0 & \text{if } I = \emptyset. \end{cases}$$

Let $\mathcal{I}$ be the algebra of subsets of $\mathbb{R}$ that can be written as a finite union of pairwise disjoint intervals. If $A \in \mathcal{I}$ and $A = \bigcup_{i=1}^n I_i$ for some pairwise disjoint intervals $I_1, \dots, I_n$, we define the length of A as

$$\ell(A) := \sum_{i=1}^n \ell(I_i).$$

For instance, if $a < b$, then $\ell((a, b)) = \ell((a, b]) = \ell([a, b)) = \ell([a, b]) = b - a$. If $a_1 < b_1 < a_2 < b_2 < a_3 < \dots < a_n < b_n$, then

$$\ell\left(\bigcup_{i=1}^n (a_i, b_i)\right) = \sum_{i=1}^n (b_i - a_i).$$

It can be shown that ℓ is well-defined on $\mathcal{I}$, in the sense that if $A \in \mathcal{I}$ is such that

$$A = \bigcup_{i=1}^n I_i = \bigcup_{j=1}^m I'_j,$$

where (I_i) and (I'_j) are pairwise disjoint intervals, then

$$\ell(A) = \sum_{i=1}^n \ell(I_i) = \sum_{j=1}^m \ell(I'_j).$$

The following Lemma follows immediately from the above definition.

Lemma 2.3.1. The function ℓ is finitely additive on $\mathcal{I}$. That is, if $A_1, \dots, A_n \in \mathcal{I}$ are pairwise disjoint, then

$$\ell\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \ell(A_i).$$

Proof. For each $i = 1, 2, \dots, n$, there exist pairwise disjoint intervals $I_1^i, \dots, I_{n_i}^i$ such that

$$A_i = \bigcup_{j=1}^{n_i} I_j^i, \quad \ell(A_i) = \sum_{j=1}^{n_i} \ell(I_j^i).$$

The intervals (I_j^i) are all pairwise disjoint and

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \bigcup_{j=1}^{n_i} I_j^i.$$

Hence

$$\ell\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \sum_{j=1}^{n_i} \ell(I_j^i) = \sum_{i=1}^n \ell(A_i).$$

■

Exercise 2.3.1. Show that ℓ is translation-invariant on $\mathcal{I}$; that is,

$$\ell(x + A) = \ell(A)$$

for any $x \in \mathbb{R}$ and $A \in \mathcal{I}$, where $x + A := \{x + a : a \in A\}$. ■

We now extend ℓ to all subsets of $\mathbb{R}$.

Definition 2.3.2. We define the Lebesgue outer measure $\lambda_1^* : 2^{\mathbb{R}} \rightarrow \bar{\mathbb{R}}$ by

$$\lambda_1^*(A) := \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ is a bounded open interval in } \mathbb{R} \right\}, \quad A \subseteq \mathbb{R}.$$

That is, to calculate $\lambda_1^*(A)$, we first find the sum of the lengths of countably many bounded open intervals that cover A , and then take the infimum across all such covers of A . We insist that the intervals used to cover A must be countably many bounded open intervals (this is not necessary though).

Theorem 2.3.1. The outer measure λ_1^* satisfies the following:

1. $\lambda_1^*(\emptyset) = 0$.
2. λ_1^* is increasing, i.e., if $A \subseteq B \subseteq \mathbb{R}$, then $\lambda_1^*(A) \leq \lambda_1^*(B)$.
3. λ_1^* is σ -sub-additive, i.e., if (A_n) is a countable collection of subsets of $\mathbb{R}$, then

$$\lambda_1^* \left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \lambda_1^*(A_n).$$

Proof. 1. Any collection of intervals covers $\emptyset$, hence the result follows from taking intervals with arbitrarily small total length.

2. Let $A \subseteq B \subseteq \mathbb{R}$. If (I_n) covers B , then it also covers A . Hence the set

$$\mathcal{C}_B := \left\{ \sum_{n=1}^{\infty} \ell(I_n) : B \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ is a bounded open interval in } \mathbb{R} \right\}$$

is a subset of

$$\mathcal{C}_A := \left\{ \sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n, I_n \text{ is a bounded open interval in } \mathbb{R} \right\}.$$

Thus

$$\lambda_1^*(A) = \inf \mathcal{C}_A \leq \inf \mathcal{C}_B = \lambda_1^*(B).$$

3. Let (A_n) be a countable collection of subsets of $\mathbb{R}$ and let $\epsilon > 0$. Using the definition of infimum, for each $n \in \mathbb{N}^+$ we can find a countable collection of bounded open intervals $(I_k^n)_{k=1}^{\infty}$ such that

$$A_n \subseteq \bigcup_{k=1}^{\infty} I_k^n \text{ and } \sum_{k=1}^{\infty} \ell(I_k^n) < \lambda_1^*(A_n) + \frac{\epsilon}{2^n}.$$

Thus, since $(I_k^n)_{k,n=1}^{\infty}$ is a cover of $\bigcup_{n=1}^{\infty} A_n$ by bounded open intervals, it follows that

$$\lambda_1^* \left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{(l,k) \in \mathbb{N}^+ \times \mathbb{N}^+} \ell(I_k^n) = \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \ell(I_k^n) \leq \sum_{n=1}^{\infty} \left(\lambda_1^*(A_n) + \frac{\epsilon}{2^n} \right) = \sum_{n=1}^{\infty} \lambda_1^*(A_n) + \epsilon.$$

Since this is true for any $\epsilon > 0$, we conclude that

$$\lambda_1^* \left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \lambda_1^* (A_n).$$

■

Proposition 2.3.1. *If I is an interval, then $\lambda_1^*(I) = \ell(I)$. That is, λ_1^* extends ℓ to all subsets of $\mathbb{R}$.*

Proof. 1. First suppose that $I = [a, b]$ is the closed and bounded interval, where $a, b \in \mathbb{R}$ and $a \leq b$. Since $\{(a - \epsilon/2, b + \epsilon/2), \emptyset, \emptyset, \dots\}$ is a cover of I , we have $\lambda_1^*(I) \leq \ell((a - \epsilon/2, b + \epsilon/2)) = \ell(I) + \epsilon$ for every $\epsilon > 0$, implying that $\lambda_1^*(I) \leq \ell(I)$.

We now show the reverse inequality. Assume (for a contradiction) that $\lambda_1^*(I) < \ell(I)$. This means that there exists a countable collection of open intervals $\{(a_n, b_n) : a_n < b_n, n = 1, 2, \dots\}$ such that

$$I \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n) \text{ and } \sum_{n=1}^{\infty} (b_n - a_n) < \ell(I) = b - a.$$

If the interval (a_1, b_1) covers I , then $a_1 < a < b < b_1$, which implies that

$$b - a < b_1 - a_1 \leq \sum_{n=1}^{\infty} (b_n - a_n),$$

a contradiction since λ_1^* is increasing. Thus, (a_1, b_1) does not cover I . Likewise, assume that $(a_1, b_1) \cup (a_2, b_2)$ covers I . We can assume (without loss of generality) that $a_1 < a$. If $b_1 > b$, then we get a contradiction like before, so assume that $b_1 < b$. This implies that $a_2 < b_1$ and $b_2 > b$. Hence

$$b - a < b_2 - a_1 < b_2 - a_2 + b_1 - a_1 \leq \sum_{n=1}^{\infty} (b_n - a_n),$$

a contradiction again. So $(a_1, b_1) \cup (a_2, b_2)$ does not cover I . In general, we can show that for each $n \in \mathbb{N}^+$, $I_n := (a_1, b_1) \cup \dots \cup (a_n, b_n)$ does not cover I . For each $n \in \mathbb{N}^+$ we pick $x_n \in I \setminus I_n$. The sequence (x_n) is bounded (between a and b), thus by the Bolzano-Weierstrass Theorem, it has a subsequence (x_{n_k}) that converges to some $x \in [a, b]$. For each $m \in \mathbb{N}^+$, $I \setminus I_m$ is a closed set (since I_m is open and I is closed) and (x_n) is eventually in $I \setminus I_m$, thus the limit x must also be in $I \setminus I_m$ for each $m \in \mathbb{N}^+$. Thus,

$$x \in I \setminus \left(\bigcup_{m=1}^{\infty} I_m \right) = I \setminus \bigcup_{m=1}^{\infty} (a_m, b_m),$$

contradicting the fact that

$$I \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n).$$

Hence there is no such collection of intervals, which implies that $\lambda_1^*(I) \geq \ell(I)$.

2. Now let I be any bounded interval. Then $[\inf I + \epsilon/2, \sup I - \epsilon/2] \subseteq I$ for every $\epsilon > 0$. Hence $\lambda_1^*(I) \geq \lambda_1^*([\inf I + \epsilon/2, \sup I - \epsilon/2]) = \ell([\inf I + \epsilon/2, \sup I - \epsilon/2]) = \ell(I) - \epsilon$ for each $\epsilon > 0$, implying that $\lambda_1^*(I) \geq \ell(I)$. Again noting that for each $\epsilon > 0$, $\{(\inf I - \epsilon/2, \sup I + \epsilon/2), \emptyset, \emptyset, \dots\}$ is a cover of I , we get that $\lambda_1^*(I) \leq \ell(I) + \epsilon \forall \epsilon > 0$, which implies that $\lambda_1^*(I) \leq \ell(I)$. Hence $\lambda_1^*(I) = \ell(I)$.

3. Assume that I is unbounded. If $I = (b, \infty)$ for some $b \in \mathbb{R}$, then for each $n \in \mathbb{N}^+$,

$$\lambda_1^*(I) \geq \lambda_1^*(b, b+n) = \ell(b, b+n) = n \rightarrow \infty.$$

Hence $\lambda_1^*(I) = \infty = \ell(I)$. The other cases are treated in the same way. ■

Remark 2.3.1. λ_1^* is not finitely additive on $2^{\mathbb{R}}$; thus, it is not a measure on the power set of $\mathbb{R}$. ■

Proposition 2.3.2. *If $A \subseteq \mathbb{R}$ is countable, then $\lambda_1^*(A) = 0$.*

Proof. First consider the singleton $\{x\}$, where $x \in \mathbb{R}$. For each $\epsilon > 0$, the bounded open intervals $I_n = (x - \epsilon/2^{n+1}, x + \epsilon/2^{n+1})$ satisfy

$$\{x\} \subseteq \bigcup_{n=1}^{\infty} I_n \text{ and } \sum_{n=1}^{\infty} \ell(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon.$$

Hence $\lambda_1^*(\{x\}) \leq \epsilon$ for each $\epsilon > 0$, which implies that $\lambda_1^*(\{x\}) = 0$. Now let $A = \{x_1, x_2, \dots\}$ be a countable subset of $\mathbb{R}$. By σ -sub-additivity we get

$$\lambda_1^*(A) = \lambda_1^*\left(\bigcup_{n=1}^{\infty} \{x_n\}\right) \leq \sum_{n=1}^{\infty} \lambda_1^*(\{x_n\}) = 0. \quad \blacksquare$$

The converse of this result is not true. That is, not all sets of outer measure zero are countable.

Example 2.3.1 (Cantor Set). We construct the Cantor set. Start with the interval $C_0 := [0, 1]$. The first step is to remove the middle-third $(\frac{1}{3}, \frac{2}{3})$ and remain with $C_1 := [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Repeat the process again and remove the middle thirds of each of the remaining intervals to get $C_2 := [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$. The process continues indefinitely. We can define it more precisely as follows:

$$C_0 := [0, 1], \quad C_{n+1} := \frac{C_n}{3} \cup \left(\frac{2}{3} + \frac{C_n}{3}\right) \quad n \in \mathbb{N}.$$

The *Cantor set*, $\mathcal{C}$, consists of the elements of $[0, 1]$ that are never removed:

$$\mathcal{C} := \bigcap_{n \in \mathbb{N}} C_n.$$

We show that $\lambda_1^*(\mathcal{C}) = 0$. Since λ_1^* is additive on intervals and $\frac{C_n}{3}$ and $(\frac{2}{3} + \frac{C_n}{3})$ are disjoint for each n , it follows that

$$\lambda_1^*(C_{n+1}) = \frac{1}{3}\lambda_1^*(C_n) + \frac{1}{3}\lambda_1^*(C_n) = \left(\frac{2}{3}\right)\lambda_1^*(C_n).$$

Hence,

$$\lambda_1^*(C_n) = \left(\frac{2}{3}\right)^n \lambda_1^*(C_0) = \left(\frac{2}{3}\right)^n.$$

Since $\mathcal{C} \subseteq C_n$ for each $n \in \mathbb{N}$, it follows that

$$\lambda_1^*(\mathcal{C}) \leq \limsup_{n \rightarrow \infty} \left(\frac{2}{3}\right)^n = 0 \implies \lambda_1^*(\mathcal{C}) = 0. \quad \blacksquare$$

However, the set $\mathcal{C}$ is uncountable.

Exercise 2.3.2. Show that the Cantor set $\mathcal{C}$ is uncountable by following these steps:

1. Given an integer $b \geq 2$, show that every real number $x \in [0, 1]$ can be written as

$$x = \sum_{n=1}^{\infty} \frac{x_n}{b^n},$$

where $x_n \in \{0, 1, \dots, b-1\}$. In this case, we write $x = 0.x_1x_2x_3\dots_b$ and call it the b -ary expansion of x . When $b = 2$ we call it *binary*, for $b = 3$ we call it the *ternary* expansion of x , and $b = 10$ corresponds to the usual *decimal* expansion of x .

2. Show that the Cantor set $\mathcal{C}$ is the set of numbers in $[0, 1]$ whose ternary expansion only contains the digits 0 and 2; i.e.,

$$\mathcal{C} = \{x = 0.x_1x_2x_3\dots_3 : x_n \in \{0, 2\} \ \forall n \in \mathbb{N}^+\}.$$

3. By considering the mapping $f : \mathcal{C} \rightarrow [0, 1]$ defined by

$$f(0.x_1x_2x_3\dots_3) = \frac{1}{2} \sum_{n=1}^{\infty} \frac{x_n}{2^n},$$

show that $\mathcal{C}$ is uncountable. ■

As we have mentioned before, λ_1^* is not σ -additive on $2^{\mathbb{R}}$ (it is not even additive). That is, there are disjoint subsets A and B of $\mathbb{R}$ such that $\lambda_1^*(A \cup B) < \lambda_1^*(A) + \lambda_1^*(B)$ (we already know that $\lambda_1^*(A \cup B) \leq \lambda_1^*(A) + \lambda_1^*(B)$ always holds since λ_1^* is σ -sub-additive). A construction of such sets is available in many standard textbooks on measure theory (e.g., the *Vitali set* can be used to construct such ill-behaved sets). We are going to restrict the domain of λ_1^* by “removing” these bad sets.

Definition 2.3.3. A set $A \subseteq \mathbb{R}$ is Lebesgue measurable if and only if for every $E \subseteq \mathbb{R}$, $\lambda_1^*(E) = \lambda_1^*(E \cap A) + \lambda_1^*(E \cap A^c)$. We will denote by $\mathcal{L}(\mathbb{R})$ the set of all Lebesgue measurable sets.

A Lebesgue measurable set does what we would expect every “reasonable” set to do, thus it makes sense to remove all the sets that do not behave in this way. The following results show that the Lebesgue measurable sets form a σ -algebra on $\mathbb{R}$ and that the restriction of λ_1^* to $\mathcal{L}(\mathbb{R})$ is σ -additive (i.e., λ_1^* is a measure on $\mathcal{L}(\mathbb{R})$). Note that since λ_1^* is sub-additive, to show that a set A is Lebesgue measurable, we only need to show that for every E , $\lambda_1^*(E) \geq \lambda_1^*(E \cap A) + \lambda_1^*(E \cap A^c)$.

We first show that $\mathcal{L}(\mathbb{R})$ is non-empty.

Proposition 2.3.3. If $A \subseteq \mathbb{R}$ and $\lambda_1^*(A) = 0$, then $A \in \mathcal{L}(\mathbb{R})$. That is, all sets of outer measure zero are Lebesgue measurable.

Proof. Let $A \subseteq \mathbb{R}$ be a set of outer measure 0. We want to show that for every $E \subseteq \mathbb{R}$, $\lambda_1^*(E) = \lambda_1^*(E \cap A) + \lambda_1^*(E \cap A^c)$. As noted above, we only need to show that $\lambda_1^*(E) \geq \lambda_1^*(E \cap A) + \lambda_1^*(E \cap A^c)$. Since λ_1^* is increasing, we have $\lambda_1^*(E \cap A) \leq \lambda_1^*(A) = 0$ and $\lambda_1^*(E \cap A^c) \leq \lambda_1^*(E)$. Hence

$$\lambda_1^*(E) \geq \lambda_1^*(E \cap A^c) + 0 = \lambda_1^*(E \cap A^c) + \lambda_1^*(E \cap A).$$
■

Theorem 2.3.2. *The collection $\mathcal{L}(\mathbb{R})$ is a σ -algebra on $\mathbb{R}$ that contains $\mathcal{I}$, and the restriction of λ_1^* to $\mathcal{L}(\mathbb{R})$ is a measure called the Lebesgue measure on $\mathbb{R}$. We denote this measure by λ_1 . Furthermore, the measure space $(\mathbb{R}, \mathcal{L}(\mathbb{R}), \lambda_1)$ is complete.*

We omit the proof of this result.

Corollary 2.3.1.

$$\mathcal{B}(\mathbb{R}) \subseteq \mathcal{L}(\mathbb{R}).$$

Proof. Since $\mathcal{I} \subseteq \mathcal{L}(\mathbb{R})$ and $\sigma(\mathcal{I}) = \mathcal{B}(\mathbb{R})$, this follows immediately. ■

Remark 2.3.2. There exist sets which belong to $\mathcal{L}(\mathbb{R})$ but not in $\mathcal{B}(\mathbb{R})$. That is, $\mathcal{L} \setminus \mathcal{B}(\mathbb{R}) \neq \emptyset$. ■

Remark 2.3.3. It is common to define the Lebesgue measure λ_1 only on $\mathcal{B}(\mathbb{R})$ instead of $\mathcal{L}(\mathbb{R})$. We will do this at times. However, the measure space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda_1)$ is not complete; its completion is $(\mathbb{R}, \mathcal{L}(\mathbb{R}), \lambda_1)$. Thus, every Lebesgue measurable set $A \in \mathcal{L}(\mathbb{R})$ can be written as $A = B \cup N$, where $B \in \mathcal{B}(\mathbb{R})$ and N is a set of outer measure zero (a λ_1 -null set). ■

We end this section with a remark on the construction of the N -dimensional Lebesgue measure, λ_N . First, we define a rectangle to be a subset of $\mathbb{R}^n$ of the form

$$R = I_1 \times \dots \times I_N,$$

where I_i is an interval for each $i = 1, 2, \dots, N$. We call R an *open rectangle* if each I_i is an open interval. We define the *volume* of a rectangle as

$$\text{vol}(R) := \prod_{i=1}^N \ell(I_i).$$

We also define the *Lebesgue outer measure* of a set $A \subseteq \mathbb{R}^N$ to be

$$\lambda_N^*(A) := \inf \left\{ \sum_{n=1}^{\infty} \text{vol}(R_n) : A \subseteq \bigcup_{n=1}^{\infty} R_n, R_n \text{ is a bounded open rectangle in } \mathbb{R}^N \right\}$$

and say that $A \subseteq \mathbb{R}^N$ is Lebesgue-measurable if and only if for every $E \subseteq \mathbb{R}^N$,

$$\lambda_N^*(E) = \lambda_N^*(E \cap A) + \lambda_N^*(E \cap A^c).$$

We will denote by $\mathcal{L}(\mathbb{R}^N)$ the set of all Lebesgue measurable sets.

The following theorem summarizes the main results in the construction of the Lebesgue measure in $\mathbb{R}^N$.

Theorem 2.3.3. *The collection $\mathcal{L}(\mathbb{R}^N)$ is a σ -algebra such that $\mathcal{B}(\mathbb{R}^N) \subseteq \mathcal{L}(\mathbb{R}^N)$, and the restriction of λ_N^* to $\mathcal{L}(\mathbb{R}^N)$ is a measure called the N -dimensional Lebesgue measure. We denote it by λ_N .*

Proof. Omitted. ■

Chapter 3

Measurable Functions and Random Variables

In this chapter we introduce the notion of measurability of a function $f : (\Omega, \mu) \rightarrow (E, \mathcal{E})$ between two measure spaces $(\Omega, \mathcal{F}, \mu)$ and $(E, \mathcal{E})$. We will then define a random variable as a measurable function $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We also define the probability distribution of a random variable and use it to classify random variables.

3.1 Measurable Functions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. In previous courses, we defined a random variable as a function $X : \Omega \rightarrow \mathbb{R}$. However, in Chapter 1 we saw how this view is inadequate when the sigma algebra $\mathcal{F}$ is not the power set of Ω . This naturally led us to impose an additional *measurability* condition on X — that $X^{-1}(B) \in \mathcal{F}$ for every Borel set B . Let us look at an example that will further motivate this idea.

Suppose we toss two fair coins. The sample space for this random experiment is

$$\Omega = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}.$$

We first choose the sigma algebra $\mathcal{F}$ to be the power set of Ω and let $X : \Omega \rightarrow \mathbb{R}$ count the number of heads of the outcome. That is, $X(\text{HH}) = 2$, $X(\text{HT}) = X(\text{TH}) = 1$ and $X(\text{TT}) = 0$. Since the coin is fair, we will choose the probability measure $\mathbb{P}$ to be

$$\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{4} \delta_{\omega}.$$

We can then calculate probabilities about X by taking pre-images and then using $\mathbb{P}$. For instance, the probability that X takes the value 1 is

$$\Pr(X = 1) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = 1\}) = \mathbb{P}(X^{-1}(\{1\})) = \mathbb{P}(\{\text{HT}, \text{TH}\}) = \frac{2}{4}.$$

In general, if $B \subseteq \mathbb{R}$ is a *Borel set* (yes, these are the subsets of $\mathbb{R}$ that are of interest to us!), then

$$\Pr(X \in B) = \mathbb{P}(X^{-1}(B)).$$

Let us now change the sigma algebra. Suppose the only information available is the outcome of the first experiment. Then the sigma algebra generated by this information is

$$\mathcal{G} := \sigma(\{\{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}\}) = \{\{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}, \emptyset, \Omega\}.$$

That is, the event $\{HH, HT\}$ occurs if the outcome of the first toss is a head, while its complement occurs if the outcome of the first toss is a tail. The sigma algebra $\mathcal{G}$ cannot distinguish between the elements HH and HT, for instance. Calculating $\Pr(X = 1)$ is now hopeless since $X^{-1}(\{1\}) = \{HT, TH\}$ is not in $\mathcal{G}$. Another way to view this is that the values of X cannot be completely determined from the outcome of the random experiment. For instance, if the outcome is such that $\{HH, HT\}$ occurs, then the value of X could either be 1 or 2. We say that X is not $\mathcal{G}$ -measurable. That is, X contains information that is not available in $\mathcal{G}$.

This problem is solved as soon as we impose that $X^{-1}(B) \in \mathcal{G}$ for any $B \in \mathcal{B}(\mathbb{R})$. This additional condition will ensure that $\Pr(X \in B)$ can be calculated for any $B \in \mathcal{B}(\mathbb{R})$ (via $\Pr(X \in B) = \mathbb{P}(X^{-1}(B))$) and also ensure that the value of X , after the experiment is completed, can be uniquely determined from the information contained in the σ -algebra. That is,

X is $\mathcal{G}$ -measurable if the information in $\mathcal{G}$ is sufficient to determine each value of X .

This leads to the following definition.

Definition 3.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be a function. We say that X is a random variable if $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$.

The definition of a random variable does not require a measure to be defined on $(\Omega, \mathcal{F})$, all we need is Ω and the sigma algebra $\mathcal{F}$. We now extend this definition to general measurable spaces.

Definition 3.1.2. Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces and $f : \Omega \rightarrow E$ be a function (we will write $f : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$). We say that f is $\mathcal{F}/\mathcal{E}$ -measurable if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{E}$.

1. If $(E, \mathcal{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we say that f is Borel measurable, or simply, measurable.
2. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $(E, \mathcal{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then f is called a random variable and we usually use the letter X instead of f .
3. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $(E, \mathcal{E}) = (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, then f is called a random vector and we also use the letter X instead of f .

Whenever we say f is measurable without mentioning $(E, \mathcal{E})$, we will assume that we mean Borel measurable (i.e., $(E, \mathcal{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$).

Here are some examples.

Example 3.1.1. 1. Let $(\Omega, \mathcal{F})$ be any measurable space and $A \subseteq \Omega$. We define the indicator function $I_A : \Omega \rightarrow \mathbb{R}$ of A as

$$I_A(\omega) := \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A. \end{cases}$$

Then I_A is measurable (Borel measurable) if and only if $A \in \mathcal{F}$. To see this, note that for any $B \in \mathcal{B}(\mathbb{R})$,

$$I_A^{-1}(B) = \begin{cases} \emptyset & 0, 1 \notin B \\ A & 1 \in B, 0 \notin B \\ A^c & 0 \in B, 1 \notin B \\ \Omega & 0, 1 \in B, \end{cases}$$

which is always in $\mathcal{F}$ if and only if $A \in \mathcal{F}$.

2. On $(\Omega, 2^\Omega)$, every function $f : \Omega \rightarrow \mathbb{R}$ is measurable, since $f^{-1}(B) \subseteq \Omega$ for each $B \in \mathcal{B}(\mathbb{R})$.

3. On $(\Omega, \{\emptyset, \Omega\})$, the only (Borel) measurable functions $f : \Omega \rightarrow \mathbb{R}$ are constants.
4. On $(\Omega, \{\emptyset, \Omega, A, A^c\})$ $A \subseteq \Omega$, the only measurable functions $f : \Omega \rightarrow \mathbb{R}$ are constant on A and constant on A^c . That is, $f : \Omega \rightarrow \mathbb{R}$ is measurable if and only if $f = \alpha I_A + \beta I_{A^c}$ for some constants $\alpha, \beta \in \mathbb{R}$. ■

Let us now look at a concrete example.

Example 3.1.2. Consider the example where two fair coins are tossed and only the outcome of the first coin is recorded. Let A be the event that the outcome of the first coin is a head and B be the event that the outcome of the first coin is a tail. Then $\Omega = \{HH, HT, TH, TT\}$ and

$$\mathcal{F} := \sigma(\{A, B\}) = \sigma(\{\{HH, HT\}, \{TH, TT\}\}) = \{\{HH, HT\}, \{TH, TT\}, \emptyset, \Omega\}.$$

Notice that since $A^c = B$, then $\mathcal{B} = \{A, B\}$ are the blocks that generate $\mathcal{F}$. Both I_A and I_B are random variables. In fact, the only random variables are linear combinations of the two. In other words, X is a random variable if it is constant on both A and B . This simply means that X is only dependent on the outcome of the first toss. ■

This example *suggests* that if $\mathcal{F}$ is generated by (countably many) blocks, then the only $\mathcal{F}$ -measurable functions are the ones that are constant on the blocks. The following result makes this more precise.

Theorem 3.1.1. Let $(\Omega, \mathcal{F})$ be a measurable space, where $\mathcal{F}$ is generated by countably many blocks $\mathcal{B} = \{B_\alpha : \alpha \in I\}$. Then $f : \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}$ -measurable if and only if f is constant on each block B_α ($\alpha \in I$). In this case, f can be written as

$$f(\omega) = \sum_{\alpha \in I} a_\alpha I_{B_\alpha}(\omega), \quad \omega \in \Omega,$$

where a_α is the value of f on B_α (i.e., $B_\alpha \subseteq f^{-1}(\{a_\alpha\})$).

Proof. Exercise. ■

Example 3.1.3. Let $\Omega = \mathbb{R}$ and choose $\mathcal{F} = \sigma(\{\{\omega\} : \omega \in \mathbb{Z}\})$. Note that $\mathcal{F}$ is generated by the following (countably many) blocks:

$$\mathcal{B} = \{\{\omega\} : \omega \in \mathbb{Z}\} \cup \{\mathbb{R} \setminus \mathbb{Z}\} = \{\dots, \{-2\}, \{-1\}, \{0\}, \{1\}, \{2\}, \dots, \mathbb{R} \setminus \mathbb{Z}\}.$$

Therefore, a function $f : \Omega \rightarrow \mathbb{R}$ is measurable if it is constant on the non-integers $\mathbb{R} \setminus \mathbb{Z}$. That is, $f : \Omega \rightarrow \mathbb{R}$ is measurable if and only if there exist a function $g : \mathbb{Z} \rightarrow \mathbb{R}$ and a constant $c \in \mathbb{R}$ such that

$$f(\omega) = \begin{cases} g(\omega) & \omega \in \mathbb{Z} \\ c & \omega \in \mathbb{R} \setminus \mathbb{Z}. \end{cases}$$
■

If the sigma algebra $\mathcal{F}$ is not generated by countably many blocks, it is generally difficult to prove that a function f is measurable. The difficulty is caused by the fact that we don't have a general form of an arbitrary Borel set B (B could be an interval, a countably infinite set or even the irrational numbers). Thus, checking the pre-image condition for arbitrary Borel sets might be challenging. The following results tell us that we do not need to check this condition for every Borel set, but it is enough to check the condition for a sub-collection that generates $\mathcal{B}(\mathbb{R})$.

Theorem 3.1.2. Let $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$ be measurable spaces and $\mathcal{A} \subseteq \mathcal{E}$ be a collection of subsets of E such that $\sigma(\mathcal{A}) = \mathcal{E}$. A function $f : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$ is $\mathcal{F}/\mathcal{E}$ -measurable if and only if for each $B \in \mathcal{A}$, $f^{-1}(B) \in \mathcal{F}$.

Proof. ($\implies$:) Since $\mathcal{A} \subseteq \mathcal{E}$, it is clear that if f is $\mathcal{F}/\mathcal{E}$ -measurable then for each $B \in \mathcal{A}$, $f^{-1}(B) \in \mathcal{F}$.

($\impliedby$:) We now show the converse. Assume that for each $B \in \mathcal{A}$, $f^{-1}(B) \in \mathcal{F}$. We want to show that for each $B \in \mathcal{E}$, $f^{-1}(B) \in \mathcal{F}$. We use the “principle of appropriate sets”. Define

$$\mathcal{G} := \{B \in \mathcal{E} : f^{-1}(B) \in \mathcal{F}\}.$$

It is enough to show that $\mathcal{G}$ is a σ algebra that contains $\mathcal{A}$. We have $\emptyset \in \mathcal{G}$ since $\emptyset \in \mathcal{E}$ and $f^{-1}(\emptyset) = \emptyset \in \mathcal{F}$ ($\mathcal{F}$ is a σ -algebra). If $B \in \mathcal{G}$, then $f^{-1}(B^c) = (f^{-1}(B))^c \in \mathcal{F}$ since $\mathcal{F}$ is a σ -algebra. Thus $B^c \in \mathcal{G}$. Now let $(B_n) \subseteq \mathcal{G}$. Then $f^{-1}(\bigcup_{n=1}^{\infty} B_n) = \bigcup_{n=1}^{\infty} f^{-1}(B_n) \in \mathcal{F}$. Thus $\mathcal{G}$ is a σ -algebra that contains $\mathcal{A}$, which implies that $\mathcal{E} = \sigma(\mathcal{A}) \subseteq \mathcal{G}$. However, by definition, $\mathcal{G} \subseteq \mathcal{E}$. Hence $\mathcal{E} = \mathcal{G}$. $\blacksquare$

If $f : \Omega \rightarrow \mathbb{R}$ and $B \subseteq \mathbb{R}$, we will use the notation

$$\{f \in B\} := \{\omega \in \Omega : f(\omega) \in B\} = f^{-1}(B),$$

for the pre-image.

Corollary 3.1.1. Let $(\Omega, \mathcal{F})$ be a measurable space and $f : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a function. Then following are equivalent:

1. f is Borel measurable
2. $\{f < c\} := f^{-1}((-\infty, c)) \in \mathcal{F}$ for every $c \in \mathbb{R}$
3. $\{f \leq c\} := f^{-1}((-\infty, c]) \in \mathcal{F}$ for every $c \in \mathbb{R}$
4. $\{f > c\} := f^{-1}((c, \infty)) \in \mathcal{F}$ for every $c \in \mathbb{R}$
5. $\{f \geq c\} := f^{-1}([c, \infty)) \in \mathcal{F}$ for every $c \in \mathbb{R}$
6. $\{a < f < b\} := f^{-1}((a, b)) \in \mathcal{F}$ for every $a, b \in \mathbb{R}$
7. $\{a \leq f < b\} := f^{-1}([a, b)) \in \mathcal{F}$ for every $a, b \in \mathbb{R}$
8. $\{a < f \leq b\} := f^{-1}((a, b]) \in \mathcal{F}$ for every $a, b \in \mathbb{R}$
9. $\{a \leq f \leq b\} := f^{-1}([a, b]) \in \mathcal{F}$ for every $a, b \in \mathbb{R}$

Proof. Each of the collections generate $\mathcal{B}(\mathbb{R})$, so we can apply the previous result. $\blacksquare$

Example 3.1.4. Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1]) = \sigma(\{(a, b) : 0 \leq a < b \leq 1\})$, and $\mathbb{P}(A) = \lambda_1(A)$ for each $A \in \mathcal{F}$. Define $X : \Omega \rightarrow \mathbb{R}$ by

$$X(\omega) := \begin{cases} \frac{1}{4} & 0 \leq \omega < \frac{1}{2} \\ \omega & \frac{1}{2} \leq \omega \leq 1. \end{cases}$$

Then X is a random variable. For each $c \in \mathbb{R}$, we have

$$\{X \leq c\} = \begin{cases} \Omega & c \geq 1 \\ [0, c] & \frac{1}{2} \leq c < 1 \\ [0, \frac{1}{2}] & 0 \leq c < \frac{1}{2} \\ \emptyset & c < 0 \end{cases}$$

which is always a Borel subset of $[0, 1]$. $\blacksquare$

We will also deal with functions taking values on the extended real numbers $\bar{\mathbb{R}}$. We define the Borel σ -algebra on $\bar{\mathbb{R}}$ as

$$\mathcal{B}(\bar{\mathbb{R}}) := \{B \subseteq \bar{\mathbb{R}} : B \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\} = \{B \cup B' : B \in \mathcal{B}(\mathbb{R}), B' \subseteq \{-\infty, \infty\}\}.$$

If $(\Omega, \mathcal{F})$ is a measurable space, we define

$$m\mathcal{F} := \{f : (\Omega, \mathcal{F}) \rightarrow (\bar{\mathbb{R}}, \mathcal{B}(\bar{\mathbb{R}})) : f \text{ is } \mathcal{F}/\mathcal{B}(\bar{\mathbb{R}})\text{-measurable}\}.$$

Note that if $f : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable, then it is also $\mathcal{F}/\mathcal{B}(\bar{\mathbb{R}})$ -measurable as function $f : (\Omega, \mathcal{F}) \rightarrow (\bar{\mathbb{R}}, \mathcal{B}(\bar{\mathbb{R}}))$. If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space, we will call a measurable function $X : \Omega \rightarrow \bar{\mathbb{R}}$ an *extended random variable*.

We say that a sequence (f_n) of functions $f_n : \Omega \rightarrow \mathbb{R}$ *converges pointwise* to a function f (written as $f_n \rightarrow f$) if for every $\omega \in \Omega$, the sequence of real numbers $(f_n(\omega))$ converges to $f(\omega)$. That is,

$$f_n \rightarrow f \iff \forall \omega \in \Omega \forall \epsilon > 0 \exists N \in \mathbb{N} \forall n \in \mathbb{N} [n \geq N \implies |f_n(\omega) - f(\omega)| < \epsilon].$$

We also define functions like $\sup_n f_n, \inf_n f_n, \limsup f_n$, and $\liminf f_n$ in a similar fashion. Note that even if f is real-valued, these functions take values in $\bar{\mathbb{R}}$.

We end this section by noting that all the “usual” combinations (sums, differences, etc) of measurable functions are also measurable.

Theorem 3.1.3. *Let $(\Omega, \mathcal{F})$ be a measurable space and $f, g, f_n \in m\mathcal{F}$ be real-valued Borel measurable functions. Then*

1. $\alpha f, f \pm g, f^2, fg, f/g \in m\mathcal{F}$ ($\alpha \in \mathbb{R}$), when they are well-defined.
2. $\sup_n f_n, \inf_n f_n \in m\mathcal{F}$
3. $\limsup_{n \rightarrow \infty} f_n, \liminf_{n \rightarrow \infty} f_n \in m\mathcal{F}$
4. If $\lim_{n \rightarrow \infty} f_n$ exists, then $\lim_{n \rightarrow \infty} f_n \in m\mathcal{F}$.

Proof. We give a sketch of the proof and leave the rest for the reader.

1. First assume that $\alpha > 0$. Then for any $c \in \mathbb{R}$ we have $\{\alpha f > c\} = \{f > c/\alpha\} \in \mathcal{F}$. The other cases are treated in a similar way.
2. Notice that for any $c \in \mathbb{R}$,

$$\left\{ \sup_n f_n \leq c \right\} = \bigcap_{n=1}^{\infty} \{f_n \leq c\} \in \mathcal{F}.$$

3. The easy way is to notice that

$$\limsup_{n \rightarrow \infty} f_n = \inf_n \sup_{k \geq n} f_k \in m\mathcal{F}.$$

Alternatively, notice that for any $c \in \mathbb{R}$,

$$\begin{aligned} \omega \in \left\{ \limsup_{n \rightarrow \infty} f_n > c \right\} &\iff \forall n \geq 1 \exists k \geq n \text{ such that } f_k(\omega) > c \\ &\iff \forall n \geq 1 \exists k \geq n \text{ such that } \omega \in \{f_k > c\} \iff \omega \in \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} \{f_k > c\} \\ &\iff \omega \in \limsup_{n \rightarrow \infty} \{f_n > c\} \in \mathcal{F}. \end{aligned}$$

4. Note that $\lim_{n \rightarrow \infty} f_n$ exists if and only if $\limsup_{n \rightarrow \infty} f_n = \liminf_{n \rightarrow \infty} f_n =: \lim_{n \rightarrow \infty} f_n$.

■

If X is a random variable, we took it for granted that $g(X)$ is also a random variable when $g : \mathbb{R} \rightarrow \mathbb{R}$. The following results tell us when this is the case.

Theorem 3.1.4. *Let $(\Omega, \mathcal{F})$, $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measurable spaces and $f : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$, $g : (E, \mathcal{E}) \rightarrow (G, \mathcal{G})$ be measurable functions. Then $h := g \circ f : (\Omega, \mathcal{F}) \rightarrow (G, \mathcal{G})$ is $\mathcal{F}/\mathcal{G}$ -measurable.*

Proof. Let $B \in \mathcal{G}$. We need to show that $h^{-1}(B) \in \mathcal{F}$. Since g is $\mathcal{E}/\mathcal{G}$ -measurable, $g^{-1}(B) \in \mathcal{E}$. Similarly, since f is $\mathcal{F}/\mathcal{E}$ -measurable, $f^{-1}(g^{-1}(B)) \in \mathcal{F}$. Thus

$$h^{-1}(B) = (g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)) \in \mathcal{F}.$$

■

Corollary 3.1.2. *Let $(\Omega, \mathcal{F})$ be a measurable space, $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable and $g : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a Borel measurable function. Then $g(X)$ is also a random variable.*

Proof. Apply the above theorem to $(E, \mathcal{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $(G, \mathcal{G}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, $f = X$ and $g = g$. ■

3.2 The σ -algebra $\sigma(X)$

We begin with a Lemma.

Lemma 3.2.1. *Let Ω be a non-empty set, $\mathcal{F}$ and $\mathcal{G}$ be sigma-algebras on Ω with $\mathcal{G} \subseteq \mathcal{F}$, and $X : \Omega \rightarrow \mathbb{R}$. If X is $\mathcal{G}$ -measurable, then X is $\mathcal{F}$ -measurable. In other words, $m\mathcal{G} \subseteq m\mathcal{F}$.*

Proof. Let $B \in \mathcal{B}(\mathbb{R})$. Then $X^{-1}(B) \in \mathcal{G} \subseteq \mathcal{F}$. ■

Now suppose $(\Omega, \mathcal{F})$ is a measurable space and $X : \Omega \rightarrow \mathbb{R}$ is a function that is not a random variable. This simply means that there exists $B \in \mathcal{B}(\mathbb{R})$ such that $X^{-1}(B) \notin \mathcal{F}$. In this section we want to define the smallest sigma algebra $\sigma(X)$ that will make X a random variable.

Definition 3.2.1. *Let Ω be a set and $X : \Omega \rightarrow \mathbb{R}$ be a function. We define the σ -algebra generated by X , $\sigma(X)$, to be the smallest σ -algebra with respect to which X is a random variable.*

Corollary 3.2.1. *Let $(\Omega, \mathcal{F})$ be a measurable space and $X : \Omega \rightarrow \mathbb{R}$ be a function. Then*

$$X \in m\mathcal{F} \iff \sigma(X) \subseteq \mathcal{F}.$$

Let us look for a better description of $\sigma(X)$. By definition, X must be $\sigma(X)$ -measurable, so $X^{-1}(B) \in \sigma(X)$ for all $B \in \mathcal{B}(\mathbb{R})$. Hence we have

$$\{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} \subseteq \sigma(X).$$

It turns out that $\sigma(X)$ consists of only these types of sets.

Theorem 3.2.1. *Let Ω be a set and $X : \Omega \rightarrow \mathbb{R}$ be a function. Then $\sigma(X)$ exists and is given by*

$$\sigma(X) = \{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}.$$

Proof. Let $\Gamma := \{\mathcal{F} \subseteq 2^\Omega : \mathcal{F} \text{ is a sigma-algebra and } X \in m\mathcal{F}\}$. Then $\Gamma \neq \emptyset$ (since $2^\Omega \in \Gamma$) and $\sigma(X) = \bigcap_{\mathcal{F} \in \Gamma} \mathcal{F}$. By definition, each $\mathcal{F} \in \Gamma$ satisfies $\{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} \subseteq \mathcal{F}$, hence $\{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} \subseteq \sigma(X)$. The proof is completed by showing that $\{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$ is a σ -algebra (i.e., $\{X^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\} \in \Gamma$), which is left as an exercise. ■

Example 3.2.1. Consider the random experiment of tossing two fair coins. The sample space is $\Omega = \{HH, HT, TH, TT\}$. Define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) := \text{number of heads in } \omega$. We have $X(HH) = 2, X(HT) = X(TH) = 1$ and $X(TT) = 0$. If $B \in \mathcal{B}(\mathbb{R})$, then $X^{-1}(B)$ be one of $\{HH\}, \{HT, TH\}, \{TT\}$ or a union of these sets, depending on which of the numbers 0, 1 and 2 the set B contains. Thus

$$\sigma(X) = \sigma(\{\{HH\}, \{HT, TH\}, \{TT\}\}).$$

That is, $\sigma(X)$ is generated by the blocks $\mathcal{B} = \{X^{-1}(\{2\}), X^{-1}(\{1\}), X^{-1}(\{0\})\}$.

We show that this is no accident (in the ‘discrete’ case).

Theorem 3.2.2. Let Ω be a set and $X : \Omega \rightarrow \mathbb{R}$ be a function that has a countable range: $\text{Range}(X) = \{x_n : n = 1, 2, 3, \dots\}$. Then

$$\sigma(X) = \sigma(\{X^{-1}(\{x_n\}) : n = 1, 2, 3, \dots\})$$

is generated by countably many blocks.

Proof. It is clear that $\sigma(\{X^{-1}(\{x_n\}) : n = 1, 2, 3, \dots\}) \subseteq \sigma(X)$. On the other hand, if $B \in \mathcal{B}(\mathbb{R})$, then

$$X^{-1}(B) = \bigcup_{n: x_n \in B} X^{-1}(\{x_n\}),$$

which is a (countable) union of $\{X^{-1}(\{x_n\}) : n = 1, 2, 3, \dots\}$. The fact that $X^{-1}(\{x_n\}) \cap X^{-1}(\{x_m\}) = \emptyset$ if $n \neq m$ and $\bigcup_{n=1}^{\infty} X^{-1}(\{x_n\}) = \Omega$ concludes the proof. ■

For the case of a general function X , we have the following:

Theorem 3.2.3. Let $X : \Omega \rightarrow \mathbb{R}$ and $\mathcal{A} \subseteq \mathcal{B}(\mathbb{R})$ be such that $\sigma(\mathcal{A}) = \mathcal{B}(\mathbb{R})$. Then $\sigma(X) = \sigma(\{X^{-1}(B) : B \in \mathcal{A}\})$.

Proof. Let $\mathcal{F} := \sigma(\{X^{-1}(B) : B \in \mathcal{A}\})$.

($\supseteq$): $\mathcal{F} \subseteq \sigma(X)$ is obvious.

($\subseteq$): Let $\mathcal{G} := \{B \subseteq \mathbb{R} : X^{-1}(B) \in \mathcal{F}\}$. It is an exercise (Tutorial 8) to show that $\mathcal{G}$ is a σ -algebra. Since $\mathcal{A} \subseteq \mathcal{G}$, we also have $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{A}) \subseteq \mathcal{G} \implies X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$. Hence $\sigma(X) \subseteq \mathcal{F}$. ■

This result implies that (for instance) $\sigma(X)$ is generated by $\{\{X \leq c\} : c \in \mathbb{R}\}$ or $\{\{X > c\} : c \in \mathbb{R}\}$ or $\{\{a < X \leq b\} : a, b \in \mathbb{R}\}$, or ... etc. Let us see it action.

Example 3.2.2. Let $\Omega = [0, 1]$ and define $X : \Omega \rightarrow \mathbb{R}$ by

$$X(\omega) := \begin{cases} 2 & 0 \leq \omega < \frac{1}{2} \\ \omega^2 & \frac{1}{2} \leq \omega \leq 1. \end{cases}$$

For each $c \in \mathbb{R}$ we have

$$\{X \leq c\} = \begin{cases} \Omega & c \geq 2 \\ [\frac{1}{2}, 1] & 1 \leq c < 2 \\ [\frac{1}{2}, \sqrt{c}] & \frac{1}{4} \leq c \leq 1 \\ \emptyset & c < \frac{1}{4}. \end{cases}$$

Thus

$$\sigma(X) = \sigma(\{\{X \leq c\} : c \in \mathbb{R}\}) = \sigma\left(\left\{\Omega, \left[\frac{1}{2}, 1\right], \emptyset\right\} \cup \left\{\left[\frac{1}{2}, b\right] : \frac{1}{2} \leq b \leq 1\right\}\right)$$

$$\begin{aligned}
 &= \sigma \left\{ \left\{ \left[\frac{1}{2}, b \right] : \frac{1}{2} \leq b \leq 1 \right\} \right\} \\
 &= \left\{ B \subseteq [0, 1] : B \in \mathcal{B} \left(\left[\frac{1}{2}, 1 \right] \right) \text{ or } B = \left[0, \frac{1}{2} \right) \cup B' \text{ where } B' \in \mathcal{B} \left(\left[\frac{1}{2}, 1 \right] \right) \right\} \\
 &= \left\{ B \cup B' : B \in \mathcal{B} \left(\left[\frac{1}{2}, 1 \right] \right), B' \subseteq \left[0, \frac{1}{2} \right) \right\}
 \end{aligned}$$

The set $\left[0, \frac{1}{2} \right)$ is a block of $\sigma(X)$ since X does not distinguish between elements of this set, implying that if a function $Y : \Omega \rightarrow \mathbb{R}$ is $\sigma(X)$ -measurable, then Y must also be constant on $\left[0, \frac{1}{2} \right)$.

Definition 3.2.2. Let Ω be a set and $\mathcal{X} = \{X_\alpha : \alpha \in I\}$ be a collection of real-valued functions on Ω . We define the σ -algebra generated by $\mathcal{X}$ as the smallest σ -algebra such that all members of $\mathcal{X}$ are random variables. We also denote this σ -algebra by $\sigma(\mathcal{X})$.

We end with the following powerful result.

Theorem 3.2.4 (Doob-Dynkin Lemma). Let Ω be a set and $X : \Omega \rightarrow \mathbb{R}^N$, $Y : \Omega \rightarrow \mathbb{R}$ be functions. Then Y is $\sigma(X)$ -measurable if and only if there exists a Borel measurable function $g : (\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $Y = g(X)$.

Proof. See Tutorial. ■

3.3 The Law of a Random Variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable (i.e., $X \in m\mathcal{F}$) and $B \in \mathcal{B}(\mathbb{R})$. To calculate the probability that $X \in B$, we first find $X^{-1}(B)$ and then evaluate $\mathbb{P}(X^{-1}(B))$. This means that we can define a set function $\mathbb{P}_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ by

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)), \quad B \in \mathcal{B}(\mathbb{R}).$$

The function $\mathbb{P}_X$ is a probability measure on $\mathcal{B}(\mathbb{R})$. Indeed,

1. $\mathbb{P}_X(\emptyset) = \mathbb{P}(X^{-1}(\emptyset)) = 0$.
2. $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) \geq 0, \quad B \in \mathcal{B}(\mathbb{R})$.
3. If $B_1, B_2, \dots \in \mathcal{B}(\mathbb{R})$ are pairwise disjoint elements of $\mathcal{B}(\mathbb{R})$, then $X^{-1}(B_1), X^{-1}(B_2), \dots$ are pairwise disjoint elements of $\mathcal{F}$ and

$$\mathbb{P}_X \left(\bigcup_{n=1}^{\infty} B_n \right) = \mathbb{P} \left(X^{-1} \left(\bigcup_{n=1}^{\infty} B_n \right) \right) = \mathbb{P} \left(\bigcup_{n=1}^{\infty} X^{-1}(B_n) \right) = \sum_{n=1}^{\infty} \mathbb{P}(X^{-1}(B_n)) = \sum_{n=1}^{\infty} \mathbb{P}_X(B_n).$$

4. $\mathbb{P}_X(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$.

Thus, each random variable induces a measure on $\mathcal{B}(\mathbb{R})$. We will call $\mathbb{P}_X$ the *probability distribution* or *Law* of X .

Example 3.3.1. Consider the random experiment of tossing 2 fair coins and let X count the number of heads. Let $\mathcal{F} = 2^\Omega$ and define the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{F})$ by

$$\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{4} \delta_\omega.$$

Then X is automatically a random variable. It can be shown that

$$\mathbb{P}_X = \frac{1}{4}\delta_0 + \frac{1}{2}\delta_1 + \frac{1}{4}\delta_2.$$

We will later see that if X is a ‘discrete’ random variable, then $\mathbb{P}_X$ is a convex combination¹ of Dirac measures.

Proposition 3.3.1. *Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}$ be a random variable with $\text{Range}(X) \in \mathcal{B}(\mathbb{R})$. Then $\mathbb{P}_X$ is concentrated on $\text{Range}(X)$.*

Proof. Note that $B = \text{Range}(X)^c \in \mathcal{B}(\mathbb{R})$ and $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\emptyset) = 0$. ■

Example 3.3.2. Let $\Omega = (0, 1]$, $\mathcal{F} = \mathcal{B}((0, 1])$ and $\mathbb{P} = \lambda_1$. Then $X(\omega) = 4\omega \forall \omega \in \Omega$ is a random variable. We now calculate $\mathbb{P}_X$. First, if $B = (-\infty, x]$, then

$$\begin{aligned} \mathbb{P}_X((-\infty, x]) &= \mathbb{P}(\{X \leq x\}) = \mathbb{P}(\{\omega \in (0, 1] : 4\omega \leq x\}) = \mathbb{P}((0, 1] \cap (-\infty, x/4]) \\ &= \begin{cases} 0 & x \leq 0 \\ \frac{x}{4} & 0 < x < 4 \\ 1 & x \geq 4. \end{cases} \end{aligned}$$

Now, since $\text{Range}(X) = (0, 4]$, $\mathbb{P}_X$ is concentrated on $(0, 4]$ and

$$\mathbb{P}_X(B) = \frac{\lambda_1((0, 4] \cap B)}{4}, \quad B \in \mathcal{B}(\mathbb{R})$$

by the $\pi - \lambda$ theorem (since $\{(-\infty, x] : x \in \mathbb{R}\}$ is a π -system that generates $\mathcal{B}(\mathbb{R})$).

Here is a way to characterise $\mathbb{P}_X$ without the need to find $\mathbb{P}_X(B)$ for every $B \in \mathcal{B}(\mathbb{R})$.

Definition 3.3.1. *Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow \mathbb{R}$ be a random variable. The Cumulative Distribution Function (CDF) of X is the function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ defined by*

$$F_X(x) := \mathbb{P}_X((-\infty, x]) = \mathbb{P}(\{X \leq x\}), \quad x \in \mathbb{R}.$$

Theorem 3.3.1. *The CDF characterizes the distribution of a random variable. That is, if $X, Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ are random variables, then*

$$\mathbb{P}_X = \mathbb{P}_Y \iff F_X = F_Y.$$

Proof. ($\implies$:) Assume that $\mathbb{P}_X = \mathbb{P}_Y$, then it easily follows that $F_X = F_Y$ from the definition.
 ($\impliedby$:) Assume that $F_X = F_Y$. This means that $\mathbb{P}_X$ and $\mathbb{P}_Y$ agree on the π -system $\{(-\infty, x] : x \in \mathbb{R}\}$ generating $\mathcal{B}(\mathbb{R})$, hence they agree everywhere on $\mathcal{B}(\mathbb{R})$. ■

Theorem 3.3.2. *The CDF of any random variable X satisfies the following properties*

1. F_X is increasing; i.e., if $x \leq y$ then $F_X(x) \leq F_X(y)$.
2. F_X is right continuous; i.e., $F_X(x^+) := \lim_{y \rightarrow x^+} F_X(y) = F_X(x)$.
3. $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$.

The converse of the above is also true. That is, if you start with an arbitrary function that satisfies the above properties, then that function is a CDF of some random variable.

¹A convex combination is a linear combination whose coefficients are non-negative and add up to 1

Proof. Exercise. ■

Let $I \subseteq \mathbb{R}$ be an interval. We say that $F : I \rightarrow \mathbb{R}$ is *absolutely continuous* if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $(a_1, b_1), \dots, (a_n, b_n)$ is a finite collection of disjoint intervals in I with

$$\sum_{i=1}^n |b_i - a_i| < \delta,$$

then

$$\sum_{i=1}^n |F(b_i) - F(a_i)| < \epsilon.$$

Properties of absolutely continuous functions are discussed in the appendix.

Definition 3.3.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X \in m\mathcal{F}$ be a random variable.

1. We say that X is discrete iff $\mathbb{P}_X$ is concentrated on a countable set $\{x_n : n = 1, 2, 3, \dots\}$.
2. We say that X is continuous iff F_X is a continuous function.
3. We say that X is absolutely continuous iff F_X is an absolutely continuous function.

Exercise 3.3.1. Show that if X is absolutely continuous, then it is continuous. ■

Remark 3.3.1. There are continuous CDFs that are not absolutely continuous, and one example is given by the *Cantor-Lebesgue function*. ■

Theorem 3.3.3. If X is a discrete random variable with $\mathbb{P}_X$ concentrated on $\{x_n : n = 1, 2, 3, \dots\}$, then

$$\mathbb{P}_X = \sum_{n=1}^{\infty} p_X(x_n) \delta_{x_n},$$

where the function $p_X : \mathbb{R} \rightarrow \mathbb{R}$, called the probability mass function (PMF) of X , is defined by

$$p_X(x) := \mathbb{P}_X(\{x\}), \quad x \in \mathbb{R}.$$

Proof. Exercise. ■

Example 3.3.3. 1. The coin toss random variable counting the number of heads is obviously discrete. In fact, every random variable with a countable range is discrete.

2. Let $\Omega = (0, 1]$, $\mathcal{F} = \mathcal{B}((0, 1])$, $\mathbb{P} = \lambda_1$ and $X(\omega) = 4\omega \forall \omega \in \Omega$. We showed that

$$F_X(x) = \mathbb{P}_X((-\infty, x]) = \begin{cases} 0 & x < 0 \\ \frac{x}{4} & 0 \leq x < 4 \\ 1 & x \geq 4. \end{cases}$$

We show that F_X is absolutely continuous on $\mathbb{R}$. First note that for all $x, y \in \mathbb{R}$, $|F_X(x) - F_X(y)| \leq |x - y|$ (consider cases), hence f is *Lipschitz continuous*. Now let $\epsilon > 0$ and choose $\delta = \epsilon$. If $(a_1, b_1), \dots, (a_n, b_n)$ is a finite collection of disjoint intervals in I with

$$\sum_{i=1}^n |b_i - a_i| < \delta,$$

then

$$\sum_{i=1}^n |F_X(b_i) - F_X(a_i)| \leq \sum_{i=1}^n |b_i - a_i| < \delta = \epsilon.$$

■

Chapter 4

Integration

In this chapter we develop the Lebesgue integral of an extended real-valued measurable function f with respect to a measure μ . In the special case when $\mu = \mathbb{P}$ is a probability measure and $f = X$ is a random variable, this integral is the expected value of X . We also prove the Lebesgue-Radon-Nikodym theorem and use it to characterise all probability distributions.

Our intuition about integration is that it represents ‘aggregation’ or ‘taking a weighted average’. We will see that by an appropriate choice of μ and Ω , the integral will cover the many forms of aggregation we are used to in mathematics, including summation, expectation, and calculating area/volume using the Riemann integral. The advantage now is that we can look at all these forms of aggregation as essentially doing the same thing: integrating with respect to a measure. This point of view will help us prove a single theorem that can be applied to all these special cases. In applications to probability theory, it will enable us to treat discrete, continuous and other random variables the same way. There will be no need, for instance, to prove a theorem for discrete random variables using summation and then prove the same theorem again for absolutely continuous random variables using integrals.

4.1 Integration and Expectation on $m\mathcal{F}^+$

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $f : \Omega \rightarrow \mathbb{R}$ be a measurable function. In this section we want to define the integral of f with respect to the measure μ , denoted by

$$\int_{\Omega} f d\mu \quad \text{or} \quad \int_{\Omega} f(x) d\mu(x).$$

When $\mu = \mathbb{P}$ is a probability measure and $f = X$ is a random variable, we will define the expectation of X as

$$\mathbb{E}(X) := \int_{\Omega} X d\mathbb{P}.$$

For reasons we will see later, it will be useful to define the integral for *extended measurable function* $f : \Omega \rightarrow \bar{\mathbb{R}}$. So throughout this chapter we will assume that all measurable functions are extended measurable functions.

We will carry out the definition in four steps:

1. We first define the integral for the case when $f = I_A$ is an indicator function of a measurable set $A \in \mathcal{F}$.
2. We will then extend the definition to when f is positive¹ and is a (finite) linear combination of indicators. Such functions are called (positive) *simple* functions.

¹We should really be saying ‘non-negative’.

3. We then extend the definition to all positive measurable functions and prove useful results that aid with the calculation of the integral in many concrete examples.
4. Finally, we extend the definition to general measurable (sign-changing) functions f and define the notion of integrability. This will be covered in the next section.

We begin with indicators. Throughout the rest of the chapter, $(\Omega, \mathcal{F}, \mu)$ is a measure space and $f : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is $\mathcal{F}/\mathcal{B}(\mathbb{R})$ -measurable.

Definition 4.1.1. *If $f = I_A$ for some $A \in \mathcal{F}$, then we define the integral of f with respect to the measure μ to be*

$$\int_{\Omega} f d\mu = \int_{\Omega} I_A d\mu := \mu(A).$$

In the case of a probability space, we have

$$\mathbb{E}(I_A) = \int_{\Omega} I_A d\mathbb{P} := \mathbb{P}(A).$$

There is nothing surprising about this definition; it is consistent with our intuition of aggregation. We move on to simple functions.

Definition 4.1.2. *A measurable function $s : \Omega \rightarrow \bar{\mathbb{R}}$ is called simple if it has a finite range. We will denote by $\mathcal{S}$ the set of all (measurable) simple functions. We will also write $\mathcal{S}^+$ for the class of positive simple functions and $m\mathcal{F}^+$ for the class of positive measurable functions.*

Note that if $s \in \mathcal{S}$ with $\text{Range}(s) = \{s_1, \dots, s_n\}$, then s can be written as

$$s = \sum_{k=1}^n s_k I_{A_k},$$

where $A_k = s^{-1}(\{s_k\}) \in \mathcal{F}$ for $k = 1, 2, \dots, n$ are pairwise disjoint. We will call this the canonical representation of a simple function and assume this representation when we talk about a simple function.

Definition 4.1.3. *Let $f = s$, where $s \in \mathcal{S}^+$ is a positive simple function with canonical representation*

$$s = \sum_{k=1}^n s_k I_{A_k}.$$

We define the integral of f with respect to the measure μ as

$$\int_{\Omega} f d\mu = \int_{\Omega} s d\mu := \sum_{k=1}^n s_k \mu(A_k).$$

Note that the value of this integral could be ∞ , even if f is finite.

The probability space equivalent of this definition is that if X is a simple random variable with

$$X = \sum_{k=1}^n s_k I_{A_k},$$

then

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P} := \sum_{k=1}^n s_k \mathbb{P}(A_k).$$

Remark 4.1.1. Note that if $s \in \mathcal{S}^+$ is a positive simple function with canonical representation

$$s = \sum_{k=1}^n s_k I_{A_k},$$

then

$$\int_{\Omega} s \, d\mu = \sum_{k=1}^n s_k \int_{\Omega} I_{A_k} \, d\mu.$$

Example 4.1.1. Show that if $s \in \mathcal{S}^+$ and $\alpha > 0$, then

$$\int_{\Omega} \alpha s \, d\mu = \alpha \int_{\Omega} s \, d\mu.$$

We extend this definition to all positive measurable functions.

Definition 4.1.4. Let $f \in m\mathcal{F}^+$ be a positive measurable function. We define the integral of f with respect to μ as

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq f \right\}.$$

When $\mu = \mathbb{P}$ is a probability measure and $f = X$ is a random variable, then this definition means

$$\mathbb{E}(X) = \int_{\Omega} X \, d\mathbb{P} := \sup \left\{ \mathbb{E}(s) : s \in \mathcal{S}^+, 0 \leq s \leq X \right\}.$$

Remark 4.1.2. If $f \in \mathcal{S}^+$, then this definition gives the same result as the previous definition of the integral for positive simple functions. That is,

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq f \right\}$$

even when $f \in \mathcal{S}^+$. Prove this.

Theorem 4.1.1. If $f, g \in m\mathcal{F}^+$ are such that $f(\omega) \leq g(\omega)$ for every $\omega \in \Omega$, then

$$\int_{\Omega} f \, d\mu \leq \int_{\Omega} g \, d\mu.$$

Proof. This simply follows from the fact that

$$\left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq f \right\} \subseteq \left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq g \right\}.$$

Hence

$$\int_{\Omega} f \, d\mu := \sup \left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq f \right\} \leq \sup \left\{ \int_{\Omega} s \, d\mu : s \in \mathcal{S}^+, 0 \leq s \leq g \right\} =: \int_{\Omega} g \, d\mu.$$

This definition is not user-friendly. We would like to avoid having to use it to calculate integrals in concrete cases. The following two results will help us do just that.

Theorem 4.1.2 (Monotone Convergence Theorem (MON), Beppo-Levi). *Let $f \in m\mathcal{F}^+$ and (f_n) be an increasing sequence of positive measurable functions that converges to f point-wise. Then*

$$\int_{\Omega} f \, d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu.$$

That is,

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu = \int_{\Omega} \lim_{n \rightarrow \infty} f_n \, d\mu.$$

The probability space equivalent statement is that if (X_n) is an increasing sequence of non-negative random variables that converges point-wise to a non-negative random variable X , then

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}\left(\lim_{n \rightarrow \infty} X_n\right) = \mathbb{E}(X).$$

Proof. 1. Since $f_n \leq f_{n+1} \leq f$ for every $n \geq 1$, we have (for each n)

$$\int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f_{n+1} \, d\mu \leq \int_{\Omega} f \, d\mu.$$

Hence

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \leq \int_{\Omega} f \, d\mu$$

by the monotone convergence theorem for real sequences.

2. We now show the reverse inequality. Let $\epsilon \in (0, 1)$ and pick $s \in \mathcal{S}^+$ such that $0 \leq s \leq f$. We assume that

$$s = \sum_{k=1}^m s_k I_{A_k},$$

where $s_k > 0$. Define (g_n) by

$$g_n := \sum_{k=1}^m (1 - \epsilon) s_k I_{A_k \cap \{f_n \geq (1 - \epsilon) s_k\}}, \quad n = 1, 2, 3, \dots,$$

so that $g_n \leq f_n$. Then

$$\begin{aligned} \int_{\Omega} f_n \, d\mu &\geq \int_{\Omega} g_n \, d\mu = \sum_{k=1}^m (1 - \epsilon) s_k \mu(A_k \cap \{f_n \geq (1 - \epsilon) s_k\}) \\ &\rightarrow \sum_{k=1}^m (1 - \epsilon) s_k \mu(A_k) = \int_{\Omega} (1 - \epsilon) s \, d\mu = (1 - \epsilon) \int_{\Omega} s \, d\mu. \end{aligned}$$

Hence by sending $\epsilon \rightarrow 0^+$, we get

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} s \, d\mu.$$

Since $s \in \mathcal{S}^+$, $0 \leq s \leq f$ was arbitrary, we get

$$\lim_{n \rightarrow \infty} \int_{\Omega} f_n \, d\mu \geq \int_{\Omega} f \, d\mu.$$

■

This result says that we can interchange between taking limits and evaluating integrals in the case when the sequence of functions is increasing and positive. We will see other theorems of this nature in the next section. The result will be helpful in calculating the integral of $f \in m\mathcal{F}^+$ if we already know how to calculate the integrals of the f_n 's. In other words, we need to be able to find a sequence (f_n) such that

1. (f_n) is increasing;
2. $f_n \in m\mathcal{F}^+$ for each $n \geq 1$;
3. (f_n) converges point-wise to f ; AND
4. $\int_{\Omega} f_n d\mu$ is easy to calculate.

Here is one case where all these requirements are met.

Theorem 4.1.3 (Simple Function Approximation). *If $f \in m\mathcal{F}^+$, then there exists an increasing sequence of positive simple functions (s_n) such that (s_n) converges point-wise to f .*

Remark 4.1.3. It then follows that

$$\int_{\Omega} f d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} s_n d\mu$$

by (MON). ■

Proof. Let² $s_n := 2^{-n} \lfloor 2^n f \rfloor \wedge n$. This expression can also be written as

$$s_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} I_{\{\frac{k}{2^n} \leq f < \frac{k+1}{2^n}\}} + n I_{\{f \geq n\}}.$$

Now let $\omega \in \Omega$. If $f(\omega) = \infty$, then

$$s_n(\omega) = n \rightarrow \infty = f(\omega).$$

Assume that $f(\omega) < \infty$. Then there exists $N \in \mathbb{N}^+$ such that for each $n \geq N$, there is a $k \in \mathbb{N}$ such that $\frac{k}{2^n} \leq f(\omega) < \frac{k+1}{2^n}$. Hence

$$0 \leq f(\omega) - s_n(\omega) \leq \frac{1}{2^n} \rightarrow 0.$$

As an exercise, check that (s_n) is increasing. ■

Let us see this theorem in action.

Example 4.1.2. Consider the measure space $([0, 1], \mathcal{B}([0, 1]), \lambda_1)$ and $f : [0, 1] \rightarrow \bar{\mathbb{R}}$ defined by $f(x) = x$. We have

$$s_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} I_{\{\frac{k}{2^n} \leq f < \frac{k+1}{2^n}\}} + n I_{\{f \geq n\}},$$

where

$$\left\{ \frac{k}{2^n} \leq f < \frac{k+1}{2^n} \right\} = \left\{ x \in [0, 1] : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\} = [0, 1] \cap \left[\frac{k}{2^n}, \frac{k+1}{2^n} \right).$$

²Here $x \wedge y = \min\{x, y\}$

which is the empty set when $k \geq 2^n$. Hence

$$s_n = \sum_{k=0}^{2^n-1} \frac{k}{2^n} I_{\{\frac{k}{2^n} \leq f < \frac{k+1}{2^n}\}} = \sum_{k=0}^{2^n-1} \frac{k}{2^n} I_{[\frac{k}{2^n}, \frac{k+1}{2^n})}$$

$$\int_{[0,1)} s_n d\lambda_1 = \sum_{k=0}^{2^n-1} \frac{k}{2^n} \lambda_1 \left(\left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) \right) = \sum_{k=0}^{2^n-1} \frac{k}{2^{2n}} = \frac{1}{2} \left(1 - \frac{1}{2^n} \right).$$

Hence (MON) gives

$$\int_{[0,1)} f d\mu = \lim_{n \rightarrow \infty} \int_{[0,1)} s_n d\mu = \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 - \frac{1}{2^n} \right) = \frac{1}{2}.$$

We will see later that in general, if $f : [a, b] \rightarrow \mathbb{R}$ is Riemann-integrable, then

$$\int_{[a,b]} f d\lambda_1 = \int_a^b f(x) dx.$$

That is, integrating with respect to the Lebesgue measure is the same as ordinary Riemann integration when f is Riemann integrable on a bounded interval.

Also note that $([0, 1), \mathcal{B}([0, 1)), \lambda_1)$ is actually a probability space, therefore f is random variable. It is easy to show that $f \sim U(0, 1)$ (check this!), which is consistent with the fact the the integral represents the expected value of f . ■

Here is another form of aggregation.

Example 4.1.3. Let us look at the counting measure on $\mathbb{N}$. That is, the measure space $(\mathbb{N}, 2^{\mathbb{N}}, \#)$. A function $f : \mathbb{N} \rightarrow \bar{\mathbb{R}}$ is simply a sequence of (extended) real numbers $f(0), f(1), f(2), \dots$. Every function is measurable in this space since $\mathcal{F}$ is the power set. Now let f be a positive measurable function. By this we simply mean that $f(0), f(1), \dots$ is a sequence of non-negative extended real numbers. We want to find the integral of f with respect to $\#$. Define the sequence of positive simple functions (s_n) by

$$s_n := f I_{(-\infty, n] \cap \mathbb{N}} = \sum_{k=0}^n f(k) I_{\{k\}} = \sum_{k=0}^n f(k) I_{\{k\}} + 0 I_{[n, \infty) \cap \mathbb{N}}.$$

For each $x \in \mathbb{N}$, we have

$$\lim_{n \rightarrow \infty} s_n(x) = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(k) I_{\{k\}}(x) = \sum_{k=0}^{\infty} f(k) I_{\{k\}}(x) = f(x).$$

So (s_n) converges point-wise to f . It is also easy to see that (s_n) is an increasing sequence, hence (MON) gives

$$\int_{\mathbb{N}} f d\# = \lim_{n \rightarrow \infty} \int_{\mathbb{N}} s_n d\# = \lim_{n \rightarrow \infty} \sum_{k=0}^n f(k) 1 = \sum_{k=0}^{\infty} f(k).$$

Thus, summation is a type of integration, where the measure is the counting measure. ■

The next result also gives a way of calculating an integral (with respect to any measure) by converting it to an ordinary Riemann improper integral. Some textbooks use it as the *definition* of integration with respect to a measure.

Theorem 4.1.4 (Layer Cake Representation, Darth Vader Law). *If $f \in m\mathcal{F}^+$, then*

$$\int_{\Omega} f d\mu = \int_0^{\infty} \mu(\{f > t\}) dt.$$

If $\mu = \mathbb{P}$ is a probability measure and $f = X$ is a positive random variable, this reads:

$$\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(\{X > t\}) dt = \int_0^{\infty} \mathbb{P}_X((t, \infty]) dt.$$

Proof. Exercise. Show this first for indicators, then positive simple functions and then use (MON) to extend it to every element of $m\mathcal{F}^+$. ■

Theorem 4.1.5 (Markov's Inequality). *If $f \in m\mathcal{F}^+$ and $t > 0$, then*

$$\mu(\{f > t\}) \leq \frac{1}{t} \int_{\Omega} f d\mu.$$

Proof. Note that $tI_{\{f > t\}} \leq f$, hence

$$t\mu(\{f > t\}) = \int_{\Omega} tI_{\{f > t\}} d\mu \leq \int_{\Omega} f d\mu.$$

Corollary 4.1.1. *If $f \in m\mathcal{F}^+$ is such that*

$$\int_{\Omega} f d\mu = 0,$$

then $f = 0$ μ -a.e..

Proof. Let $A = \{f > 0\}$. We want to show that A is a μ -null set. For each $n \geq 1$, define $A_n := \{f > \frac{1}{n}\}$. Then $A = \bigcup_{n=1}^{\infty} A_n$. By Markov's inequality,

$$\mu(A_n) \leq n \int_{\Omega} f d\mu = 0,$$

hence $\mu(A) = 0$ by σ -sub-additivity. ■

We will end this section by looking at the question of interchanging limits and integration again. We saw that (MON) allows us to do this when the sequence (f_n) is increasing and positive. What happens when we drop these assumptions? In general, one cannot interchange limits and integrals.

Example 4.1.4. Consider the measure space $([0, 1], \mathcal{B}([0, 1]), \lambda_1)$ and $f_n = nI_{[0, \frac{1}{n}]}$ for $n = 1, 2, 3, \dots$. For each $x \in [0, 1]$, there exists $N \geq 1$ such that $\frac{1}{n} < x$ for every $n \geq N$. Hence $\lim_{n \rightarrow \infty} f_n(x) = 0 =: f(x)$. On the other hand we have

$$\int_{[0,1]} f d\lambda_1 = 0 < 1 = \lim_{n \rightarrow \infty} \int_{[0,1]} f_n d\lambda_1.$$

This can be written as

$$\int_{[0,1]} \lim_{n \rightarrow \infty} f_n d\lambda_1 < \lim_{n \rightarrow \infty} \int_{[0,1]} f_n d\lambda_1.$$

■

Theorem 4.1.6 (Fatou's Lemma). *Let (f_n) be a sequence of positive measurable functions. Then*

$$\int_{\Omega} \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} f_n d\mu.$$

If (X_n) is a sequence of positive random variables, then

$$\mathbb{E}(\liminf_{n \rightarrow \infty} X_n) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n).$$

Proof. Exercise. Apply (MON) to $g_n := \inf_{k \geq n} f_k$. ■

We also have a reverse version.

Theorem 4.1.7 (Reverse Fatou's Lemma). *Let (f_n) be a sequence of positive measurable functions such that there exists $g \in m\mathcal{F}^+$ such that*

$$\int_{\Omega} g d\mu < \infty \text{ and } 0 \leq f_n(\omega) \leq g(\omega) \quad \forall \omega \in \Omega, n \geq 1$$

. Then

$$\int_{\Omega} \limsup_{n \rightarrow \infty} f_n d\mu \geq \limsup_{n \rightarrow \infty} \int_{\Omega} f_n d\mu.$$

Proof. Exercise. Apply the previous result to $g_n = g - f_n$. ■

4.2 Extension and Further Properties

We now extend the definition of the integral to functions in $m\mathcal{F}$ that may change sign. If $f \in m\mathcal{F}$ we define the positive part of f as the function $f^+ := \max\{f, 0\}$, and the negative part of f as $f^- := \max\{-f, 0\}$. It is easy to show that both $f^+, f^- \in m\mathcal{F}^+$, $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Definition 4.2.1. *Let $f \in m\mathcal{F}$. We define the integral of f with respect to μ as*

$$\int_{\Omega} f d\mu := \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu,$$

provided at least one of the integrals on the right is finite. In that case, we say that the integral exists, otherwise, we say that the integral does not exist. We will say that f is integrable if both integrals on the right are finite, i.e., if

$$\int_{\Omega} |f| d\mu < \infty.$$

We will denote by $\mathcal{L}^1 := \mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ the set of all integrable functions. In general, if $p \in [1, \infty)$, we define

$$\mathcal{L}^p := \mathcal{L}^p(\Omega, \mathcal{F}, \mu) := \left\{ f \in m\mathcal{F} : \int_{\Omega} |f|^p d\mu < \infty \right\}.$$

Remark 4.2.1. As an exercise, formulate these definitions for probability spaces and random variables. ■

Lemma 4.2.1. *If $f, g \in m\mathcal{F}$ are such that $f(\omega) \leq g(\omega)$ for every $\omega \in \Omega$, then*

$$\int_{\Omega} f d\mu \leq \int_{\Omega} g d\mu,$$

provided both integrals exist.

Proof. Write $f = f^+ - f^-$ and $g = g^+ - g^-$. Then $f^+ \leq g^+$ and $g^- \leq f^-$, hence by Theorem 4.1.1,

$$\int_{\Omega} f^+ d\mu \leq \int_{\Omega} g^+ d\mu \text{ and } \int_{\Omega} g^- d\mu \leq \int_{\Omega} f^- d\mu.$$

Rearranging gives

$$\int_{\Omega} f d\mu = \int_{\Omega} f^+ d\mu - \int_{\Omega} f^- d\mu \leq \int_{\Omega} g^+ d\mu - \int_{\Omega} g^- d\mu = \int_{\Omega} g d\mu.$$

■

Corollary 4.2.1. *If $f \in m\mathcal{F}$, then*

$$\left| \int_{\Omega} f d\mu \right| \leq \int_{\Omega} |f| d\mu,$$

provided the integral exists.

Proof. Exercise. Use the fact that $-|f| \leq f \leq |f|$. ■

We will also be interested in integrating functions over subsets of Ω . If $A \in \mathcal{F}$, we define the integral of f over A as

$$\int_A f d\mu := \int_{\Omega} f I_A d\mu,$$

provided the integral on the right hand side exists. If $\mu = \mathbb{P}$ is a probability space and $f = X$ is a random variable, we will write this as

$$\mathbb{E}(X; A) := \mathbb{E}(X I_A).$$

Lemma 4.2.2. *If $A \in \mathcal{F}$ and $\mu(A) = 0$, then*

$$\int_A f d\mu = 0$$

for all f such that the integral exists.

Proof. 1. First assume that $f = I_F$ is an indicator function, where $F \in \mathcal{F}$. Then

$$\int_A f d\mu = \int_{\Omega} I_A I_F d\mu = \int_{\Omega} I_{A \cap F} d\mu = \mu(A \cap F) = 0.$$

2. Now assume that f is a positive simple function with canonical representation

$$f = \sum_{k=1}^n s_k I_{A_k}.$$

Then

$$\int_A f d\mu = \int_{\Omega} \sum_{k=1}^n s_k I_{A \cap A_k} d\mu = \sum_{k=1}^n s_k \mu(A \cap A_k) = 0.$$

3. Let $f \in m\mathcal{F}^+$. Take (s_n) to be an increasing sequence of positive simple functions that converge point-wise to f . Since the sequence $(I_A s_n)$ is also increasing and converges point-wise to f , by (MON), we have

$$\int_A f d\mu = \int_{\Omega} I_A f d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} I_A s_n d\mu = \lim_{n \rightarrow \infty} \int_A s_n d\mu = 0.$$

4. Finally, let $f \in m\mathcal{F}$ be such that the integral exists. Then

$$\int_A f d\mu = \int_{\Omega} I_A f d\mu = \int_{\Omega} I_A f^+ d\mu - \int_{\Omega} I_A f^- d\mu = 0 - 0 = 0.$$

■

Remark 4.2.2. This technique of going through the four steps in proving a result involving integrals is very common in measure theory, and some authors call it *The Standard Machine* (SM). ■

Theorem 4.2.1. Let $f \in m\mathcal{F}$ be such that the integral

$$\int_{\Omega} f d\mu$$

exists. If $A, B \in \mathcal{F}$ are disjoint, then

$$\int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu.$$

Remark 4.2.3. In general, we have

$$\int_{A \cup B} f d\mu = \int_A f d\mu + \int_B f d\mu - \int_{A \cap B} f d\mu,$$

provided at least one of the integrals is finite. ■

Proof. We apply the standard machine (SM).

1. If $f = I_F$ for $F \in \mathcal{F}$, then

$$\int_{A \cup B} f d\mu = \int_{\Omega} I_{F \cap (A \cup B)} d\mu = \mu(F \cap (A \cup B)) = \mu(F \cap A) + \mu(F \cap B) = \int_A I_F d\mu + \int_B I_F d\mu.$$

2. This is easily extended to the case when f is simple.

3. If $f \in m\mathcal{F}^+$, then we take an increasing sequence (s_n) such that $s_n \in \mathcal{S}^+$ and $s_n \rightarrow f$. Then the sequences $(I_{A \cup B} s_n)$, $(I_A s_n)$ and $(I_B s_n)$ are increasing and converge to $I_{A \cup B} f$, $I_A f$ and $I_B f$, respectively. Hence by (MON) we get

$$\begin{aligned} \int_{A \cup B} f d\mu &= \int_{\Omega} I_{A \cup B} f d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} I_{A \cup B} s_n d\mu = \lim_{n \rightarrow \infty} \int_{A \cup B} s_n d\mu = \lim_{n \rightarrow \infty} \left(\int_A s_n d\mu + \int_B s_n d\mu \right) \\ &= \lim_{n \rightarrow \infty} \int_A s_n d\mu + \lim_{n \rightarrow \infty} \int_B s_n d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} I_A s_n d\mu + \lim_{n \rightarrow \infty} \int_{\Omega} I_B s_n d\mu = \int_A f d\mu + \int_B f d\mu. \end{aligned}$$

4. Finally let $f \in m\mathcal{F}$ be such that the integral

$$\int_{\Omega} f d\mu$$

exists. This implies that for each $F \in \mathcal{F}$, at least one of the integrals

$$\int_{\Omega} I_F f^+ d\mu \quad \text{or} \quad \int_{\Omega} I_F f^- d\mu$$

is finite. The following calculations are then justified

$$\begin{aligned} \int_{A \cup B} f d\mu &= \int_{\Omega} I_{A \cup B} f d\mu = \int_{\Omega} I_{A \cup B} f^+ d\mu - \int_{\Omega} I_{A \cup B} f^- d\mu = \int_{A \cup B} f^+ d\mu - \int_{A \cup B} f^- d\mu \\ &= \int_A f^+ d\mu + \int_B f^+ d\mu - \int_A f^- d\mu - \int_B f^- d\mu = \int_A f d\mu + \int_B f d\mu. \end{aligned}$$

Theorem 4.2.2. *Let $f, g \in m\mathcal{F}$ be such that $f = g$ μ -a.e.. Then*

$$\int_{\Omega} f d\mu = \int_{\Omega} g d\mu,$$

in the sense that if one integral exists, then the other one exists and they are equal.

Proof. Let $N = \{f \neq g\} = \{\omega \in \Omega : f(\omega) \neq g(\omega)\}$. Then N is a μ -null set, hence there exists $N' \in \mathcal{F}$ such that $N \subseteq N'$ and $\mu(N') = 0$. We then have

$$\int_{\Omega} f d\mu = \int_{\Omega \setminus N'} f d\mu + \int_{N'} f d\mu = \int_{\Omega \setminus N'} g d\mu + 0 = \int_{\Omega} g d\mu.$$

Remark 4.2.4. Note that we had to assume that $f, g \in m\mathcal{F}$. If $f \in m\mathcal{F}$ and $f = g$ μ -a.e., it does not follow that $g \in m\mathcal{F}$, unless $(\Omega, \mathcal{F}, \mu)$ is complete. ■

Example 4.2.1. We now evaluate integrals with respect to the Dirac measure. Let Ω be a set and $a \in \Omega$. We consider the measure space $(\Omega, 2^{\Omega}, \delta_a)$ and a function $f : \Omega \rightarrow \mathbb{R}$. We want to evaluate

$$\int_{\Omega} f d\delta_a.$$

Define the simple function g by $g = f(a)I_{\{a\}}$. Then $f = g$ δ_a -a.e.. To see this, notice that since $f(a) = g(a)$, then $\{\omega \in \Omega : f(\omega) \neq g(\omega)\} \subseteq \{a\}^c$, which has measure zero. Hence,

$$\int_{\Omega} f d\delta_a = \int_{\Omega} g d\delta_a = f(a)\delta_a(\{a\}) = f(a).$$

Theorem 4.2.3. *If $f \in \mathcal{L}^1$, then f is finite μ -a.e. That is, $\mu(\{|f| = \infty\}) = 0$.*

Proof. We apply Markov's inequality. Define $A_n := \{|f| > n\}$ for $n = 1, 2, 3, \dots$. The sequence (A_n) is decreasing and

$$\bigcap_{n=1}^{\infty} A_n = \{|f| = \infty\}.$$

By Markov's inequality,

$$\mu(A_n) \leq \frac{1}{n} \int_{\Omega} |f| d\mu < \infty,$$

hence by continuity from above, we have

$$0 \leq \mu(\{|f| = \infty\}) = \lim_{n \rightarrow \infty} \mu(A_n) \leq \lim_{n \rightarrow \infty} \frac{1}{n} \int_{\Omega} |f| d\mu = 0.$$

This result is very useful since it allows us to define $f + g$ when $f, g \in \mathcal{L}^1$ by ignoring a null set where (possibly) one of them is ∞ and the other is $-\infty$.

Theorem 4.2.4. *If $f, g \in \mathcal{L}^1$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g \in \mathcal{L}^1$ and*

$$\int_{\Omega} \alpha f + \beta g d\mu = \alpha \int_{\Omega} f d\mu + \beta \int_{\Omega} g d\mu.$$

Proof. Exercise. A tedious application of the standard machine. ■

We end this section with a powerful theorem that allows us to interchange limits and integrals.

Theorem 4.2.5 (Lebesgue Dominated Convergence Theorem (DCT)). *Let (f_n) be a sequence of measurable functions such that there exists $g \in \mathcal{L}^1 \cap m\mathcal{F}^+$ such that $|f_n(\omega)| \leq g(\omega)$ for every $\omega \in \Omega$ and $n \geq 1$. If $f \in m\mathcal{F}$ and $\lim_{n \rightarrow \infty} f_n(\omega) = f(\omega)$ for every $\omega \in \Omega$, then*

1. $f \in \mathcal{L}^1$,

- 2.

$$\lim_{n \rightarrow \infty} \int_{\Omega} |f_n - f| d\mu = 0$$

and

- 3.

$$\int_{\Omega} f d\mu = \lim_{n \rightarrow \infty} \int_{\Omega} f_n d\mu.$$

Proof. 1. By Fatou's Lemma

$$\int_{\Omega} |f| d\mu = \int_{\Omega} \lim_{n \rightarrow \infty} |f_n| d\mu = \int_{\Omega} \liminf_{n \rightarrow \infty} |f_n| d\mu \leq \liminf_{n \rightarrow \infty} \int_{\Omega} |f_n| d\mu \leq \int_{\Omega} g d\mu < \infty.$$

2. Note that $|f| = \lim_{n \rightarrow \infty} |f_n| \leq g$. Thus $|f_n - f| \leq |f_n| + |f| \leq 2g$. Hence we can apply reverse Fatou's lemma to $g_n := |f_n - f| \leq 2g \in \mathcal{L}^1$ to get

$$0 \leq \limsup_{n \rightarrow \infty} \int_{\Omega} |f_n - f| d\mu \leq \int_{\Omega} \limsup_{n \rightarrow \infty} |f_n - f| d\mu = 0.$$

3. This follows from above since

$$0 \leq \left| \int_{\Omega} f_n d\mu - \int_{\Omega} f d\mu \right| \leq \int_{\Omega} |f_n - f| d\mu.$$

Remark 4.2.5. In the language of probability, the theorem says that if (X_n) is a sequence of random variables such that there exists a random variable $Y \in \mathcal{L}^1 \cap m\mathcal{F}^+$ such that $|X_n(\omega)| \leq Y(\omega)$ for every $\omega \in \Omega$, $n \geq 1$, and $X \in m\mathcal{F}$ is such that $\lim_{n \rightarrow \infty} X_n(\omega) = X(\omega)$ for every $\omega \in \Omega$, then

1. $X \in \mathcal{L}^1$,

- 2.

$$\lim_{n \rightarrow \infty} \mathbb{E}(|X_n - X|) = 0$$

and

- 3.

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}\left(\lim_{n \rightarrow \infty} X_n\right) = \mathbb{E}(X).$$

■

■

4.3 Some Special Cases

We now look at how to evaluate the integral in some important special cases. We start with the Lebesgue measure on a closed and bounded interval. Let $\Omega = [a, b]$ where $a < b$ and $\mathcal{F} = \mathcal{L}([a, b]) := \{A \cap [a, b] : A \in \mathcal{L}(\mathbb{R})\}$ be the sigma algebra of Lebesgue measurable subsets of $[a, b]$. If $f : [a, b] \rightarrow \mathbb{R}$, we want to find simpler ways of evaluating

$$\int_{\Omega} f d\lambda_1.$$

Let us recall the Riemann integral. A set $P = \{x_0, \dots, x_n\}$ where $a = x_0 < x_1 < x_2 < \dots < x_n = b$ is called a *partition* of $[a, b]$. We call $\|P\| := \max\{x_i - x_{i-1} : i = 1, 2, \dots, n\}$ the *mesh* of P . A partition P' is called a *refinement* of P if $P \subseteq P'$. A *tagged partition* P^* of $[a, b]$ is a partition together with a choice $x_i^* \in [x_{i-1}, x_i]$ for each $i = 1, 2, 3, \dots, n$. For a bounded function $f : [a, b] \rightarrow \mathbb{R}$, we define the *Riemann sum* corresponding to a tagged partition P^* by

$$R_{P^*}(f) := \sum_{i=1}^n f(x_i^*)(x_i - x_{i-1}).$$

We say that f is *Riemann integrable* if there is an $L \in \mathbb{R}$ such that

$$\lim_{\|P^*\| \rightarrow 0} R_{P^*}(f) = L;$$

that is, if for each $\epsilon > 0$ there exists $\delta > 0$ such that if P^* is a tagged partition with $\|P^*\| < \delta$, then $|R_{P^*}(f) - L| < \epsilon$. We then call L the *Riemann integral* of f on $[a, b]$, and denote it by

$$\int_a^b f(x) dx := L.$$

In particular, if f is Riemann integrable and (P_n^*) is a sequence of tagged partitions such that $\|P_n^*\| \rightarrow 0$ as $n \rightarrow \infty$, then

$$\lim_{n \rightarrow \infty} R_{P_n^*}(f) = \int_a^b f(x) dx.$$

Theorem 4.3.1. *Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is Riemann-integrable if and only if f is continuous λ_1 -a.e.*

Proof. Omitted. ■

Theorem 4.3.2. *Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. If f is Riemann integrable, then it is Lebesgue integrable and the integrals are the same:*

$$\int_a^b f(x) dx = \int_{[a,b]} f d\lambda_1.$$

Proof. Let $M := \sup_{x \in [a,b]} |f(x)| < \infty$. Pick a sequence (P_n^*) of tagged partitions with $\|P_n^*\| \rightarrow 0$

(such a sequence can be easily constructed. For example, take $P_n^* = \{x_0, x_1, \dots, x_n\}$ and $x_i^* = x_{i-1}$ where $x_i = a + i \frac{b-a}{n}$ for $i = 0, 1, 2, \dots, n$). Let $J \subseteq [a, b]$ be the set of discontinuity points of f , so that $\lambda_1(J) = 0$. Define the sequence (s_n) by

$$s_n := \sum_{i=1}^n f(x_i^*) I_{[x_{i-1}, x_i]}.$$

Note that $|s_n(x)| \leq M$ and $s_n(x) \rightarrow f(x)$ for each $x \in [a, b] \setminus J$. Since $\int_{[a,b]} M d\lambda_1 < \infty$, we can apply (DCT) to get that $f \in \mathcal{L}^1$ and

$$\int_{[a,b]} f d\lambda_1 = \lim_{n \rightarrow \infty} \int_{[a,b]} s_n d\lambda_1 = \lim_{n \rightarrow \infty} R_{P_n^*}(f) = \int_a^b f(x) dx.$$

■

It is possible for a function to be Lebesgue integrable but not Riemann integrable.

Example 4.3.1. Let $f : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \cap [0, 1] \\ 0 & x \in [0, 1] \setminus \mathbb{Q}. \end{cases}$$

Then $f = 0$ λ_1 -a.e., hence

$$\int_{[0,1]} f d\lambda_1 = 0.$$

However, f is not Riemann integrable. For any $\delta > 0$, it is possible to construct tagged partitions P_1^* and P_2^* with $\max\{||P_1^*||, ||P_2^*||\} < \delta$ such that P_1^* has rational tags, while P_2^* has irrational tags. We then have

$$R_{P_1^*} = 0 \text{ and } R_{P_2^*} = 1.$$

■

Next we look at linear combinations of measures. Recall that if (μ_k) is a sequence of measures and (a_k) is a sequence of non-negative numbers, then $\mu := \sum_{k=1}^{\infty} a_k \mu_k$ is also a measure.

Theorem 4.3.3. Let (μ_k) be a sequence of measures and (a_k) is a sequence of non-negative numbers. Set $\mu := \sum_{k=1}^{\infty} a_k \mu_k$. If $f \in \mathcal{L}^1(\Omega, \mathcal{F}, \mu)$, then

$$\int_{\Omega} f d\mu = \sum_{k=1}^{\infty} a_k \int_{\Omega} f d\mu_k.$$

Proof. Exercise. Use the standard machine. ■

Example 4.3.2. Let $f : ([0, 1], \mathcal{B}([0, 1])) \rightarrow \mathbb{R}$ be given by $f(x) = x^2$ and let the measure μ be given by $\mu = 2\delta_0 + 3\lambda_1$. Since f is continuous (everywhere), it is Riemann integrable, hence

$$\int_{[0,1]} f d\mu = 2 \int_{[0,1]} f d\delta_0 + 3 \int_{[0,1]} f d\lambda_1 = 2f(0) + 3 \int_0^1 x^2 dx = 0 + 1 = 1.$$

■

We end the section with the following change of variables formula.

Theorem 4.3.4. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $X : \Omega \rightarrow \mathbb{R}$ be a random variable and $\mathbb{P}_X$ be the Law of X . If $g : \mathbb{R} \rightarrow \mathbb{R}$ is $\mathcal{B}(\mathbb{R})/\mathcal{B}(\mathbb{R})$ -measurable with $g \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$, then

$$\mathbb{E}(g(X)) := \int_{\Omega} g(X) d\mathbb{P} = \int_{\mathbb{R}} g d\mathbb{P}_X.$$

Proof. Standard machine. ■

Example 4.3.3. Consider the usual coin toss experiment, where we toss two fair coins and let X be the number of heads. We take $\Omega = \{HH, HT, TH, TT\}$, $\mathcal{F} = 2^\Omega$ and $\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{4} \delta_\omega$. We have shown that

$$\mathbb{P}_X = \frac{1}{4} \delta_0 + \frac{1}{2} \delta_1 + \frac{1}{4} \delta_2.$$

Now let $g : \mathbb{R} \rightarrow \mathbb{R}$ be $g(x) = x^2$. We want to find $\mathbb{E}(X^2) = \mathbb{E}(g(X))$. First, if we do it directly, then

$$\mathbb{E}(g(X)) = \int_{\Omega} X^2 d\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{4} \int_{\Omega} X^2 d\delta_\omega = \sum_{\omega \in \Omega} \frac{1}{4} X^2(\omega) = \frac{1}{4} (2^2 + 1^2 + 1^2 + 0^2) = 3/2.$$

Using the change of variables formula we also get

$$\mathbb{E}(g(X)) = \int_{\mathbb{R}} g d\mathbb{P}_X = \int_{\mathbb{R}} g d\left(\frac{1}{4} \delta_0 + \frac{1}{2} \delta_1 + \frac{1}{4} \delta_2\right) = \frac{1}{4} g(0) + \frac{1}{2} g(1) + \frac{1}{4} g(2) = 3/2.$$

■

4.4 The Lebesgue-Radon-Nikodym Theorem

Let $f \in m\mathcal{F}^+$ and define $\nu_f^\mu : \mathcal{F} \rightarrow \bar{\mathbb{R}}$ by

$$\nu_f^\mu(A) := \int_A f d\mu, \quad A \in \mathcal{F}.$$

Then ν_f^μ is a positive measure (exercise). Also, if $\mu(A) = 0$, then $\nu_f^\mu(A) = 0$. Likewise, if $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $X \in m\mathcal{F}^+$ is a non-negative random variable with $\mathbb{E}(X) = 1$, then the function $\mathbb{P}^* : \mathcal{F} \rightarrow [0, 1]$ defined by

$$\mathbb{P}^*(A) := \mathbb{E}(X; A), \quad A \in \mathcal{F}$$

is a probability measure such that for every $A \in \mathcal{F}$, $\mathbb{P}(A) = 0 \implies \mathbb{P}^*(A) = 0$.

Definition 4.4.1. Let μ, ν be positive measures on $(\Omega, \mathcal{F})$.

1. We say that ν is absolutely continuous with respect to μ (written as $\nu \ll \mu$) if for each $A \in \mathcal{F}$, $\mu(A) = 0 \implies \nu(A) = 0$.
2. We say that μ and ν are equivalent (written as $\mu \equiv \nu$) if $\nu \ll \mu$ and $\mu \ll \nu$.
3. We say that ν and μ are mutually singular (written as $\nu \perp \mu$) if there exists $A \in \mathcal{F}$ such that $\nu(A) = \mu(A^c) = 0$.

Example 4.4.1. 1. On $(\mathbb{R}, \mathcal{L}(\mathbb{R}))$, $\delta_0 \perp \lambda_1$ since $\lambda_1(\{0\}) = \delta_0(\mathbb{R} \setminus \{0\}) = 0$.

2. If $(\Omega, \mathcal{F}, \mu)$ is a measurable space, then $\mu \ll \#$, since $\#(A) = 0 \implies A = \emptyset \implies \mu(A) = 0$.
3. If $(\Omega, \mathcal{F}, \mu)$ is a measure space, then for any constants $a, b > 0$, $a\mu \equiv b\mu$.
4. Note that in general, if $\mu = \sum_{k=1}^{\infty} a_k \mu_k$ where $a_k > 0$ for each $k \geq 1$, then $\mu_k \ll \mu$ for every $k \geq 1$.

■

We have seen above that $\nu_f^\mu \ll \mu$. The Radon-Nikodym Theorem tells us that all measures that are absolutely continuous with respect to μ arise in this way (at least in the σ -finite case).

Theorem 4.4.1 (Radon-Nikodym). *Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, ν be two σ -finite measures on $(\Omega, \mathcal{F})$ such that $\nu \ll \mu$. Then there exists $f \in m\mathcal{F}^+$ such that $\nu = \nu_f^\mu$, in the sense that*

$$\nu(A) = \int_A f d\mu, \quad A \in \mathcal{F}.$$

Furthermore, if $g \in m\mathcal{F}^+$ is such that $\nu = \nu_g^\mu$, then $f = g$ μ -a.e.. Thus, f is unique up to a μ -null set.

Proof. See appendix. ■

Definition 4.4.2. *The function f such that $\nu = \nu_f^\mu$ is called the Radon-Nikodym derivative or density of ν with respect to μ and is denoted by*

$$f =: \frac{d\nu}{d\mu}.$$

Example 4.4.2. 1. If $(\Omega, \mathcal{F}, \mu)$ is a measure space, then define (for any constant $a > 0$), $\nu := a\mu$. We have that $\nu \ll \mu$ and

$$\frac{d\nu}{d\mu} = a, \quad \mu - a.e.$$

To see this, note that for each $A \in \mathcal{F}$,

$$\nu(A) = a\mu(A) = a \int_A 1 d\mu = \int_A a d\mu.$$

2. If $(\Omega, \mathcal{F}, \mu)$ is a measure space and $F \in \mathcal{F}$, define $\nu(A) := \mu(A \cap F)$ for every $A \in \mathcal{F}$. Then ν is also a measure on $(\Omega, \mathcal{F})$ and $\nu \ll \mu$ with

$$\frac{d\nu}{d\mu} = I_F, \quad \mu - a.e.$$

Theorem 4.4.2. *Let $(\Omega, \mathcal{F})$ be a measurable space and ν, μ be measures with $\nu \ll \mu$. If $g \in \mathcal{L}^1(\Omega, \mathcal{F}, \nu)$, then $gd\nu/d\mu \in \mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ and*

$$\int_A g d\nu = \int_A g \frac{d\nu}{d\mu} d\mu, \quad \text{for every } A \in \mathcal{F}.$$

Proof. Fix $A \in \mathcal{F}$. First let $g = I_B$ for some $B \in \mathcal{F}$. Then

$$\text{LHS} = \int_A I_B d\nu = \nu(A \cap B) = \int_{A \cap B} \frac{d\nu}{d\mu} d\mu = \int_A I_B \frac{d\nu}{d\mu} d\mu = \text{RHS}.$$

The rest follows by the standard machine. ■

Proposition 4.4.1. *Let $(\Omega, \mathcal{F})$ be a measurable space and μ, ν and λ be measures.*

1. *If $\nu \ll \mu$ and $\lambda \ll \mu$, then $a\nu + b\lambda \ll \mu$ for every $a, b \geq 0$. Furthermore,*

$$\frac{d(a\nu + b\lambda)}{d\mu} = a \frac{d\nu}{d\mu} + b \frac{d\lambda}{d\mu}, \quad \mu - a.e..$$

2. *If $\lambda \ll \nu$ and $\nu \ll \mu$, then $\lambda \ll \mu$ and*

$$\frac{d\lambda}{d\mu} = \frac{d\lambda}{d\nu} \frac{d\nu}{d\mu}.$$

3. If $\nu \equiv \mu$, then $d\nu/d\mu > 0$ μ -a.e., $d\mu/d\nu > 0$ ν -a.e., and

$$\frac{d\mu}{d\nu} = \frac{1}{\frac{d\nu}{d\mu}} \quad \nu\text{-a.e.}$$

Proof. Exercise. ■

If $(\Omega, \mathcal{F}, \mu)$ is a measure space, there are measures on $(\Omega, \mathcal{F})$ that are neither absolutely continuous with respect to μ nor mutually singular. For instance, on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the measure $\nu := \lambda_1 + \delta_0$ is not absolutely continuous with respect to λ_1 since $\lambda_1(\{0\}) = 0$ while $\nu(\{0\}) = 1$. The measures ν and λ_1 are also not mutually singular, since if there was a set $A \in \mathcal{B}(\mathbb{R})$ such that $\nu(A) = \lambda_1(A^c) = 0$, then we would also have $\lambda_1(A) = 0$, which implies that $\infty = \lambda_1(\mathbb{R}) = \lambda_1(A \cup A^c) = \lambda_1(A) + \lambda_1(A^c) = 0 + 0 = 0$, a contradiction.

The notions of absolute continuity and singularity are in some sense orthogonal properties. This is motivated by the fact that if $\nu \ll \mu$ and $\nu \perp \mu$, then for some $A \in \mathcal{F}$, $\nu(A) = \mu(A^c) = 0 \implies \nu(A^c) = 0 \implies \nu(\Omega) = 0 \implies \nu = 0$. Now recall from Linear Algebra that if $\mathbf{u} \in \mathbb{R}^n$ is a vector, then every vector $\mathbf{v} \in \mathbb{R}^n$ can be decomposed into a sum $\mathbf{v} = \mathbf{v}^{\parallel} + \mathbf{v}^{\perp}$, where $\mathbf{v}^{\parallel}$ is parallel to $\mathbf{u}$ and $\mathbf{v}^{\perp}$ is orthogonal to $\mathbf{u}$. We do something similar for measures.

Theorem 4.4.3 (Lebesgue Decomposition). *Let $(\Omega, \mathcal{F})$ be a measurable space and μ, ν be σ -finite measures on $(\Omega, \mathcal{F})$. Then there exist two unique mutually singular σ -finite measures ν_{ac} and ν_s on $(\Omega, \mathcal{F})$ such that $\nu = \nu_{ac} + \nu_s$, $\nu_{ac} \ll \mu$ and $\nu_s \perp \mu$.*

Proof. Set $\lambda := \mu + \nu$. Then λ is also σ -finite and $\mu \ll \lambda$, thus there exists a density $d\mu/d\lambda$. Define $E := \{\frac{d\mu}{d\lambda} > 0\}$ so that $E^c = \{\frac{d\mu}{d\lambda} = 0\}$ since $\frac{d\mu}{d\lambda} \geq 0$. It is an exercise to check that the measures ν_{ac} and ν_s defined by

$$\nu_{ac}(A) := \nu(E \cap A), \quad \nu_s(A) := \nu(E^c \cap A), \quad A \in \mathcal{F}$$

satisfy the desired conditions. ■

Remark 4.4.1. We call the (ν_{ac}, ν_s) the *Lebesgue decomposition* of ν with respect to μ .

We are going to use this result in the special case of Lebesgue measure on the real line. For the rest of this section we will be working on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We want to study the Lebesgue decomposition of a measure μ with respect to the Lebesgue measure. We start with the following definitions. We will restrict our attention only to measures μ such that $\mu([a, b]) < \infty$ for any $a, b \in \mathbb{R}$. Note that this implies that μ is σ -finite.

Definition 4.4.3. A measure $\mu : \mathcal{B}(\mathbb{R}) \rightarrow [0, \infty]$ is called

1. Discrete if μ is concentrated on a countable set
2. Continuous if $\mu(\{x\}) = 0$ for every $x \in \mathbb{R}$
3. Singular continuous if $\mu \perp \lambda_1$ and μ is continuous
4. Absolutely continuous if $\mu \ll \lambda_1$.

Discrete measures are simple. If μ is a discrete measure that is concentrated on the countable set $A = \{x_1, x_2, \dots\}$, then

$$\mu = \sum_{n=1}^{\infty} \mu(\{x_n\}) \delta_{x_n}.$$

Likewise, absolutely continuous measures are characterised by the existence of a density $\frac{d\mu}{d\lambda_1}$. Furthermore, absolutely continuous measures are continuous, since for each $x \in \mathbb{R}$, $\lambda_1(\{x\}) = 0 \implies \mu(\{x\}) = 0$. The only mysterious class is that of the singular continuous measures. Luckily, we will not encounter them in this course. Here is an improvement of the Lebesgue decomposition on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Theorem 4.4.4. *Let μ be a σ -finite measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then there exist three unique σ -finite measures μ_{ac} , μ_d and μ_{sc} on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that*

1. $\mu = \mu_{ac} + \mu_d + \mu_{sc}$
2. μ_{ac} is absolutely continuous (with respect to λ_1)
3. μ_d is discrete
4. μ_{sc} is singular continuous.

Proof. 1. Let (μ_{ac}, μ_s) be the Lebesgue decomposition of μ with respect to λ_1 . All we need to do is show that we can further decompose μ_s into two measures μ_d and μ_{sc} , where μ_d is discrete and μ_{sc} is singular continuous.

2. We first show that μ has countably many atoms (i.e., points $x \in \mathbb{R}$ such that $\mu(\{x\}) > 0$). Since μ is σ -finite, we can find $A_1, A_2, \dots \in \mathcal{B}(\mathbb{R})$ such that $\bigcup_{n=1}^{\infty} A_n = \mathbb{R}$ and $\mu(A_n) < \infty$ for each $n \geq 1$. For each $n \geq 1$, define

$$C_n := \{x \in A_n : \mu(\{x\}) > 0\}$$

to be the set of all atoms in A_n , and for each $k \geq 1$

$$C_n^k := \left\{x \in A_n : \mu(\{x\}) > \frac{1}{k}\right\}.$$

Then

$$C_n = \bigcup_{k=1}^{\infty} C_n^k.$$

Each C_n^k is finite since

$$\infty > \mu(A_n) \geq \mu(C_n^k) \geq \#(C_n^k) \times \frac{1}{k}.$$

Hence C_n is countable (countable union of finite sets). But we also have

$$\{x \in \mathbb{R} : \mu(\{x\}) > 0\} = \bigcup_{n=1}^{\infty} C_n,$$

implying that the set of atoms is countable.

3. Now let $\{x_1, x_2, \dots\}$ be the set of atoms. Define the measures

$$\mu_d := \sum_{n=1}^{\infty} \mu(\{x_n\}) \delta_{x_n} \text{ and } \mu_{sc} := \mu_s - \mu_d \geq 0.$$

It is an exercise to check that these measures indeed satisfy the required conditions. ■

Theorem 4.4.5. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be a random variable with law $\mathbb{P}_X$.*

1. X is discrete if and only if $\mathbb{P}_X$ is discrete.
2. X is continuous if and only if $\mathbb{P}_X$ is continuous.
3. X is absolutely continuous if and only if $\mathbb{P}_X$ is absolutely continuous, and in that case, the Radon-Nikodym derivative $d\mathbb{P}_X/d\lambda_1$ is called the probability density function (PDF) of X and is denoted by f_X ; that is,

$$f_X := \frac{d\mathbb{P}_X}{d\lambda_1}, \quad \lambda_1 - a.e.$$

Proof. 1. This is just a restatement of the definition.

2. The same here.

3. See appendix. ■

Corollary 4.4.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be a random variable with law $\mathbb{P}_X$ (on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$). Then*

1. $\mathbb{P}_X = \mathbb{P}_X^{ac} + \mathbb{P}_X^d + \mathbb{P}_X^{sc}$, where $\mathbb{P}_X^{ac}$ is absolutely continuous, $\mathbb{P}_X^d$ is discrete and $\mathbb{P}_X^{sc}$ is singular continuous.
2. F_X has countably many discontinuities $\{x_n : n = 1, 2, 3, \dots\}$ and

$$\mathbb{P}_X^d = \sum_{n=1}^{\infty} \mathbb{P}_X(\{x_n\}) \delta_{x_n}.$$

3. F_X is differentiable λ_1 -almost everywhere and

$$\frac{d\mathbb{P}_X^{ac}}{d\lambda_1} = F'_X \quad \lambda_1 - a.e.$$

Proof. 1. The measure $\mathbb{P}_X$ is finite (and hence σ -finite), so we can find such a decomposition by Theorem 4.4.4.

2. Note that the discontinuities of F_X coincide with the atoms of $\mathbb{P}_X$. The rest follows from the proof of the previous result.

3. The proof of this part is beyond the scope of the course. ■

Putting all the results together we have the following.

Corollary 4.4.2. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be a random variable with law $\mathbb{P}_X = \mathbb{P}_X^{ac} + \mathbb{P}_X^d + \mathbb{P}_X^{sc}$, where $\mathbb{P}_X^{ac}$ is absolutely continuous, $\mathbb{P}_X^d$ is discrete and $\mathbb{P}_X^{sc}$ is singular continuous. If $g \in \mathcal{L}^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$, then*

$$\mathbb{E}(g(X)) = \int_{\mathbb{R}} g d\mathbb{P}_X = \int_{\mathbb{R}} g F'_X d\lambda_1 + \sum_{n=1}^{\infty} g(x_n) \mathbb{P}_X(\{x_n\}) + \int_{\mathbb{R}} g d\mathbb{P}_X^{sc}.$$

In particular,

1. if X is discrete, then

$$\mathbb{E}(g(X)) = \sum_{n=1}^{\infty} g(x_n) \mathbb{P}_X(\{x_n\}) = \sum_{n=1}^{\infty} g(x_n) p_X(x_n);$$

2. if X is absolutely continuous, then

$$\mathbb{E}(g(X)) = \int_{\mathbb{R}} g d\mathbb{P}_X = \int_{\mathbb{R}} g F'_X d\lambda_1 = \int_{\mathbb{R}} g f_X d\lambda_1.$$

Proof. This follows from the change of variables formula. ■

Example 4.4.3. Take the probability space $([0, 1], \mathcal{B}([0, 1]), \mathbb{P})$ where $\frac{d\mathbb{P}}{d\lambda_1}(\omega) = 2\omega$, $\omega \in [0, 1]$. Define $X : [0, 1] \rightarrow \mathbb{R}$ by

$$X(\omega) = \begin{cases} 2 & 0 \leq \omega < \frac{1}{3} \\ \omega & \frac{1}{3} \leq \omega < \frac{2}{3} \\ 0 & \frac{2}{3} \leq \omega \leq 1. \end{cases}$$

It is easy to see that X is a random variable and $\sigma(X)$ is the Borel sigma algebra on $[1/3, 2/3]$ together with the blocks $[0, 1/3]$ and $[2/3, 1]$. The CDF of X is given by

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \mathbb{P}([2/3, 1]) & 0 \leq x < 1/3 \\ \mathbb{P}([2/3, 1] \cup [1/3, x]) & 1/3 \leq x < 2/3 \\ \mathbb{P}([2/3, 1] \cup [1/3, 2/3]) & 2/3 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$$

Now

$$\mathbb{P}([2/3, 1]) = \int_{[2/3, 1]} \frac{d\mathbb{P}}{d\lambda_1} d\lambda_1 = \int_{2/3}^1 2t dt = 5/9.$$

Similarly, for $1/3 \leq x < 2/3$ we have

$$\mathbb{P}([1/3, x]) = \int_{[1/3, x]} \frac{d\mathbb{P}}{d\lambda_1} d\lambda_1 = \int_{1/3}^x 2t dt = x^2 - \frac{1}{9}.$$

Therefore,

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 5/9 & 0 \leq x < 1/3 \\ 5/9 + x^2 - 1/9 & 1/3 \leq x < 2/3 \\ 5/9 + 3/9 & 2/3 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$$

We see that

$$F'_X(x) = \begin{cases} 2x & 1/3 \leq x < 2/3 \\ 0 & \text{elsewhere.} \end{cases}$$

This gives $\frac{d\mathbb{P}_X^{ac}}{d\lambda_1}(x) = F'_X(x) = 2xI_{[1/3, 2/3]}(x)$, $x \in \mathbb{R}$. We also see that $\mathbb{P}_X^d = 5/9\delta_0 + 1/9\delta_2$ and $\mathbb{P}_X^{sc} = 0$. Putting it all together we have

$$\mathbb{P}_X(B) = \int_B 2xI_{[1/3, 2/3]}(x) d\lambda_1 + \frac{5}{9}\delta_0(B) + \frac{1}{9}\delta_2(B), \quad B \in \mathcal{B}(\mathbb{R}).$$

We can then calculate

$$\begin{aligned} \mathbb{E}(X) &= \int_{\mathbb{R}} x d\mathbb{P}_X = \int_{\mathbb{R}} x(2x)I_{[1/3, 2/3]}(x) d\lambda_1 + \frac{5}{9} \int_{\mathbb{R}} x d\delta_0 + \frac{1}{9} \int_{\mathbb{R}} x d\delta_2 \\ &= \int_{1/3}^{2/3} 2x^2 dx + \frac{5}{9}(0) + \frac{1}{9}(2) = \frac{14}{81} + \frac{2}{9} = \frac{32}{81}. \end{aligned}$$

We can also calculate $\mathbb{E}(X)$ directly using $\mathbb{P}$ as

$$\begin{aligned}\mathbb{E}(X) &= \int_{\Omega} X(\omega) d\mathbb{P} = \int_{[0,1]} X(\omega) \frac{d\mathbb{P}}{d\lambda_1}(\omega) d\lambda_1 = \int_{[0,1]} (2I_{[0,1/3)}(\omega) + \omega I_{[1/3,2/3)}(\omega)) 2\omega d\lambda_1 \\ &= 4 \int_0^{1/3} \omega d\omega + 2 \int_{1/3}^{2/3} \omega^2 d\omega = \frac{2}{9} + \frac{14}{81} = \frac{32}{81}.\end{aligned}$$

■

Chapter 5

The Story So Far

It is now a good time to pause and reflect on what we have accomplished so far. The purpose of this chapter is twofold. First, we want to provide a summary of the main takeaways from the course so far, in the context of probability spaces (which was really the reason we did measure theory in the first place). Secondly, we will give a real world example that illustrates the intuitive meaning behind many of the concepts we have introduced.

5.1 Where Are We? What Have We Done?

Our study began with a random experiment which is modelled by a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The sample space Ω is assumed to be non-empty. The σ -algebra $\mathcal{F}$ is a collection of subsets of Ω (called events or measurable sets) that is closed under countably many set operations (unions, intersections and compliments) and contains both $\emptyset$ and Ω . The measure $\mathbb{P}$ is a set function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ that satisfies, among others, the crucial σ -additivity property: if $A_1, A_2, \dots \in \mathcal{F}$ are pairwise disjoint, then

$$\mathbb{P} \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

After looking at a few examples, we quickly realized that σ -algebras can get quite complicated. Generally, one does not know the form of an arbitrary element of a σ -algebra, except in a few cases. Fortunately, in many concrete cases, there is a smaller, manageable collection $\mathcal{A} \subseteq \mathcal{F}$ that summarises the information in $\mathcal{F}$ (i.e., $\mathcal{F}$ can be uniquely recovered from knowledge of $\mathcal{A}$). This collection $\mathcal{A}$ generates $\mathcal{F}$, in the sense that $\mathcal{F}$ is the smallest σ -algebra containing $\mathcal{A}$.

The most important σ -algebra in the course is the Borel σ -algebra on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$. It is defined to be the smallest σ -algebra containing all the intervals. It has the advantage that it is generated by any collection of all intervals of a certain type. It also contains every set we can imagine (countable sets, co-countable sets, open sets, etc), yet it is not equal to the power set of $\mathbb{R}$. We also defined the Borel σ -algebra on any subset of $\mathbb{R}$ as the restriction of $\mathcal{B}(\mathbb{R})$ to that subset.

In probability theory, σ -algebras have an additional intuitive meaning. When the random experiment is conducted, the full outcome of the experiment is not usually determined completely. A σ -algebra is a collection of events for which we can determine whether or not they have occurred after the experiment is conducted, **for all the possible (sometimes unknown) outcomes**.

The case of a finite σ -algebra is simple. We saw that the full description of a finite σ -algebra is given by the ‘minimal’ elements of $\mathcal{F}$, called *blocks*. In that case, $\mathcal{F}$ is generated by these blocks, and every element of $\mathcal{F}$ is a union of some of these blocks, with the empty set corresponding to the empty union. This elegant description also extends to the case when $\mathcal{F}$ is generated by

countably many blocks.

We defined a random variable X as a function $X : \Omega \rightarrow \mathbb{R}$ that satisfies the measurability condition: $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$. This definition helps us achieve two goals. First, it allows us to calculate probability statements about possible values of X after the experiment is conducted. This is done via

$$\text{Probability that } X \in B = \mathbb{P}(X^{-1}(B)).$$

Note that $\mathbb{P}(X^{-1}(B))$ only makes sense if $X^{-1}(B) \in \mathcal{F}$ for each $B \in \mathcal{B}(\mathbb{R})$. Secondly, it allows us to determine – unambiguously – the value of X after the experiment is conducted. Indeed, since $\mathcal{F}$ consists of the events whose occurrence or non-occurrence can be decided, the fact that $X^{-1}(\{x\}) \in \mathcal{F}$ for every $x \in \mathbb{R}$ implies that we can decide, for each $x \in \mathbb{R}$, whether or not $X = x$ after the experiment is conducted. Doing this for all $x \in \mathbb{R}$ will make us determine the value of X . Thus, saying that X is $\mathcal{F}$ -measurable is equivalent to saying that knowing the status of the events in $\mathcal{F}$ lets us determine the value of X . This point of view is captured elegantly when $\mathcal{F}$ is generated by countably many blocks:

X is measurable if and only if it is constant on blocks.

In general, if $\mathcal{F}$ does not distinguish between two elements of Ω , then any $\mathcal{F}$ -measurable function should not do so as well. Another way of saying this is that if X is $\mathcal{F}$ -measurable, then X must not contain more information than $\mathcal{F}$.

We associated to each random variable X a probability measure $\mathbb{P}_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$, called the Probability Distribution or Law of X . This probability measure is defined by $\mathbb{P}_X(B) := \mathbb{P}(X^{-1}(B))$ for every $B \in \mathcal{B}(\mathbb{R})$. It allows us to “talk about the *values* of X ”, rather than the elements of Ω . This means that we can completely¹ disregard the domain probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and solely focus on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$ to do any calculations involving X . For instance, to calculate the probability that the value of X lies in the interval $(0, 2)$, instead of first finding $X^{-1}((0, 2))$, we can simply calculate $\mathbb{P}_X((0, 2))$. This is in line with how we have been calculating probabilities since STA1006S — by using densities and mass functions on the real line, rather than using a measure on the sample space. In fact, we didn’t even mention the sample space in most cases.

The Cumulative Distribution Function (CDF) of X is the function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ defined by $F_X(x) := \mathbb{P}_X((-\infty, x])$ for each $x \in \mathbb{R}$. It completely determines $\mathbb{P}_X$, in the sense that we can recover $\mathbb{P}_X$ from the knowledge of F_X . We also used F_X to classify random variables as discrete, continuous or absolutely continuous. The CDF of a discrete random variable X is a step function and, in that case, $\mathbb{P}_X$ is concentrated on a countable set $\{x_1, x_2, x_3, \dots\}$, with

$$\mathbb{P}_X = \sum_{n=1}^{\infty} \mathbb{P}_X(\{x_n\})\delta_{x_n}.$$

We then looked at integration and expectation. The main idea is that the abstract Lebesgue integral

$$\int_{\Omega} f d\mu$$

represents some form of aggregation or addition of values of f , weighted by the measure μ . Integration over $\mathbb{N}$ with respect to the counting measure recovered summation as a type of integration, while integration (on $\mathbb{R}$) with respect to the Lebesgue measure was shown to be equivalent to,

¹For the most part.

mutatis mutandis, ordinary Riemann integration. We then defined the expected value of a random variable X as an integral of the function X with respect to the probability measure $\mathbb{P}$:

$$\mathbb{E}(X) := \int_{\Omega} X d\mathbb{P}.$$

A special measure worth mentioning in this context is the Dirac measure, δ_a , at a point $a \in \Omega$. This represents the ‘ultimate’ discrete measure, and integration with respect to it is straight forward:

$$\int_{\Omega} f d\delta_a = f(a).$$

The change of variables formula again rescues us from having to deal with Ω by evaluating the expectation of X as an integral over $\mathbb{R}$:

$$\int_{\Omega} X d\mathbb{P} = \int_{\mathbb{R}} x d\mathbb{P}_X.$$

To get an intuitive interpretation of this formula, assume that Ω is finite, so that X is a discrete random variable taking finitely many values (a simple function). Then the integral for evaluating $\mathbb{E}(X)$ can be written as

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P} = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\{\omega\}).$$

We can reduce the number of terms in the sum by collecting all outcomes ω that X maps to the same real number $x \in \text{Range}(X) := \{x_1, \dots, x_m\}$. That is, if we define (for $i = 1, 2, \dots, m$) $A_i := X^{-1}(\{x_i\}) = \{\omega \in \Omega : X(\omega) = x_i\}$, then

$$\mathbb{E}(X) = \int_{\Omega} X d\mathbb{P} = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\{\omega\}) = \sum_{i=1}^m x_i \mathbb{P}(A_i) = \sum_{i=1}^m x_i \mathbb{P}_X(\{x_i\}) = \int_{\mathbb{R}} x d\mathbb{P}_X.$$

The last thing we considered was absolute continuity of measures. The main result in this section is that $\mathbb{P}_X$ can be written as $\mathbb{P}_X = \mathbb{P}_X^{ac} + \mathbb{P}_X^d + \mathbb{P}_X^{sc}$, where $\mathbb{P}_X^{ac} \ll \lambda_1$, $\mathbb{P}_X^d$ is discrete (i.e., $\mathbb{P}_X^d$ is concentrated on a countable set) and $\mathbb{P}_X^{sc}$ is continuous (i.e., $\mathbb{P}_X^{sc}(\{x\}) = 0 \forall x \in \mathbb{R}$) and singular (to λ_1). We also saw how to recover $\mathbb{P}_X$ from F_X . Indeed, the countably many discontinuities of F_X enable us to construct $\mathbb{P}_X^d$, while we also have $d\mathbb{P}_X^{ac}/d\lambda_1 = F'_X$ λ_1 -a.s. This also gave a new characterisation of random variables in terms of $\mathbb{P}_X$.

How does this connect with what we know from second year? In second year, when we said “let $X \sim N(0, 1)$ ”, what we meant was that there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X on this space such that $\mathbb{P}_X \ll \lambda_1$ with

$$\frac{d\mathbb{P}_X}{d\lambda_1}(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, \quad x \in \mathbb{R}.$$

In that case, we could then calculate probabilities as

$$\mathbb{P}_X(B) = \int_B \frac{d\mathbb{P}_X}{d\lambda_1}(x) d\lambda_1 = \int_B \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} d\lambda_1$$

for each $B \in \mathcal{B}(\mathbb{R})$. The expected value was also calculated using the change of variables formula by

$$\mathbb{E}(X) = \int_{\mathbb{R}} x d\mathbb{P}_X = \int_{\mathbb{R}} x \frac{d\mathbb{P}_X}{d\lambda_1}(x) d\lambda_1 = \int_{\mathbb{R}} x \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} d\lambda_1.$$

The story of discrete random variables is similar. If X has a Poisson distribution with parameter $\lambda > 0$, this means that there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X such that $\mathbb{P}_X$ is discrete and

$$\mathbb{P}_X = \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} \delta_x.$$

The change of variables formula again gives

$$\mathbb{E}(X) = \int_{\mathbb{R}} x d\mathbb{P}_X = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!}.$$

The point, though, is that we don't even need to know what Ω is to make these calculations; it is enough to know $\mathbb{P}_X$.

5.2 Average Age

We now look at a real world application of the concepts and ideas that we have developed. Suppose we are interested in calculating the average age of everyone in South Africa. This will involve finding everyone's age, calculating the sum of the ages and then dividing that sum by the population of SA. We will describe the whole procedure using the language of probability. We will assume that all possible ages are integers between 0 and 120 (both inclusive).

The first step is to collect the data. To do this, we call each citizen for an interview to get a few details about them. These details may include their name or ID number, province, age, or other information that we may deem relevant. We consider the experiment of randomly picking one person and retrieving this information from them (name, ID, etc). The sample space for this experiment is

$$\Omega = \{\text{All South Africans}\}.$$

Then Ω is a finite set with $|\Omega| =: n$. Each outcome $\omega \in \Omega$ is a citizen of this country (e.g., $\omega = \text{Oupa Manyisa}$). We want to define a random variable $X : \Omega \rightarrow \mathbb{R}$ by

$$X(\omega) = \text{age of } \omega.$$

But we first need a σ -algebra $\mathcal{F}$ on Ω to even talk about random variables. Here, a σ -algebra represents the information we get after interviewing the randomly chosen person. It is clear that we need $\mathcal{F}$ to at least contain information about age in order for X to be $\mathcal{F}$ -measurable. If this is the case, then we will be able to get, for instance, the names of all teenagers via $X^{-1}(\{13, 14, \dots, 19\})$. Note that $\sigma(X)$ simply groups everyone according to their age (i.e., the age groups are the blocks of $\sigma(X)$). So we must have $\sigma(X) \subseteq \mathcal{F}$. We choose $\mathcal{F} = 2^\Omega$, which simply means that the information from the interview helps us distinguish between all individuals. The choice of $\mathbb{P}$ is obvious:

$$\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{n} \delta_\omega.$$

The Law of X represents probabilities about the selected individual being a certain age (or set of ages). It is therefore a measure on $\mathcal{B}(\mathbb{R})$ that is concentrated on the finite set $\{0, 1, 2, 3, \dots, 120\}$. Therefore we have,

$$\mathbb{P}_X = \sum_{x=0}^{120} \mathbb{P}_X(\{x\}) \delta_x.$$

For each $x \in \{0, 1, 2, \dots, 120\}$, $\mathbb{P}_X(\{x\})$ is the proportion of citizens aged x .

With these definitions sorted, we can now calculate the average age, which is simply $\mathbb{E}(X)$. The first method of calculating this is

$$\mathbb{E}(X) = \int_{\Omega} X \, d\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{n} X(\omega) = \frac{1}{n} \sum_{\omega \in \Omega} X(\omega).$$

This means that we add the ages $X(\omega)$ and then divide by the population size n . This is too much work if n is large ($n \approx 50$ million in SA). There is also a lot of repetitive adding. For instance, if there are 1 million 20 year olds, then the same number 20 appears 1 million times in the sum.

An obvious way to solve this problem is to group individuals according to age and just take a weighted average of the ages themselves (i.e., work in $\mathbb{R}$, instead of Ω). For each $x \in \mathbb{R}$, the set $X^{-1}(\{x\})$ is the set of all the individuals aged x . To calculate the expected value of X we can then evaluate the shorter sum

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} \frac{1}{n} X(\omega) = \sum_{x=0}^{120} x \mathbb{P}(X^{-1}(\{x\})) = \sum_{x=0}^{120} x \mathbb{P}_X(\{x\}) = \int_{\mathbb{R}} x \, d\mathbb{P}_X,$$

which is just the change of variables formula. Again, we have avoided working directly with the individuals (on Ω) but instead worked with the ages (on $\mathbb{R}$).

Chapter 6

Conditional Expectation

In this chapter we put a rigorous foundation on another fundamental concept of probability: conditional expectation. We will introduce this concept with the example of finding the average age that was introduced in the previous chapter to motivate the seemingly abstract *partial averaging property*.

6.1 Intuition

We continue with our project of finding the average age in SA. Suppose that we decide to split the task of finding the average age by province. That is, we ask each of the nine provinces to first calculate their own average; we would then combine these averages to get a national average. In terms of our random experiment, this means that the information available should also allow us to determine the province of the randomly chosen individual. Let us label the provinces with numbers as follows:

Province	Code
WC	1
EC	2
NC	3
FS	4
NW	5
KZN	6
GP	7
MP	8
L	9

Define the random variable $Y : \Omega \rightarrow \mathbb{R}$ by $Y(\omega) = \text{province code of } \omega$. For each $y \in \{1, 2, 3, \dots, 9\}$ define $A_y := \{Y = y\} = \{\text{Individuals belonging to province } y\}$. These events are the blocks that generate $\sigma(Y)$. The average age for province y is

$$\mathbb{E}(X|Y = y) = \mathbb{E}(X|A_y) := \frac{1}{|A_y|} \sum_{\omega \in A_y} X(\omega) = \frac{n}{|A_y|} \sum_{\omega \in A_y} \frac{1}{n} X(\omega) = \frac{1}{\mathbb{P}(A_y)} \int_{A_y} X d\mathbb{P}. \quad (6.1)$$

We can think of this number as a representative age for each province. It is the single number we would produce if we were asked for the age of an individual for whom the only thing we knew about was their province. To recover the national average from the provincial averages, we simply take a weighted average of the provincial averages, weighted by the proportion of the total population in each province; that is,

$$\mathbb{E}(X) = \sum_{y=1}^9 \mathbb{E}(X|Y = y) \mathbb{P}(\{Y = y\}) = \sum_{y=1}^9 \int_{A_y} X d\mathbb{P} = \int_{\Omega} X d\mathbb{P} \quad (6.2)$$

since the events $A_1, \dots, A_9$ form a partition of Ω . Now here is the crucial point. The averages vary with each province, so we can define a new random variable, denoted $\mathbb{E}(X|Y)$, that represents the provisional average. This random variable equals $\mathbb{E}(X|Y = y)$ when $Y = y$, so $\mathbb{E}(X|Y)$ is constant on each block A_y , and this constant equals the provincial average. Hence $\mathbb{E}(X|Y)$ is $\sigma(Y)$ -measurable (it is a function of Y). We can think of $\mathbb{E}(X|Y)$ as the best $\sigma(Y)$ -measurable random variable that estimates X . That is, if each person $\omega \in \Omega$ only gave us information about their province, then $\mathbb{E}(X|Y)(\omega)$ would be the ‘best’ guess for their age. This guess is constant on each province, of course. Equation (6.2) can now be written as

$$\mathbb{E}(X) = \mathbb{E}(\mathbb{E}(X|Y)),$$

which is the Law of Total Expectation.

Let us get another characterisation of $\mathbb{E}(X|Y)$. We have seen that

$$\mathbb{E}(X|Y)(\omega) = \frac{1}{\mathbb{P}(A_y)} \int_{A_y} X d\mathbb{P}, \quad \omega \in A_y.$$

Multiplying both sides by $\mathbb{P}(A_y)$ gives

$$\int_{A_y} X d\mathbb{P} = \mathbb{E}(X|Y)(\omega) \times \mathbb{P}(A_y) = \mathbb{E}(X|Y)(\omega) \int_{A_y} 1 d\mathbb{P} = \int_{A_y} \mathbb{E}(X|Y) d\mathbb{P},$$

hence

$$\int_{A_y} X d\mathbb{P} = \int_{A_y} \mathbb{E}(X|Y) d\mathbb{P} \quad \text{for every } y \in \{1, 2, \dots, 9\}. \quad (6.3)$$

By dividing (6.3) by $\mathbb{P}(A_y)$ we get an intuitive interpretation of this equation. It says that the average age in province y is equal to the average of the average age in province y (which is a constant). For a union of two provinces z and y we have

$$\int_{A_y \cup A_z} X d\mathbb{P} = \int_{A_y} X d\mathbb{P} + \int_{A_z} X d\mathbb{P} = \int_{A_y} \mathbb{E}(X|Y) d\mathbb{P} + \int_{A_z} \mathbb{E}(X|Y) d\mathbb{P} = \int_{A_y \cup A_z} \mathbb{E}(X|Y) d\mathbb{P}.$$

The interpretation here is that the average age of the population in provinces y and z is equal to the average of the average ages of the two provinces ($\mathbb{E}(X|Y = y)$ and $\mathbb{E}(X|Y = z)$). Since each $A \in \sigma(Y)$ can be written as a union of blocks A_y , we therefore have

$$\int_A X d\mathbb{P} = \int_A \mathbb{E}(X|Y) d\mathbb{P}, \quad \text{for every } A \in \sigma(Y). \quad (6.4)$$

This is called the *partial averaging property*.

Finally, we notice that to define $\mathbb{E}(X|Y)$ we only used $\sigma(Y)$, and not any other properties of Y . This suggests that the concept of conditional expectation can be defined with respect to a general σ -algebra.

Let us summarise. We first defined the ‘conditional expectation’ of X given an event as an average of X over that event. We then observed that definition can be extended to events that form a partition of Ω , and in that case, the conditional expectation will be a random variable that is constant on each partition set, with the value of the constant being the average of X over that set. This conditional expectation is measurable with respect to the σ -algebra generated by the partition and also satisfies the partial averaging property.

In the next section we are going to formalise these ideas and define the conditional expectation of X given a σ -algebra $\mathcal{G}$.

6.2 Abstraction

The discussion in the previous section was to ensure a smooth transition into the following:

Definition 6.2.1 (Kolmogorov, 1933). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X \in m\mathcal{F}$ be a random variable. If $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of $\mathcal{F}$, then any random variable Z that satisfies the following:*

1. *Z is $\mathcal{G}$ -measurable;*
2. *Z satisfies the partial averaging property:*

$$\int_A Z d\mathbb{P} = \int_A X d\mathbb{P}, \quad \text{for every } A \in \mathcal{G}$$

is called a version of the conditional expectation of X given $\mathcal{G}$.

We should really be saying *the* conditional expectation, but we can only do so after the following theorem.

Theorem 6.2.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X \in \mathcal{L}^1$ be a random variable. If $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of $\mathcal{F}$, then a version of the conditional expectation of X given $\mathcal{G}$ exists and is unique $\mathbb{P}$ -almost surely. We will denote it by $\mathbb{E}(X|\mathcal{G})$, and simply call it the (a.s. defined) conditional expectation of X given $\mathcal{G}$.*

Definition 6.2.2. *If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and $X, Y \in m\mathcal{F}$ are random variables, then we define*

$$\mathbb{E}(X|Y) := \mathbb{E}(X|\sigma(Y)),$$

if it exists.

Proof. Existence is proved by applying the Radon-Nikodym Theorem. First assume that $X \geq 0$ and define the measure $\mu : \mathcal{G} \rightarrow [0, \infty]$ by

$$\mu(A) = \int_A X d\mathbb{P}, \quad A \in \mathcal{G}.$$

Then μ is finite (since $X \in \mathcal{L}^1$) and $\mu \ll \mathbb{P}$. Thus there exists an almost surely unique Radon-Nikodym derivative $Z \in m\mathcal{G}^+$ such that

$$\int_A X d\mathbb{P} = \mu(A) = \int_A Z d\mathbb{P}, \quad \text{for every } A \in \mathcal{G},$$

which is the partial averaging property. Complete the proof to the general case by writing $X = X^+ - X^-$. ■

Here is the interpretation. You have conducted a random experiment whose random outcome $\omega \in \Omega$ is not known. The σ -algebra $\mathcal{F}$ consists of all the events for which we can decide whether or not they have occurred once the experiment is conducted (note that this information might still not enable us to uniquely determine ω). The random variable X is $\mathcal{F}$ -measurable, which means that its value can be determined only from the knowledge of the occurrence or non-occurrence of the events in $\mathcal{F}$ (without knowing ω). Suppose that at the current stage, we are only given partial information contained in the σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. Then $\mathbb{E}(X|\mathcal{G})$ is the $\mathcal{G}$ -measurable random variable that best approximates X (in some sense). That is, the best estimator for X that can be determined from the information in $\mathcal{G}$.

Having established existence, we now look at some examples.

Example 6.2.1. Suppose a fair coin is tossed twice and X represents the number of heads in the two tosses. Choose $\Omega = \{HH, HT, TH, TT\}$, $\mathcal{F} = 2^\Omega$ and $\mathbb{P} = \sum_{\omega \in \Omega} \frac{1}{4} \delta_\omega$. Now let $\mathcal{G} \subseteq \mathcal{F}$ be the σ -algebra that contains information about the outcome of the first toss. Then $\mathcal{G} = \sigma(\{\{HH, HT\}, \{TH, TT\}\})$. We want to find $\mathbb{E}(X|\mathcal{G})$. Since this random variable must be $\mathcal{G}$ -measurable, it must be constant on the blocks that generate $\mathcal{G}$; that is,

$$\mathbb{E}(X|\mathcal{G}) = aI_{\{HH, HT\}} + bI_{\{TH, TT\}}$$

for some $a, b \in \mathbb{R}$. To find a and b , we use the partial averaging property. If $A = \{HH, HT\}$, then

$$\int_A \mathbb{E}(X|\mathcal{G}) d\mathbb{P} = a\mathbb{P}(\{HH, HT\}) = \int_A X d\mathbb{P} = \frac{1}{4}(X(HH) + X(HT)) = \frac{3}{4}.$$

Hence $a = \frac{3}{2}$. A similar calculation on $A = \{TH, TT\}$ gives $b = \frac{1}{2}$.

Here is a more intuitive approach to finding $\mathbb{E}(X|\mathcal{G})$. The (constant) value of the random variable $\mathbb{E}(X|\mathcal{G})$ on the block $\{HH, HT\}$ is given by the weighted average of X over that block:

$$\mathbb{E}(X|\mathcal{G})(\omega) = X(HH) \times \frac{\mathbb{P}(\{HH\})}{\mathbb{P}(\{HH, HT\})} + X(HT) \times \frac{\mathbb{P}(\{HT\})}{\mathbb{P}(\{HH, HT\})} = \frac{3}{2}, \quad \omega \in \{HH, HT\}.$$

Similarly,

$$\mathbb{E}(X|\mathcal{G})(\omega) = X(TH) \times \frac{\mathbb{P}(\{TH\})}{\mathbb{P}(\{TH, TT\})} + X(TT) \times \frac{\mathbb{P}(\{TT\})}{\mathbb{P}(\{TH, TT\})} = \frac{1}{2}, \quad \omega \in \{TH, TT\}.$$

So the random variable $\mathbb{E}(X|\mathcal{G})$ can be determined only from the outcome of the first toss. It is equal to $\frac{3}{4}$ if the first toss is a head and $\frac{1}{2}$ if the first toss is a tail. In contrast, X cannot be determined from this information: if, for instance, the first toss results in a head, the value of X could be 1 or 2 with equal probability. The value of $\mathbb{E}(X|\mathcal{G})$ on this scenario is simply the weighted average of these possibilities (partial average).

Example 6.2.2. Consider the probability space $([0, 1], \mathcal{B}([0, 1]), \mathbb{P} = \lambda_1)$. Define $X, Y : [0, 1] \rightarrow \mathbb{R}$ by

$$X(\omega) := \omega^2, \quad Y(\omega) = (\omega + 1)I_{[0, 1/2]}(\omega), \quad \omega \in [0, 1].$$

We want $\mathbb{E}(X|Y) = \mathbb{E}(X|\sigma(Y))$. First we have

$$\begin{aligned} \sigma(Y) &= \sigma(\{Y \leq c\} : c \in \mathbb{R}) = \sigma\left(\left\{[0, c] : 0 \leq c \leq \frac{1}{2}\right\}\right) \\ &= \left\{B \subseteq [0, 1] : B \in \mathcal{B}\left(\left[0, \frac{1}{2}\right]\right) \text{ or } B = \left(\frac{1}{2}, 1\right] \cup B', B' \in \mathcal{B}\left(\left[0, \frac{1}{2}\right]\right)\right\}. \end{aligned}$$

Thus, a necessary condition for a function Z to be $\sigma(Y)$ -measurable is that it must be constant on $(\frac{1}{2}, 1]$. Hence $\mathbb{E}(X|Y)(\omega) = a$ for $\omega \in (\frac{1}{2}, 1]$. To find a , we use partial averaging

$$\begin{aligned} \int_{(\frac{1}{2}, 1]} \mathbb{E}(X|Y) d\mathbb{P} &= a\mathbb{P}\left(\left(\frac{1}{2}, 1\right]\right) = \int_{(\frac{1}{2}, 1]} \omega^2 d\lambda_1 = \frac{1}{4} \\ \implies a &= \frac{\frac{1}{4}}{\mathbb{P}\left(\left(\frac{1}{2}, 1\right]\right)} = \frac{1}{2}. \end{aligned}$$

For $\omega \in [0, \frac{1}{2}]$, $X(\omega) = \omega^2 = (Y(\omega) - 1)^2$, hence X is $\sigma(Y)$ -measurable on that interval. So we have

$$\mathbb{E}(X|Y)(\omega) = \begin{cases} X(\omega) & \omega \in [0, \frac{1}{2}] \\ \frac{1}{2} & \omega \in (\frac{1}{2}, 1] \end{cases}$$

Check that this random variable does indeed satisfy the required properties.

Here are some useful properties of the conditional expectation. We say that two σ -algebras $\mathcal{F}$ and $\mathcal{G}$ are *independent* if for any $A \in \mathcal{F}$ and $B \in \mathcal{G}$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Proposition 6.2.1. *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X, Y \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P})$. If $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of $\mathcal{F}$, then*

1. (Total Expectation) $\mathbb{E}(\mathbb{E}(X|\mathcal{G})) = \mathbb{E}(X)$
2. If X is $\mathcal{G}$ -measurable, then $\mathbb{E}(X|\mathcal{G}) = X$ a.s.
3. If $X \geq 0$, then $\mathbb{E}(X|\mathcal{G}) \geq 0$ a.s.
4. $\mathbb{E}(aX + bY|\mathcal{G}) = a\mathbb{E}(X|\mathcal{G}) + b\mathbb{E}(Y|\mathcal{G})$ a.s., $a, b \in \mathbb{R}$.
5. (Taking out what is known) If Y is $\mathcal{G}$ -measurable, then $\mathbb{E}(YX|\mathcal{G}) = Y\mathbb{E}(X|\mathcal{G})$ a.s.
6. (Tower Property) If $\mathcal{H} \subseteq \mathcal{G}$, then $\mathbb{E}(\mathbb{E}(X|\mathcal{G})|\mathcal{H}) = \mathbb{E}(X|\mathcal{H})$ a.s.
7. If $\sigma(X)$ and $\mathcal{G}$ are independent, then $\mathbb{E}(X|\mathcal{G}) = \mathbb{E}(X)$.

Proof. We provide a sketch.

1. Use the partial averaging property with $A = \Omega$.
2. X is $\mathcal{G}$ -measurable and trivially satisfies the partial averaging property in this case.
3. Exercise
4. Exercise.
5. Exercise. Use the standard machine on Y .
6. Exercise.
7. Omit.

■

Part II

Stochastic Processes

Chapter 7

An Introduction to Stochastic Processes

In this chapter we introduce stochastic processes using the language of probability theory.

7.1 What is a Stochastic Process?

A random variable X is a real-valued function of the outcome of a random experiment. Each outcome $\omega \in \Omega$ leads to exactly one value $X(\omega) \in \mathbb{R}$. We now want to take this a step further and consider the case when each outcome of the random experiment determines the values of a collection of random variables $\{X_t : t \in \mathbb{I}\}$. We will call such a collection of random variables a *stochastic process* and call the set $\mathbb{I}$ the *index set*. We only consider $\mathbb{I} \subseteq [0, \infty)$ and think of each $t \in \mathbb{I}$ as a time point.

Definition 7.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\emptyset \neq \mathbb{I} \subseteq [0, \infty)$ be a set. A stochastic process $X = \{X_t : t \in \mathbb{I}\}$ is a collection of random variables. We call $\mathbb{I}$ the index set. If $\mathbb{I} \subseteq \mathbb{N}$, we call X a discrete-time stochastic process, while if $\mathbb{I} = [0, \infty)$ or $\mathbb{I} = [0, T]$ for some $T > 0$, we call X a continuous-time stochastic process.

Remark 7.1.1. In general, if $(E, \mathcal{E})$ is a measurable space, we call a collection $X = \{X_t : t \in \mathbb{I}\}$ of E -valued random variables an *E-valued stochastic process*. The set E is called the *state space*. In this course we mainly focus on $(E, \mathcal{E}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and simply call X a stochastic process. ■

Here are some real world examples.

- Example 7.1.1.**
1. Let X_t be the value of the *S&P500* Index at time t . Here, the index set could be taken to be discrete if the prices are observed at discrete time intervals (e.g., daily) or continuous. Each price X_t is a random variable, and one outcome ω determines the values (prices) of X_t for all $t \in \mathbb{I}$.
 2. In gambling, if X_n is your wealth (total earnings) after you have played the n th game, then $X = \{X_n : n = 0, 1, 2, 3, \dots\}$ is a discrete-time stochastic process. ■

The important thing to note here is that one outcome $\omega \in \Omega$ must determine the values of all the random variables $\{X_t : t \in \mathbb{I}\}$. For a fixed outcome $\omega \in \Omega$, we call the function $t \mapsto X_t(\omega)$ a *sample path* of X . Thus, a stochastic process is a ‘sample path-valued’ random variable. That is, we can think of a stochastic process as a function $X : \Omega \rightarrow \{\text{sample paths}\}$, that outputs a path (a function from $\mathbb{I}$ to $\mathbb{R}$) for each outcome, rather than one real number (as is the case for random variables).

Example 7.1.2 (Renewal Process). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $T_1, T_2, T_3, \dots$ be a sequence of positive independent and identically distributed random variables in $\mathcal{L}^1$. Think of T_n

as the time between the occurrence of the $(n-1)^{th}$ and n^{th} events (inter-arrival time). Define the arrival times $W_1, W_2, W_3, \dots$ as

$$W_n := \sum_{k=1}^n T_k, \quad n \geq 1.$$

The stochastic process $X = \{X_t : t \geq 0\}$ defined by

$$X_t := \text{Number of } W_n \text{'s less than or equal to } t = \sum_{n=1}^{\infty} I_{\{W_n \leq t\}}, \quad t \geq 0,$$

is called a *renewal process*. A *Poisson process* is a special case when each T_n has an exponential distribution with parameter $\lambda > 0$. ■

Example 7.1.3. Consider a gambling game where you start with an initial wealth of R5. A coin is tossed twice, and for each toss, you either win R1 if the outcome is a head or lose R1 otherwise. Pick

$$\Omega = \{HH, HT, TH, TT\}, \quad \mathcal{F} = 2^\Omega \text{ and } \mathbb{P} = \frac{1}{4} \sum_{\omega \in \Omega} \delta_\omega.$$

Define $X_0(\omega) = 5$ for all $\omega \in \Omega$ and

$$X_1(HH) = X_1(HT) = 5 + 1 = 6, \quad X_1(TH) = X_1(TT) = 5 - 1 = 4,$$

$$X_2(HH) = 5 + 1 + 1 = 7, \quad X_2(HT) = 5 + 1 - 1 = 5, \quad X_2(TH) = 5 - 1 + 1 = 5, \quad X_2(TT) = 5 - 1 - 1 = 3.$$

Then $X = \{X_0, X_1, X_2\}$ is a stochastic process, where X_t represents your wealth at times $t = 0, 1, 2$. The sample path corresponding to $\omega = HT$, for instance, is

$$X(HT) = \{X_0(HT), X_1(HT), X_2(HT)\} = \{5, 6, 5\}.$$

Here is a table summarising the sample paths of X :

ω	$X_0(\omega)$	$X_1(\omega)$	$X_2(\omega)$
HH	5	6	7
HT	5	6	5
TH	5	4	5
TT	5	4	3

■

Even though all the values of X are determined by one outcome, some of them can be determined without knowledge of the full outcome and are therefore known prior to others. In the above example, we do not need to know the full outcome of the experiment to determine X_1 , since X_1 depends only on the outcome of the first toss. That is, X_1 is measurable with respect to a σ -algebra that is strictly smaller than $\mathcal{F}$.

Definition 7.1.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathbb{I} \subseteq [0, \infty)$ be an index set. A filtration $\mathbb{F} = \{\mathcal{F}_t : t \in \mathbb{I}\}$ is an increasing collection of sub- σ -algebras of $\mathcal{F}$; i.e., $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for $s \leq t$, $s, t \in \mathbb{I}$. A stochastic process $X = \{X_t : t \in \mathbb{I}\}$ is adapted to a filtration $\mathbb{F}$ if for every $t \in \mathbb{I}$, X_t is $\mathcal{F}_t$ -measurable.

Intuitively, a filtration contains the information about ω that is available to us at each time point. At time $t \in \mathbb{I}$, $\mathcal{F}_t$ consists of all the events that can be decided (whether or not they have occurred) by time t . The fact that $\mathbb{F}$ is increasing implies that ‘we do not forget’. An adapted stochastic process is one whose values are known at the corresponding time point. We will call $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$

a filtered probability space.

Associated with a stochastic process $X = \{X_t : t \in \mathbb{I}\}$ is a filtration $\mathbb{F}^X = \{\mathcal{F}_t^X : t \in \mathbb{I}\}$, called the *natural filtration* of X , defined by

$$\mathcal{F}_t^X := \sigma(\{X_s : s \leq t, s \in \mathbb{I}\}), \quad t \in \mathbb{I}.$$

This is the smallest filtration with respect to which X is adapted. It represents the accumulated information gained from observing the stochastic process X .

Example 7.1.4. We continue with the example of 2 coin tosses. Define $\mathbb{F} = \{\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2\}$ by

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad \mathcal{F}_1 = \sigma(\{\{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}\}), \quad \mathcal{F}_2 = \mathcal{F} = 2^\Omega.$$

Then $\mathbb{F}$ is a filtration. The σ -algebra $\mathcal{F}_0$ contains no information, $\mathcal{F}_1$ contains information about the outcome of the first toss and $\mathcal{F}_2$ contains information on both tosses. The natural filtration $\mathbb{F}^X$ also coincides with $\mathbb{F}$ in this case. Indeed, $\mathcal{F}_0^X = \sigma(X_0) = \{\emptyset, \Omega\}$, $\mathcal{F}_1^X = \sigma(\{X_0, X_1\}) = \sigma(X_1) = \sigma(\{X_1^{-1}(\{4\}), X_1^{-1}(\{6\})\}) = \sigma(\{\{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}\})$ and

$$\mathcal{F}_2^X = \sigma(\{X_0, X_1, X_2\}) = \sigma(\{X_1^{-1}(4), X_1^{-1}(6), X_2^{-1}(3), X_2^{-1}(5), X_2^{-1}(7)\})$$

$$= \sigma(\{\{\text{TH}, \text{TT}\}, \{\text{HH}, \text{HT}\}, \{\text{TT}\}, \{\text{HT}, \text{TH}\}, \{\text{HH}\}\}) = \sigma(\{\{\text{HH}\}, \{\text{HT}\}, \{\text{TH}\}, \{\text{TT}\}\}) = 2^\Omega.$$

This can be summarised in the following table:

Blocks of $\mathcal{F}_0$	Blocks of $\mathcal{F}_1$	Blocks of $\mathcal{F}_2$
$\{\{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}\}$	$\{\{\text{HH}, \text{HT}\}, \{\text{TH}, \text{TT}\}\}$	$\{\{\text{HH}\}, \{\text{HT}\}, \{\text{TH}\}, \{\text{TT}\}\}$

■

Suppose you invest your money in an asset whose price is a stochastic process $X = \{X_t : t \in \mathbb{I}\}$. Let τ be a random variable that denotes the time when you decide to sell the asset (with $\tau = \infty$ if you never sell). So τ is a *random time*. If $\mathbb{F} = \{\mathcal{F}_t : t \in \mathbb{I}\}$ represents the information currently available at each time point, then for every $t \in \mathbb{I}$, the decision of whether or not to sell should only depend on the information available at time t , $\mathcal{F}_t$. That is, the event that you have stopped investing by time t , $\{\tau \leq t\}$, should belong to $\mathcal{F}_t$.

Definition 7.1.3. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space. A non-negative extended random variable $\tau : \Omega \rightarrow \mathbb{I} \cup \{\infty\} =: \bar{\mathbb{I}}$ is called a *stopping time* if $\{\tau \leq t\} \in \mathcal{F}_t$ for every $t \in \mathbb{I}$. If τ is a stopping time and $X = \{X_t : t \in \mathbb{I}\}$ is a stochastic process, we define the stopped process $X^\tau = \{X_t^\tau : t \in \mathbb{I}\}$ as $X_t^\tau := X_{\tau \wedge t}$; i.e.,

$$X_t^\tau(\omega) := \begin{cases} X_t(\omega) & t < \tau(\omega) \\ X_{\tau(\omega)}(\omega) & t \geq \tau(\omega) \end{cases} \quad \omega \in \Omega.$$

We also define the random variable X_τ by

$$X_\tau(\omega) := \begin{cases} X_{\tau(\omega)}(\omega) & \tau(\omega) < \infty \\ 0 & \tau(\omega) = \infty. \end{cases}$$

Example 7.1.5. In the coin toss example, say we decide to stop as soon as our wealth reaches R6. Then $\tau(\omega) := \inf\{t \in \mathbb{I} : X_t(\omega) = 6\}$, where we agree that $\inf(\emptyset) = \infty$. Then $\tau(\text{HH}) = \tau(\text{HT}) = 1$, $\tau(\text{TH}) = \tau(\text{TT}) = \infty$. The stopped process X^τ represents our wealth process if we employ this stopping strategy. We have $X_0^\tau = X_{\tau \wedge 0} = X_0 = 5$, $X_1^\tau = X_{\tau \wedge 1} = X_1$ since $\tau \geq 1$, and X_2^τ is given by

$$X_2^\tau(\text{HH}) = X_{\tau(\text{HH}) \wedge 2}(\text{HH}) = X_{1 \wedge 2}(\text{HH}) = X_1(\text{HH}) = 6.$$

A similar argument gives

$$X_2^\tau(\text{HT}) = X_1(\text{HT}) = 6, \quad X_2^\tau(\text{TH}) = X_2(\text{TH}) = 5, \quad X_2^\tau(\text{TT}) = X_2(\text{TT}) = 3.$$

The random variable X_τ is given by

$$X_\tau(\text{HH}) = X_1(\text{HH}) = 6, \quad X_\tau(\text{HT}) = X_1(\text{HT}) = 6, \quad X_\tau(\text{TH}) = 0 = X_\tau(\text{TT}).$$

A table summary:

ω	$\tau(\omega)$	$X_0(\omega)$	$X_1(\omega)$	$X_2(\omega)$	$X_0^\tau(\omega)$	$X_1^\tau(\omega)$	$X_2^\tau(\omega)$	$X_\tau(\omega)$
HH	1	5	6	7	5	6	6	6
HT	1	5	6	5	5	6	6	6
TH	∞	5	4	5	5	4	5	0
TT	∞	5	4	3	5	4	3	0

■

In most instances, stopping times can be generated as follows. Let $B \in \mathcal{B}(\mathbb{R})$ be a Borel set and $X = \{X_t : t \in \mathbb{I}\}$ be a stochastic process. Define $\tau_B : \Omega \rightarrow \bar{\mathbb{I}}$ by

$$\tau_B(\omega) := \inf \{t \in \mathbb{I} : X_t(\omega) \in B\}.$$

We call τ_B the *hitting time*.

Theorem 7.1.1. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space, $\mathbb{I} \subseteq \mathbb{N}$ be a discrete index set and $X = \{X_t : t \in \mathbb{I}\}$ be a stochastic process. If $B \in \mathcal{B}(\mathbb{R})$ and X is adapted to $\mathbb{F}$, then τ_B is an $\mathbb{F}$ -stopping time.*

Proof. Assume that $\mathbb{I} = \mathbb{N}$. For each $t \in \mathbb{N}$, we have

$$\{\tau \leq t\} = \bigcup_{s=0}^t \{X_s \in B\} \in \mathcal{F}_t$$

since $\{X_s \in B\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$ for each $s \leq t$ (because X is adapted). ■

Remark 7.1.2. The analogous result for continuous-time is true but extremely difficult to prove. It is called the *Debut Theorem*. ■

Lemma 7.1.1. *If $\mathbb{I} \subseteq \mathbb{N}$ is discrete, then $\tau : \Omega \rightarrow \bar{\mathbb{I}}$ is a stopping time if and only if $\{\tau = n\} \in \mathcal{F}_n$ for each $n \in \mathbb{I}$.*

Proof. Notice that $\{\tau = n\} = \{\tau \leq n\} \setminus \{\tau \leq n-1\} \in \mathcal{F}_n$. On the other hand,

$$\{\tau \leq n\} = \bigcup_{k=0}^n \{\tau = k\}.$$

■

7.2 Important Classes of Processes

In this section we give some examples of important classes of stochastic processes. We have already classified stochastic processes in terms of the index set (discrete-time or continuous-time). We will now classify them using properties of the sample paths, and the distributional properties of the collection of random variables $\{X_t : t \in \mathbb{I}\}$.

7.2.1 Levy Processes

Assume that $\mathbb{I} = [0, \infty)$ and let $X = \{X_t : t \geq 0\}$ be a stochastic process. For each sample outcome $\omega \in \Omega$, the sample path $\{X_t(\omega) : t \geq 0\}$ is a function from $[0, \infty)$ to $\mathbb{R}$. For each $t \geq 0$, we define the left and right limits (if they exist) as

$$X_{0-}(\omega) := X_0(\omega), \quad X_{t-}(\omega) := \lim_{s \rightarrow t-} X_s(\omega), \quad X_{t+}(\omega) := \lim_{s \rightarrow t+} X_s(\omega).$$

If $X_{t-}(\omega) = X_{t+}(\omega)$ we say the limit exists and its value is equal to the common value of the one-sided limits. The following definitions are in order:

Definition 7.2.1. Let $X = \{X_t : t \geq 0\}$ be a continuous-time stochastic process. We say that X is

1. right (resp. left) continuous if $X_{t+} = X_t$ (resp. $X_{t-} = X_t$) for all t a.s.
2. right continuous with left limits (cadlag) if X is right continuous and the left limits exist for each t . In this case, we define the jump process ΔX as $\Delta X_t := X_t - X_{t-}$ for each $t \geq 0$.
3. left continuous with right limits (caglad) if X is left continuous and the right limits exist for each t .
4. continuous (or X has continuous sample paths) if $X_{t+} = X_{t-} = X_t$ for all t a.s.. This is equivalent to $\Delta X \equiv 0$.
5. increasing if $X_s \leq X_t$ whenever $0 \leq s \leq t < \infty$.

The definition of an increasing process also makes sense in discrete time. We now look at distributional properties of X . If $s < t$, we call the random variable $X_t - X_s$ and *increment* of X . Also recall that $\mathbb{F}^X := \{\mathcal{F}_t^X : t \geq 0\}$ represents the natural filtration of X , which is the smallest filtration with respect to which X is adapted.

Definition 7.2.2. Let $X = \{X_t : t \geq 0\}$ be a continuous-time stochastic process. We say that X has

1. independent increments if for $0 \leq s < t < \infty$, $X_t - X_s$ is independent of $\mathcal{F}_s^X$.
2. stationary increments if for $0 \leq s < t < \infty$, $X_t - X_s$ has the same distribution (law) as $X_{t-s} - X_0$.

A cadlag process with $X_0 = 0$ and stationary and independent increments is called a Levy process¹.

Example 7.2.1. Here are some examples of Levy processes:

- The deterministic process $X_t = bt$, where $b \in \mathbb{R}$.
- A homogeneous Poisson process $N = \{N_t : t \geq 0\}$
- A homogeneous compound Poisson process $\left\{ \sum_{k=1}^{N_t} Y_k : t \geq 0 \right\}$, where $\{Y_k : k = 1, 2, \dots\}$ are iid random variables and N is a homogeneous Poisson process.
- A *Brownian motion* process $W = \{W_t : t \geq 0\}$. This process will be discussed in a later chapter.
- A linear combination of the above:

$$X_t = bt + \sigma W_t + \sum_{k=1}^{N_t} Y_k,$$

where $\sigma \in \mathbb{R}$. Note that a Poisson process is a compound Poisson process with $Y_k \equiv 1$.

¹Some authors replace cadlag with *stochastic continuity*: $\forall \epsilon > 0 \quad \lim_{s \rightarrow t} \mathbb{P}(|X_t - X_s| > \epsilon) = 0$.



7.2.2 Markov Processes

Let $(E, \mathcal{E})$ be a measurable space. Intuitively, an E -valued stochastic process X is a Markov process if, for $s < t$, the distribution of X_t given information about X up to and including time s , depends only on X_s . That is,

Conditional Law of X_t given information on X up to times s = function of X_s .

We need to make the above statement mathematically precise. We have not introduced the idea of a conditional distribution, but loosely speaking, we could use the conditional expectation to define it as follows. Let $\mathcal{G}$ be a σ -algebra and Z be a random variable. A version of the conditional distribution of Z given $\mathcal{G}$ is

$$\mathbb{P}_{Z|\mathcal{G}}(B) := \mathbb{E}(I_B(Z)|\mathcal{G}) = \mathbb{E}(I_{\{X \in B\}}|\mathcal{G}), \quad B \in \mathcal{E}.$$

The problem is that the conditional expectation is only defined a.s., hence there will be no guarantee that $\mathbb{P}_{Z|\mathcal{G}}$ is well-defined. However, it can be shown that for many measurable spaces $(E, \mathcal{E})$ (e.g., $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$), a ‘well-behaved’ version of $\mathbb{P}_{Z|\mathcal{G}}$ exists and can be interpreted as a conditional probability. It is called the *regular conditional probability distribution* of Z given the σ -algebra $\mathcal{G}$. We will not take this route, instead we will simply use the conditional expectation introduced in the previous chapter to define a Markov process.

Definition 7.2.3. Let $(E, \mathcal{E})$ be a measurable space and $X = \{X_t : t \in \mathbb{I}\}$ be an E -valued stochastic process. We say that X is a Markov process if for any $s \leq t$

$$\mathbb{E}(I_{\{X_t \in B\}}|\mathcal{F}_s^X) = \mathbb{E}(I_{\{X_t \in B\}}|X_s) = P(s, t, X_s, B), \quad B \in \mathcal{E},$$

for some transition function P . We call E the state space of X .

Remark 7.2.1. Note that equivalently, we can require that

$$\mathbb{E}(f(X_t)|\mathcal{F}_s^X) = \mathbb{E}(f(X_t)|X_s)$$

for any bounded measurable function $f : (E, \mathcal{E}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. ■

We can interpret the definition as follows. On the far left-hand side, $\mathbb{E}(I_{\{X_t \in B\}}|\mathcal{F}_s^X)$ is the conditional probability that $X_t \in B$ given the history of X up to time s . On the far right, $P(s, t, X_s, B)$ means that this probability is a function of the current time s , the future time t , the current state X_s , and the set B . So we can write this statement (very loosely!) as

$$“\mathbb{P}(X_t \in B|\mathcal{F}_s^X) = \mathbb{P}(X_t \in B|X_s) = P(s, t, X_s, B).”$$

Exercise 7.2.1 (Chapman-Kolmogorov). Let X be a Markov process with state space $(E, \mathcal{E})$. Prove the *Chapman-Kolmogorov equations*: for $0 \leq s \leq r \leq t$,

$$P(s, t, X_s, B) = \mathbb{E}(P(r, t, X_r, B)|\mathcal{F}_s^X) = \mathbb{E}(P(r, t, X_r, B)|X_s).$$

Hint: Condition on $\mathcal{F}_r^X$ and apply the tower property. ■

Example 7.2.2 (Homogeneous continuous-time Markov chains). Let $E = \{0, 1\}$ and $\mathcal{E} = 2^E$. Then there exists a Markov process X with state space E with transition function P given by

$$P(s, t, x, B) = \begin{cases} \frac{1}{3} [(1 + 2e^{-(t-s)}) \delta_0(B) + (2 - 2e^{-(t-s)}) \delta_1(B)] & \text{if } x = 0 \\ \frac{1}{3} [(1 - e^{-(t-s)}) \delta_0(B) + (2 + e^{-(t-s)}) \delta_1(B)] & \text{if } x = 1. \end{cases}$$

The transition probabilities do not depend on the start time s , but only on the difference $t - s$. We call such a Markov process *time-homogeneous*. We can represent the transition function as a matrix called the *transition probability matrix*:

$$P(0, t, \cdot, \cdot) = P_t := \begin{pmatrix} (1 + 2e^{-t}) & (2 - 2e^{-t}) \\ (1 - e^{-t}) & (2 + e^{-t}) \end{pmatrix}, \quad t \geq 0.$$
■

Exercise 7.2.2. Let X, Y be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra on Ω , and $f : (\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be bounded and measurable. Show that if $X \in m\mathcal{G}$ and $\sigma(Y)$ is independent of $\mathcal{G}$, then

$$\mathbb{E}(f(X, Y)|\mathcal{G}) = g(X),$$

where

$$g(x) := \mathbb{E}(f(x, Y)), \quad x \in \mathbb{R}.$$

Hint: First assume that $f = I_{B_1 \times B_2}$, where $B_i \in \mathcal{B}(\mathbb{R})$ for $i = 1, 2$. ■

Theorem 7.2.1. Let $X = \{X_t : t \geq 0\}$ be a stochastic process with state space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. If X has independent increments, then it is a Markov process.

Proof. Let $B \in \mathcal{B}(\mathbb{R})$. We need to show that $\mathbb{E}(I_{\{X_t \in B\}}|\mathcal{F}_s^X)$ is a function of X_s . Note that $\mathbb{E}(I_{\{X_t \in B\}}|\mathcal{F}_s^X) = \mathbb{E}(I_{\{X_s + (X_t - X_s) \in B\}}|\mathcal{F}_s^X)$. Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) := I_B(x + y)$ for every $x, y \in \mathbb{R}$. Then $\mathbb{E}(I_{\{X_s + (X_t - X_s) \in B\}}|\mathcal{F}_s^X) = \mathbb{E}(f(X_s, X_t - X_s)|\mathcal{F}_s^X)$, which is a function of X_s by the previous exercise. ■

This result implies that all Levy processes (e.g., Poisson process, compound Poisson process, and Brownian motion) are Markov processes.

7.2.3 Martingales

In this subsection, $\mathbb{I}$ could be either discrete or continuous and we assume that all processes are real-valued. Intuitively, a stochastic process $X = \{X_t : t \in \mathbb{I}\}$ is a *martingale* if has no upward or downward trend. If X is the price of a share at time t , this means that, on average, X is not expected to gain or lose value. Likewise, if X represents your wealth when playing a series of games, the martingale property simply means that each game is fair.

Definition 7.2.4. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a filtered probability space and $X = \{X_t : t \in \mathbb{I}\}$ be a stochastic process. We call X a *martingale* (or an $(\mathbb{F}, \mathbb{P})$ -martingale) if

1. X is adapted to $\mathbb{F}$;
2. $\mathbb{E}(|X_t|) < \infty$ for all $t \in \mathbb{I}$;
3. $\mathbb{E}(X_t|\mathcal{F}_s) = X_s$ $\mathbb{P}$ -a.s. for $s \leq t$.

We call X a *sub* (resp. *super*)-martingale if it satisfies 1, 2 and

$$3' \quad \mathbb{E}(X_t|\mathcal{F}_s) \geq X_s \text{ (resp. } \mathbb{E}(X_t|\mathcal{F}_s) \leq X_s) \text{ } \mathbb{P} \text{ a.s. for } s \leq t.$$

A martingale is a drift-less process — the best estimator of any future value is the current one. A submartingale drifts upwards, while a supermartingale drifts downwards. Furthermore, X is a martingale if and only if it is both a submartingale and a supermartingale.

Example 7.2.3. 1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $N = \{N_t : t \geq 0\}$ be a Poisson process with rate $\lambda > 0$. Let $\mathbb{F} = \mathbb{F}^N$ be the natural filtration of N . Then N is a submartingale. Indeed, N is adapted to $\mathbb{F}^N$, and $\mathbb{E}(|N_t|) = \mathbb{E}(N_t) = \lambda t < \infty$ for each $t \geq 0$. We also have (for $s < t$)

$$\mathbb{E}(N_t|\mathcal{F}_s^N) = \mathbb{E}(N_s + N_t - N_s|\mathcal{F}_s^N) = N_s + \lambda(t - s) \geq N_s.$$

On the other hand, the *compensated Poisson process* $M = \{N_t - \lambda t : t \geq 0\}$ is a martingale.

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X = (X_n)_{n=1}^\infty$ be a sequence of i.i.d random variables in $\mathcal{L}^1$ with $\mathbb{E}(X_n) = \mu \forall n \in \mathbb{N}^+$. Fix $x \in \mathbb{R}$ and define the random walk $S = \{S_n : n \in \mathbb{N}\}$ by

$$S_0 = x, \quad S_n = \sum_{k=1}^n X_k, \quad n \geq 1.$$

We pick $\mathbb{F} = \{\mathcal{F}_n : n \in \mathbb{N}\}$ as follows:

$$\mathcal{F}_0 := \{\emptyset, \Omega\}, \quad \mathcal{F}_n = \sigma(\{X_1, \dots, X_n\}) \quad n \geq 1.$$

It is easy to see that conditions 1 and 2 are both satisfied. We also have for $m \leq n$,

$$\mathbb{E}(S_n | \mathcal{F}_m) = \mathbb{E}\left(S_m + \sum_{k=m+1}^n X_k | \mathcal{F}_m\right) = S_m + (n - m)\mu.$$

Thus, S is a submartingale if $\mu \geq 0$, a supermartingale if $\mu \leq 0$ and a martingale if $\mu = 0$.

3. In a similar fashion, if $X_k \geq 0$ with $\mathbb{E}(X_k) = \mu$, then $S = \{S_n : n \in \mathbb{N}\}$ defined by $S_0 := 1$, $S_n := \prod_{k=1}^n X_k$ for $n \geq 1$ is a martingale if and only if $\mu = 1$. It is a submartingale if $\mu \geq 1$ and a supermartingale if $\mu \leq 1$. ■

The next result confirms our intuition about martingales being driftless.

Lemma 7.2.1. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered space and $X = \{X_t : t \in \mathbb{I}\}$ be a stochastic process.*

1. *If X is a martingale, then $\mathbb{E}(X_t) = \mathbb{E}(X_s) = \mathbb{E}(X_0)$ for every $s, t \in \mathbb{I}$. That is, X has a constant mean.*
2. *If X is a submartingale, then $\mathbb{E}(X_t) \geq \mathbb{E}(X_s) \geq \mathbb{E}(X_0)$ for every $s \leq t \in \mathbb{I}$. That is, X has an increasing mean*
3. *If X is a supermartingale, then $\mathbb{E}(X_t) \leq \mathbb{E}(X_s) \leq \mathbb{E}(X_0)$ for every $s \leq t \in \mathbb{I}$. That is, X has a decreasing mean.*

Proof. This is a repeated application of the Tower property. We only show the first one and leave the last two as an exercise. For $s < t$, we have

$$\mathbb{E}(X_t) = \mathbb{E}(\mathbb{E}(X_t | \mathcal{F}_s)) = \mathbb{E}(X_s) = \mathbb{E}(\mathbb{E}(X_s | \mathcal{F}_0)) = \mathbb{E}(X_0). \quad \blacksquare$$

Remark 7.2.2. The converse of 1 above is false, i.e., there are non-martingale processes with constant mean. ■

A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is *convex* if for any $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$,

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda \varphi(x) + (1 - \lambda)\varphi(y).$$

Examples of convex functions include $\varphi(x) = x^2, e^x, -\ln x, |x|$.

Lemma 7.2.2 (Jensen's inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X \in \mathcal{L}^1$. If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function such that $\varphi(X) \in \mathcal{L}^1$, and $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra of $\mathcal{F}$, then*

$$\mathbb{E}(\varphi(X) | \mathcal{G}) \geq \varphi(\mathbb{E}(X | \mathcal{G})) \quad \mathbb{P} - a.s..$$

In particular,

$$\mathbb{E}(\varphi(X)) \geq \varphi(\mathbb{E}(X)).$$

Proof. Exercise. ■

Jensen's inequality implies that $\mathbb{E}(X^2) \geq (\mathbb{E}(X))^2$.

Theorem 7.2.2. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space and $X = \{X_t : t \in \mathbb{I}\}$ be a martingale. If φ is a convex function such that $\varphi(X_t) \in \mathcal{L}^1$ for every $t \in \mathbb{I}$, then $\{\varphi(X_t) : t \in \mathbb{I}\}$ is a submartingale.*

Proof. The first two conditions are clear. For the last one, we use Jensen's inequality:

$$\mathbb{E}(\varphi(X_t) | \mathcal{F}_s) \geq \varphi(\mathbb{E}(X_t | \mathcal{F}_s)) = \varphi(X_s), \quad s \leq t.$$
■

Remark 7.2.3. This implies that if X is a martingale, then the processes $X^2, |X|, e^X$ are all submartingales. Likewise, if φ is concave (i.e., $-\varphi$ is convex), then $\varphi(X)$ is a supermartingale. ■

Theorem 7.2.3. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space and $H \in \mathcal{L}^1$. For any index set $\mathbb{I} \subseteq [0, \infty)$, define $X = \{X_t : t \in \mathbb{I}\}$ by $X_t := \mathbb{E}(H | \mathcal{F}_t)$ for each $t \in \mathbb{I}$. Then X is an $(\mathbb{F}, \mathbb{P})$ -martingale.*

Proof. Since $\mathbb{E}(H | \mathcal{F}_t)$ is $\mathcal{F}_t$ -measurable (by definition of conditional expectation), X is adapted to $\mathbb{F}$. Also, by Jensen's inequality (applied to $\varphi(x) = |x|$) we have

$$\mathbb{E}(|X_t|) = \mathbb{E}(|\mathbb{E}(H | \mathcal{F}_t)|) \leq \mathbb{E}(\mathbb{E}(|H| | \mathcal{F}_t)) = \mathbb{E}(|H|) < \infty.$$

Finally, if $s \leq t$, then (since $\mathcal{F}_s \subseteq \mathcal{F}_t$) the tower property gives

$$\mathbb{E}(X_t | \mathcal{F}_s) = \mathbb{E}(\mathbb{E}(H | \mathcal{F}_t) | \mathcal{F}_s) = \mathbb{E}(H | \mathcal{F}_s) = X_s.$$
■

Chapter 8

Discrete-time Martingales

In this chapter we will assume that $\mathbb{I} \subseteq \mathbb{N}$. The martingale definition is simpler in this case.

Lemma 8.0.1. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered space and $X = \{X_n : n \in \mathbb{N}\}$ be a stochastic process. Then X is a martingale if and only if X is adapted to $\mathbb{F}$, $X_n \in \mathcal{L}^1$ for each $n \in \mathbb{N}$ and*

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n) = X_n \quad \mathbb{P} - a.s..$$

Proof. Exercise. ■

8.1 Martingales in Discrete Time

We start with some examples of martingales in discrete time.

Example 8.1.1. Here is a list of some martingales. Let $\epsilon = \{\epsilon_n\}_{n=1}^{\infty}$ be a sequence of i.i.d. random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\epsilon_n \in \mathcal{L}^1$ for each n . We define the random walk $S = \{S_n : n \in \mathbb{N}\}$ by

$$S_0 = x, \quad S_{n+1} = S_n + \epsilon_{n+1} \quad n \in \mathbb{N}$$

for some fixed $x \in \mathbb{R}$. We work with the natural filtration of ϵ , $\mathbb{F}^\epsilon = \{\mathcal{F}_n^\epsilon : n \in \mathbb{N}\}$, defined by

$$\mathcal{F}_0^\epsilon = \{\emptyset, \Omega\}, \quad \mathcal{F}_n^\epsilon = \sigma(\{\epsilon_1, \dots, \epsilon_n\}) \quad n \geq 1.$$

In each case, we will define a new process $X = \{X_n : n \in \mathbb{N}\}$ and show that it is a martingale. In showing that X is a martingale, we are only going to show the third property and leave the first two as an exercise.

- Define $X_n = S_n - n\mu$, where $\mu := \mathbb{E}(\epsilon_n)$. Then

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n^\epsilon) = \mathbb{E}(S_{n+1} - (n+1)\mu | \mathcal{F}_n^\epsilon) = S_n - n\mu + \mathbb{E}(\epsilon_{n+1} - \mu | \mathcal{F}_n^\epsilon) = X_n.$$

In the (common) special case of a simple random walk when ϵ_n takes the value 1 with probability p and -1 with probability $q := 1 - p$, then $\mu = p - q$ and $X_n = S_n - (p - q)n$.

- Further assume that $\mu = 0$ and $\sigma^2 := \mathbb{E}(\epsilon_n^2) < \infty$ for every $n \geq 1$. Let $X_n = S_n^2 - n\sigma^2$ for every $n \in \mathbb{N}$. Then $X = \{X_n : n \in \mathbb{N}\}$ is a martingale with respect to $\mathbb{F}^\epsilon$. Indeed, if $n \in \mathbb{N}$, then

$$\begin{aligned} \mathbb{E}(X_{n+1} | \mathcal{F}_n) &= \mathbb{E}((S_n + \epsilon_{n+1})^2 - (n+1)\sigma^2 | \mathcal{F}_n) = \mathbb{E}(S_n^2 + 2\epsilon_{n+1}S_n + \epsilon_{n+1}^2 | \mathcal{F}_n) - (n+1)\sigma^2 \\ &= S_n^2 - n\sigma^2 = X_n. \end{aligned}$$

For a simple random walk with $\mathbb{P}_{\epsilon_n}(\{1\}) = 0.5 = \mathbb{P}_{\epsilon_n}(\{-1\})$, we get $\sigma^2 = 1$ and $X_n = S_n^2 - n$.

- Let $\theta \in \mathbb{R}$ and define

$$X_n := (M_\epsilon(\theta))^{-n} e^{\theta S_n}$$

where M_ϵ is the moment generating function of ϵ_n . Then

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n^\epsilon) = \mathbb{E}\left((M_\epsilon(\theta))^{-(n+1)} e^{\theta S_{n+1}} | \mathcal{F}_n^\epsilon\right) = (M_\epsilon(\theta))^{-n} e^{\theta S_n} M_\epsilon(\theta)^{-1} \mathbb{E}(e^{\theta \epsilon_{n+1}}) = X_n.$$

The case of a simple random walk with $\mathbb{P}_{\epsilon_n}(\{1\}) = p = 1 - q = 1 - \mathbb{P}_{\epsilon_n}(\{-1\})$, we get

$$M_\epsilon(\theta) = pe^\theta + qe^{-\theta} \text{ and } X_n = (pe^\theta + qe^{-\theta})^{-n} e^{\theta S_n}.$$

If we further assume that $p = q = 0.5$, then $M_\epsilon(\theta) = \cosh \theta$ and $X_n = (\cosh \theta)^{-n} e^{\theta S_n}$.

- Now assume that ϵ_n 's are discrete with $\mathbb{P}_{\epsilon_n}(\{1\}) = p = 1 - q = 1 - \mathbb{P}_{\epsilon_n}(\{-1\})$ and $p > q$. Define $X_n = \left(\frac{q}{p}\right)^{S_n}$ for $n \in \mathbb{N}$. Then

$$\mathbb{E}(X_{n+1} | \mathcal{F}_n^\epsilon) = \mathbb{E}\left(\left(\frac{q}{p}\right)^{S_{n+1}} | \mathcal{F}_n^\epsilon\right) = \left(\frac{q}{p}\right)^{S_n} \mathbb{E}\left(\left(\frac{q}{p}\right)^{\epsilon_{n+1}}\right) = X_n.$$

■

For intuition, it will be useful to imagine that you are playing a sequence of games and $X_n - X_{n-1}$ represents your winnings during the n th game if you play unit stakes. Your total winnings by time n is then equal to

$$\text{cumulative total gains} = (X_1 - X_0) + (X_2 - X_1) + \dots + (X_n - X_{n-1}) = X_n - X_0.$$

If X is a martingale, then

$$\mathbb{E}(X_n - X_{n-1} | \mathcal{F}_{n-1}) = 0,$$

implying that the game is fair. For a submartingale, the game is biased in your favour, while if X is a supermartingale, then the game is unfavourable to you.

Now suppose that you play a stake of φ_n on the n th game. Then your winnings on the n th game will be $\varphi_n(X_n - X_{n-1})$ and your total gains will be

$$\text{cumulative total gains} = \varphi_1(X_1 - X_0) + \dots + \varphi_n(X_n - X_{n-1}) = \sum_{k=1}^n \varphi_k(X_k - X_{k-1}).$$

Note that for φ to be a sensible gambling/trading strategy, we need to know the value of φ_n at time $n - 1$; i.e., φ_n must be $\mathcal{F}_{n-1}$ -measurable.

Definition 8.1.1. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space. A sequence of random variables $\varphi = \{\varphi_n : n = 1, 2, 3, \dots\}$ is called *previsible* or *predictable* if $\varphi_n \in m\mathcal{F}_{n-1}$ for each $n \in \mathbb{N}^+$.

Definition 8.1.2. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space, $X = \{X_n : n \in \mathbb{N}\}$ be a stochastic process and $\varphi = \{\varphi_n : n \in \mathbb{N}^+\}$ be a predictable sequence. We define the *martingale transform* or *discrete stochastic integral* of X by φ as the stochastic process $(\varphi \bullet X) = \{(\varphi \bullet X)_n : n \in \mathbb{N}\}$ defined by

$$(\varphi \bullet X)_0 = 0, \quad (\varphi \bullet X)_n = \sum_{k=1}^n \varphi_k(X_k - X_{k-1}), \quad n \geq 1.$$

Theorem 8.1.1. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered space, $\varphi = \{\varphi_n : n \in \mathbb{N}^+\}$ be a bounded ($|\varphi_n| \leq C < \infty$ a.a. for every $n \in \mathbb{N}^+$) predictable sequence and $X = \{X_n : n \in \mathbb{N}\}$ be a stochastic process.

1. If X is a martingale then $(\varphi \bullet X)$ is also a martingale.
2. If φ is non-negative and X is a sub (resp. super)-martingale, then $(\varphi \bullet X)$ is also a sub (resp. super)-martingale.

Proof. Since φ is predictable, it follows that

$$(\varphi \bullet X)_n = \sum_{k=1}^n \varphi_k (X_k - X_{k-1})$$

is $\mathcal{F}_n$ -measurable. For each $n \in \mathbb{N}^+$, we have

$$\mathbb{E}(|(\varphi \bullet X)_n|) \leq \mathbb{E}\left(\sum_{k=1}^n |\varphi_k| |X_k - X_{k-1}|\right) \leq C \sum_{k=1}^n \mathbb{E}(|X_k| + |X_{k-1}|) < \infty$$

since $X_k \in \mathcal{L}^1$ for every $k \in \mathbb{N}$. Note that

$$(\varphi \bullet X)_{n+1} = (\varphi \bullet X)_n + \varphi_{n+1}(X_{n+1} - X_n),$$

hence

$$\begin{aligned} \mathbb{E}((\varphi \bullet X)_{n+1} | \mathcal{F}_n) &= \mathbb{E}((\varphi \bullet X)_n | \mathcal{F}_n) + \mathbb{E}(\varphi_{n+1}(X_{n+1} - X_n) | \mathcal{F}_n) \\ &= (\varphi \bullet X)_n + \varphi_{n+1} \mathbb{E}(X_{n+1} - X_n | \mathcal{F}_n). \end{aligned}$$

The result follows. ■

The interpretation of this theorem is as follows. If you choose a gambling strategy φ , then provided that the strategy is bounded, it will not turn a fair game into an unfair one (or one that is more favourable towards you). Likewise, if the game is unfavourable towards you (i.e., X is a supermartingale), then it will remain unfavourable if you don't own the casino (i.e, if your stakes stay positive).

There are cases where $(\varphi \bullet X)$ is a martingale even when φ is not bounded. Here is an example.

Example 8.1.2 (The Martingale). The following trading strategy is the origin of the term ‘martingale’. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $X = (X_n)_{n=1}^\infty$ be a sequence of i.i.d random variables with $\mathbb{P}(\{X_n = 1\}) = \mathbb{P}(\{X_n = -1\}) = 0.5$. Define the random walk $S = \{S_n : n \in \mathbb{N}\}$ by

$$S_0 = 0, \quad S_n = \sum_{k=1}^n X_k, \quad n \geq 1.$$

Then S is a martingale with respect to $\mathbb{F}^S$. We think of S_n as the cumulative winnings (or losses) when a player invests unit stakes in a game with payoff given by X . Consider the following betting strategy:

$$\varphi_1 = 1, \quad \varphi_{n+1} = 2\varphi_n I_{\{X_n = -1\}}, \quad n \geq 1.$$

That is, you first bet R1, if you win, you quit playing ($\varphi_2 = 2\varphi_1 I_{\{X_1 = -1\}} = 0$ if $X_1 = 1$), but if you lose, then you double your next bet. If you win this time, you quit, but if you lose again, you double your bet in the next round, and so on until you win. The idea is that by the first time you win — due to the size of your bet then — you will recover all your previous losses and make a profit of R1. Here is a table that illustrates this:

Game	Stake	Cumulative Losses	Profit on winning
1	1	0	1
2	2	1	1
3	4	3	1
4	8	7	1

One can check that $(\varphi \bullet S)$ is a martingale (even though φ is not bounded). ■

Exercise 8.1.1. Show that the process in the above example is indeed a martingale, despite the fact that φ is clearly not uniformly bounded. ■

8.2 Doob Decomposition and Quadratic Variation

Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space and $X = \{X_n : n \in \mathbb{N}\}$ be an adapted stochastic process. In this section we show that if X is integrable, then we can decompose X uniquely as

$$X = X_0 + M + A,$$

where M is a martingale and A is a predictable process, both null at 0 (i.e., $M_0 = A_0 = 0$). We think of A as representing the true “signal” and M representing the “noise”, so we can write

$$X = X_0 + \text{signal} + \text{noise}.$$

We begin with a Lemma.

Lemma 8.2.1. *If M is a predictable martingale with $M_0 = 0$, then $M \equiv 0$ a.s.*

Proof. For each $n \in \mathbb{N}$,

$$M_n = \mathbb{E}(M_{n+1} | \mathcal{F}_n) = M_{n+1}.$$

Hence $M_n = M_0 = 0$ for all $n \in \mathbb{N}$. ■

Theorem 8.2.1 (Doob decomposition). *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered probability space and $X = \{X_n : n \in \mathbb{N}\}$ be a stochastic process such that*

1. X is adapted to $\mathbb{F}$;
2. $X_n \in \mathcal{L}^1(\Omega, \mathcal{F}, \mathbb{P}) \quad \forall n \in \mathbb{N}$.

Then there exist two unique stochastic processes M and A , both null at zero, such that

1. M is a martingale with respect to $\mathbb{F}$;
2. A is predictable; and
3. $X_n = X_0 + A_n + M_n$ for each $n \in \mathbb{N}$.

Furthermore, A is increasing (resp. decreasing) if and only if X is a submartingale (resp. supermartingale). We call (A, M) the Doob decomposition of X .

Proof. 1. We begin by showing uniqueness. Suppose there are two candidates for the Doob decomposition of X : (A, M) and (A', M') . Then for each $n \in \mathbb{N}$, we have

$$X_n = X_0 + M_n + A_n = X_0 + M'_n + A'_n \implies M_n - M'_n = A_n - A'_n.$$

This implies that $M_n - M'_n$ is a predictable martingale null at zero, which implies that $M_n - M'_n = 0 \implies M_n = M'_n$ and, consequently, $A_n = A'_n$ as well.

2. We now prove existence. With $A_0 = M_0 = 0$, we define (A, M) inductively as

$$A_{n+1} = A_n + \mathbb{E}(X_{n+1} - X_n | \mathcal{F}_n) \text{ and } M_{n+1} = M_n + X_{n+1} - \mathbb{E}(X_{n+1} | \mathcal{F}_n) \quad n \in \mathbb{N}.$$

Solving this gives

$$A_n = \sum_{k=1}^n \mathbb{E}(X_k - X_{k-1} | \mathcal{F}_{k-1}) \text{ and } M_n = \sum_{k=1}^n (X_k - \mathbb{E}(X_k | \mathcal{F}_{k-1})).$$

It is an exercise to check that both A and M satisfy the required conditions. ■

Remark 8.2.1. The definitions of A and M have the following interpretations: A_n is the cumulative expected jumps of X , while M_n is the correction term for the actual jumps. Thus, we can write X as

$$X = \text{Expected Jumps} + \text{Surprises}.$$

The process A is also called the *compensator* of X . ■

Remark 8.2.2. In continuous-time, such a decomposition is not always possible. However, P.A. Meyer showed that for a special class of submartingales (*submartingales of class D*), one can find a similar decomposition. The result is called *The Doob-Meyer Decomposition Theorem* and it is a cornerstone of *la théorie générale*. ■

Let us get our hands dirty.

Example 8.2.1. Consider a sequence of i.i.d. random variables $\{X_n : n = 1, 2, 3, \dots\}$ with $\mathbb{E}(X_n) = \mu$ and the random walk S_n be defined by

$$S_0 = 0, \quad S_{n+1} = S_n + X_{n+1}, \quad n \in \mathbb{N}.$$

Then it is easy to see that

$$A_n = n\mu \text{ and } M_n = S_n - n\mu$$

for every $n \in \mathbb{N}$. That is,

$$S_n = 0 + n\mu + (S_n - n\mu) \quad \forall n \in \mathbb{N},$$

where $n\mu$ is the compensator and $S_n - n\mu$ is the martingale part of S . ■

A martingale X is *square integrable* if $\mathbb{E}(X_n^2) < \infty$ for all $n \in \mathbb{N}$ (i.e., $X_n \in \mathcal{L}^2 \forall n \in \mathbb{N}$). Let X be a square integrable martingale null at 0. By Jensen's inequality, the process $X^2 = \{X_n^2 : n \in \mathbb{N}\}$ is a submartingale, and by the Doob decomposition theorem, there exists a unique *increasing* predictable process A_n such that $X^2 - A$ is a martingale. We call the process A the *predictable quadratic variation* of X and denote it by $\langle X \rangle$. That is, $\langle X \rangle$ is the unique increasing predictable process such that $X^2 - \langle X \rangle$ is a martingale.

Exercise 8.2.1. Show that

$$\langle X \rangle_n = \sum_{k=1}^n \mathbb{E}((X_k^2 - X_{k-1}^2) | \mathcal{F}_{k-1}) = \sum_{k=1}^n \mathbb{E}((X_k - X_{k-1})^2 | \mathcal{F}_{k-1}), \quad n \in \mathbb{N}^+.$$

Example 8.2.2. Consider a sequence of i.i.d. random variables $\{X_n : n = 1, 2, 3, \dots\}$ with $\mathbb{P}_{X_n}(\{1\}) = \mathbb{P}_{X_n}(\{-1\}) = 0.5$ and the random walk S_n be defined by

$$S_0 = 0, \quad S_{n+1} = S_n + X_{n+1}, \quad n \in \mathbb{N}.$$

Since we have shown that $S_n^2 - n$ is a martingale, it follows that $\langle S \rangle_n = n$ for every $n \in \mathbb{N}$. ■

Exercise 8.2.2 (Important). Throughout, we will assume that X and Y are square integrable martingales null at 0. Define the *predictable covariation* of X and Y as

$$\langle X, Y \rangle := \frac{1}{4} (\langle X + Y \rangle - \langle X - Y \rangle).$$

1. Use the Cauchy-Schwartz inequality to prove that the set of square integrable martingales is a vector space.
2. Show that $XY - \langle X, Y \rangle$ is a martingale.
Hint: For $x, y \in \mathbb{R}$, $xy = \frac{1}{4}((x+y)^2 - (x-y)^2)$.

3. Show that XY is a martingale if and only if $\langle X, Y \rangle = 0$.

■

Exercise 8.2.3 (Important). Show that if X is a square integrable martingale, then the non-overlapping increments of X are uncorrelated, i.e.

$$\mathbb{E}((X_n - X_m)(X_t - X_s)) = 0 \quad \text{for } m < n < s < t.$$

Hint: Total expectation. Condition on $\mathcal{F}_{\text{something}}$.

■

Exercise 8.2.4 (Important). Let X and Y be adapted integrable processes (also assume that $X_0 = Y_0 = 0$ for simplicity). We define the *quadratic covariation* of X and Y as

$$[X, Y]_n := \sum_{k=1}^n (X_k - X_{k-1})(Y_k - Y_{k-1}),$$

and the *quadratic variation* of X as

$$[X]_n := [X, X]_n = \sum_{k=1}^n (X_k - X_{k-1})^2.$$

1. Assume that X and Y are square integrable martingales both null at 0. Show that $X^2 - [X]$ is a martingale. Hence conclude that $[X] - \langle X \rangle$ is also a martingale.
2. Let φ be predictable. Show that

$$[\varphi \bullet X] = \varphi^2 \bullet [X].$$

3. (Product rule) For any process Z , define $\Delta Z_n := Z_n - Z_{n-1}$. Show that

$$\Delta(XY)_n = X_{n-1}\Delta Y_n + Y_{n-1}\Delta X_n + \Delta[X, Y]_n.$$

Hence, deduce the following “integration by parts formula”

$$XY = (X_- \bullet Y) + (Y_- \bullet X) + [X, Y],$$

where $X_{-0} = X_0$, $Y_{-0} = Y_0$, $X_{-n} = X_{n-1}$ and $Y_{-n} = Y_{n-1}$ for $n \geq 1$.

■

8.3 The Optional Stopping Theorem

We now look at the interaction between martingales and stopping times.

Theorem 8.3.1. *Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a filtered space, $X = \{X_n : n \in \mathbb{N}\}$ be a stochastic process and τ be a stopping time.*

1. *If X is a martingale, then the stopped process $X^\tau = \{X_n^\tau : n \in \mathbb{N}\}$ is also a martingale.*
2. *If X is a sub (resp. super)-martingale, then the stopped process $X^\tau = \{X_n^\tau : n \in \mathbb{N}\}$ is also a sub (resp. super)-martingale.*

Proof. Define $\varphi^\tau = \{\varphi_n^\tau : n \in \mathbb{N}^+\}$ by

$$\varphi_n^\tau(\omega) = \begin{cases} 1 & n \leq \tau(\omega) \\ 0 & n > \tau(\omega). \end{cases}$$

Then φ^τ is bounded, non-negative and previsible. On the other hand,

$$X_n^\tau - X_0^\tau = X_n^\tau - X_0 = (\varphi^\tau \bullet X)_n.$$

The result follows. ■

This is a remarkable result! It tells us that, if a game is fair, then no matter how we choose to quit, the game will remain fair. If X is a martingale, this implies that

$$\mathbb{E}(X_0) = \mathbb{E}(X_0^\tau) = \mathbb{E}(X_n^\tau) = \mathbb{E}(X_{\tau \wedge n}) \quad \text{for every } n \in \mathbb{N}.$$

From this equation, can we conclude that

$$\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$$

as well?

Example 8.3.1. Consider a simple random walk. Let $X = \{X_n : n \in \mathbb{N}\}$ be a sequence of i.i.d random variables with $\mathbb{P}(\{X_n = 1\}) = \mathbb{P}(\{X_n = -1\}) = 0.5$ for every $n \in \mathbb{N}^+$. Define $S_0 = 0$ and $S_{n+1} = S_n + X_{n+1}$ for $n \in \mathbb{N}$. Let $\tau : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ be defined as $\tau(\omega) := \inf \{n \in \mathbb{N} : S_n(\omega) = 1\}$. The stopped process S^τ is a martingale (with respect to $\mathbb{F}^S$), so

$$\mathbb{E}(S_{\tau \wedge n}) = \mathbb{E}(S_0) \quad \text{for every } n \in \mathbb{N},$$

but

$$\mathbb{E}(S_\tau) = 1 \neq 0 = \mathbb{E}(S_0).$$

So, in general, $\mathbb{E}(S_\tau) \neq \mathbb{E}(S_0)$. ■

We now seek to place conditions on τ and X that will allow us to conclude that $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$. Note that if $\tau(\omega) < \infty$, then

$$\lim_{n \rightarrow \infty} X_n^\tau(\omega) = \lim_{n \rightarrow \infty} X_{\tau(\omega) \wedge n}(\omega) = X_{\tau(\omega)}(\omega) = X_\tau(\omega).$$

Therefore, if $\tau < \infty$ almost surely, then the conclusion that $\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$ will hold as soon as we can justify the interchange between limits and expectation in the following:

$$\mathbb{E}(X_0) = \lim_{n \rightarrow \infty} \mathbb{E}(X_{\tau \wedge n}) = \mathbb{E}(\lim_{n \rightarrow \infty} X_{\tau \wedge n}) = \mathbb{E}(X_\tau). \quad (8.1)$$

Recall that the Dominated Convergence Theorem (DCT) allows us to do so, provided that the sequence $(X_{\tau \wedge n})$ is dominated by an integrable random variable $Y \geq 0$. That is, if there exists $Y \in m\mathcal{F}^+ \cap \mathcal{L}^1$ such that $|X_{\tau \wedge n}| \leq Y$, then (8.1) is permissible. When this is possible is provided by the *Optional Stopping Theorem*.

Theorem 8.3.2 (Doob's Optional Stopping Theorem). *Let $X = \{X_n : n \in \mathbb{N}\}$ be a martingale and τ be a stopping time. Then*

$$\mathbb{E}(X_\tau) = \mathbb{E}(X_0)$$

*if **any one** of the following conditions hold:*

1. *The stopping time is bounded a.s.; that is, there exists $N \in \mathbb{N}$ such that $\tau \leq N$ a.s..*

2. $\tau < \infty$ a.s., and $|X_n^\tau| \leq C$ for some constant $C > 0$.

3. $\mathbb{E}(\tau) < \infty$ and $|X_n^\tau - X_{n-1}^\tau| \leq C$ for some constant $C > 0$.

Proof. We will apply DCT by finding $Y \in m\mathcal{F}^+ \cap \mathcal{L}^1$ that dominates the stopped process.

1. Here we can choose

$$Y = \sum_{k=1}^N |X_k|.$$

2. Choose $Y = C$.

3. Note that the condition that $\mathbb{E}(\tau) < \infty$ also implies that $\mathbb{P}(\tau < \infty) = 1$. Since

$$X_{\tau \wedge n} = X_0 + \sum_{k=1}^{\tau \wedge n} (X_k^\tau - X_{k-1}^\tau),$$

we get

$$|X_{\tau \wedge n}| \leq |X_0| + \sum_{k=1}^{\tau \wedge n} |X_k^\tau - X_{k-1}^\tau| \leq |X_0| + C(\tau \wedge n) \leq |X_0| + C\tau.$$

Since $\mathbb{E}(|X_0| + C\tau) < \infty$, choose $Y = |X_0| + C\tau$. ■

Remark 8.3.1. The conditions placed above are not the most general. Indeed, the same conclusion holds if the stopped process is *uniformly integrable*:

$$\lim_{K \rightarrow \infty} \sup_n \int_{\{|X_n^\tau| \geq K\}} X_n^\tau d\mathbb{P} = 0.$$

For more general conditions, see Shiryaev's "Probability". ■

Example 8.3.2. Consider the symmetric random walk $S = \{S_n : n \in \mathbb{N}\}$ starting at 0 and $\mathbb{F} = \mathbb{F}^S$. Let $a, b \in \mathbb{Z}$ with $a < 0 < b$. Define $\tau = \inf \{n \in \mathbb{N} : S_n = a \text{ or } b\}$. Then τ is a stopping time with $\mathbb{E}(\tau) < \infty$ (therefore $\tau < \infty$ a.s.). Since $S_0 = 0$, we have that $|S_{\tau \wedge n}| \leq \max\{|a|, b\}$. Hence we can apply OST to get that

$$0 = \mathbb{E}(S_0) = \mathbb{E}(S_\tau).$$

We now calculate $p = \mathbb{P}(S_\tau = a)$. Since S_τ takes two possible values, we have

$$0 = \mathbb{E}(S_\tau) = pa + (1 - p)b,$$

which gives

$$p = \frac{b}{b - a}.$$
■

Example 8.3.3. In the previous example we find $\mathbb{E}(\tau)$. Define $W = \{W_n : n \in \mathbb{N}\}$ by $W_n = S_n^2 - n$. It can be shown that W is also a martingale (exercise). Now

$$|W_n^\tau - W_{n-1}^\tau| = \left| (S_n^\tau)^2 - \tau \wedge n - (S_{n-1}^\tau)^2 + \tau \wedge (n-1) \right| \leq (S_n^\tau)^2 + (S_{n-1}^\tau)^2 + 1 \leq 2 \max\{a^2, b^2\} + 1.$$

Since $\mathbb{E}(\tau) < \infty$, we can apply OST to get

$$0 = \mathbb{E}(W_0) = \mathbb{E}(W_\tau) = \mathbb{E}(S_\tau^2) - \mathbb{E}(\tau) = pa^2 + (1 - p)b^2 - \mathbb{E}(\tau)$$

which implies that

$$\mathbb{E}(\tau) = pa^2 + (1 - p)b^2 = -ab. \quad \blacksquare$$

Chapter 9

Brownian Motion and Stochastic Calculus

9.1 Brownian Motion

We begin with a definition.

Definition 9.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A stochastic process $W = \{W_t : t \geq 0\}$ is called a Brownian motion or standard Brownian motion if it satisfies the following:

1. $W_0 = 0$ $\mathbb{P}$ -a.s.
2. W has continuous sample paths.
3. W has independent increments; i.e., for each

$$0 \leq t_0 < t_1 \leq t_2 < t_3 \leq t_4 < t_5 \leq \dots \leq t_{n-1} < t_n,$$

the random variables

$$W_{t_1} - W_{t_0}, W_{t_3} - W_{t_2}, \dots, W_{t_n} - W_{t_{n-1}}$$

are independent.

4. W has normally distributed, stationary increments: $W_t - W_s \sim N(0, t - s)$ for $s < t$.

If $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ is a filtered probability space and $\mathcal{F}_t^W \subseteq \mathcal{F}_t$ for each $t \geq 0$, then we call $W = \{W_t : t \geq 0\}$ a Brownian motion with respect to $\mathbb{F}$ if it satisfies 1 - 4 and the stronger condition that

5. For $0 \leq s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$.

We state the following without proof.

Theorem 9.1.1 (Weiner). Brownian motion exists. That is, there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a stochastic process W such that W is a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$.

There are several proofs of this theorem. A popular one constructs a Brownian motion as a limit of a sequence of simple random walks as the step sizes get smaller and smaller.

Example 9.1.1. Let W be a Brownian motion. For $\alpha > 0$, define $X_t^\alpha := \frac{1}{\alpha} W_{\alpha^2 t}$ for $t \geq 0$. Then X^α is a Brownian motion. ■

Proposition 9.1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $W = \{W_t : t \geq 0\}$ be a Brownian motion. If $s, t \in (0, \infty)$ we have

1. $W_t \sim N(0, t)$
2. $\text{Cov}(W_t, W_s) = t \wedge s$
3. W_s and W_t have a Bivariate normal distribution.

Proof. 1. Note that $W_t = W_t - W_0$, so the result follows.

2. If $s < t$, then

$$\begin{aligned} \text{Cov}(W_s, W_t) &= \text{Cov}(W_s, W_s + W_t - W_s) = \text{Cov}(W_s, W_s) + \text{Cov}(W_s - W_0, W_t - W_s) \\ &= \text{Var}(W_s) + 0 = s. \end{aligned}$$

Similarly, $\text{Cov}(W_s, W_t) = t$ when $t < s$.

3. Notice that

$$\begin{pmatrix} W_t \\ W_s \end{pmatrix} = \begin{pmatrix} W_t - W_s + W_s - W_0 \\ W_s - W_0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} W_t - W_s \\ W_s - W_0 \end{pmatrix}.$$

Since $W_t - W_s$ and $W_s - W_0$ are independent normal random variables, they are jointly normal, which implies that W_t and W_s are also jointly normal. ■

Theorem 9.1.2. $W = \{W_t : t \geq 0\}$ and $M = \{M_t = W_t^2 - t : t \geq 0\}$ are martingales with respect to $\mathbb{F}^W$.

Proof. The first two properties are obvious. Let $s < t$ ($s = t$ is clear). Then

$$\mathbb{E}(W_t | \mathcal{F}_s^W) = \mathbb{E}(W_s + W_t - W_s | \mathcal{F}_s^W) = W_s + 0 = W_s.$$

Showing that M is a martingale is left as an exercise. ■

It turns out that there is a converse to this result.

Theorem 9.1.3 (Levy). *Let M be a continuous martingale such that $M_0 = 0$ and $\{M_t^2 - t : t \geq 0\}$ are martingales. Then M is a Brownian motion.*

Even though the paths of Brownian motion are continuous, it turns out that they are not differentiable.

9.2 Ito Integrals

In this section we will define the *Ito integral* of a stochastic process φ over the interval $[0, T]$ with respect to another stochastic process X . We will denote it by

$$(\varphi \bullet X)_T := \int_0^T \varphi_t dX_t,$$

and call φ the *integrand* and X the *integrator*. Throughout the section, we fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, an adapted process X , and a positive (extended) number $T \in (0, \infty]$. The stochastic integral will then be regarded as a stochastic process $(\varphi \bullet X)_t$, defined for each $0 \leq t \leq T$.

As the notation suggests, we want this integral to be the continuous-time analogue of the martingale transform from the previous chapter. That is, we want $(\varphi \bullet X)_t$ to represent the cumulative

gains when trading (in continuous time) the asset X with stakes equal to φ_t for each $t \in (0, T]$.

First, if the paths of X have finite variation, then we can define $(\varphi \bullet X)$ as a (path-wise) Stieltjes integral for appropriate processes φ . This is certainly the case, for instance, when $X_t(\omega) = t$ for every $t \in [0, T]$ and $\omega \in \Omega$. In this case, the integral will correspond to the Riemann (or more generally, Lebesgue) integral.

Definition 9.2.1. *If $X_t(\omega) = t$ for each $t \in [0, T]$, we define the integral of a stochastic process φ with respect to X as the path-wise Riemann integral*

$$(\varphi \bullet X)_t(\omega) := \int_0^t \varphi_s(\omega) ds, \quad \omega \in \Omega, \quad t \in [0, T],$$

provided the integral exists. In general, if the sample paths of X have finite variation, we define the integral of φ with respect to X as a path-wise Riemann-Stieltjes integral

$$(\varphi \bullet X)_t(\omega) := \int_0^t \varphi_s(\omega) dX_s(\omega), \quad \omega \in \Omega, \quad t \in [0, T],$$

provided the integral exists

With this definition, we can now talk about the area under the path of a Brownian motion W , as the integral

$$\int_0^T W_t(\omega) dt.$$

Unfortunately, though, many of the interesting stochastic processes we consider in practice have sample paths that have unbounded variation. A famous example is Brownian motion.

Theorem 9.2.1. *The sample paths of a Brownian motion have (a.s.) infinite variation on any interval $[0, t]$, where $t > 0$.*

This result implies that we cannot define

$$\int_0^t \varphi_s dW_s$$

as a (pathwise) Stieltjes integral. We will introduce a new type of integration — called *Ito Integration*— to deal with such problems. Let us first define this integral for *simple processes*.

Definition 9.2.2. *A stochastic process φ is called a simple process (or elementary process) if there exists a bounded partition $0 = t_0 < t_1 < t_2 < \dots < t_n = T$ of $[0, T]$ and bounded random variables $H_1, \dots, H_n$, with $H_i \in m\mathcal{F}_{t_{i-1}}$ for $i = 1, 2, \dots, n$, such that*

$$\varphi_t(\omega) = \sum_{i=1}^n H_i(\omega) I_{(t_{i-1}, t_i]}(t), \quad t \in [0, T], \quad \omega \in \Omega. \quad (9.1)$$

If φ is a simple process with representation (8.1), we will define the Ito integral of φ with respect to a Brownian motion W as the stochastic process $\{(\varphi \bullet W)_t : 0 \leq t \leq T\}$ defined by

$$(\varphi \bullet W)_t := \int_0^t \varphi_s dW_s := \sum_{i=1}^n H_i (W_{t_i \wedge t} - W_{t_{i-1} \wedge t}), \quad 0 \leq t \leq T.$$

We denote the class of simple processes by $\mathbb{S}$.

Theorem 9.2.2. *If $\varphi \in \mathbb{S}$ is a simple process, then*

1.

$$\mathbb{E} \left(\int_0^t \varphi_s dW_s \right) = 0 \quad \text{for all } 0 \leq t \leq T$$

2. (Ito isometry)

$$\mathbb{E} \left(\left(\int_0^t \varphi_s dW_s \right)^2 \right) = \mathbb{E} \left(\int_0^t \varphi_s^2 ds \right) = \int_0^t \mathbb{E}(\varphi_s^2) ds \quad \text{for all } 0 \leq t \leq T$$

3.

$$\left\{ \int_0^t \varphi_s dW_s : 0 \leq t \leq T \right\} \quad \text{is a martingale}$$

4. If φ is deterministic (i.e., $\varphi_t(\omega)$ does not depend on $\omega \in \Omega$), then

$$\int_0^t \varphi_s dW_s \sim N \left(0, \int_0^t \varphi_s^2 ds \right) \quad \text{for all } 0 \leq t \leq T.$$

Proof. Throughout, we will write

$$\varphi_t(\omega) = \sum_{i=1}^n H_i(\omega) I_{(t_{i-1}, t_i]}(t), \quad t \in [0, T], \quad \omega \in \Omega.$$

1. For $0 \leq t \leq T$, we have

$$\begin{aligned} \mathbb{E} \left(\int_0^t \varphi_s dW_s \right) &= \mathbb{E} \left(\sum_{i=1}^n H_i (W_{t_i \wedge t} - W_{t_{i-1} \wedge t}) \right) = \sum_{i=1}^n \mathbb{E} (H_i (W_{t_i \wedge t} - W_{t_{i-1} \wedge t})) \\ &= \sum_{i=1}^n \mathbb{E} (\mathbb{E} (H_i (W_{t_i \wedge t} - W_{t_{i-1} \wedge t}) | \mathcal{F}_{t_{i-1}})) = \sum_{i=1}^n \mathbb{E} (H_i \mathbb{E} ((W_{t_i \wedge t} - W_{t_{i-1} \wedge t}) | \mathcal{F}_{t_{i-1}})) = 0. \end{aligned}$$

2. We prove it for $t = T$. Note that

$$\left(\int_0^T \varphi_s dW_s \right)^2 = \sum_{j=1}^n \sum_{i=1}^n H_i H_j (W_{t_i} - W_{t_{i-1}}) (W_{t_j} - W_{t_{j-1}})$$

Since

$$\mathbb{E} (H_i H_j (W_{t_i} - W_{t_{i-1}}) (W_{t_j} - W_{t_{j-1}})) = 0$$

when $i \neq j$ (verify this), we get

$$\mathbb{E} \left(\left(\int_0^T \varphi_s dW_s \right)^2 \right) = \mathbb{E} \left(\sum_{i=1}^n H_i^2 (t_i - t_{i-1}) \right) = \sum_{i=1}^n \mathbb{E}(H_i^2)(t_i - t_{i-1}) = \int_0^T \mathbb{E}(\varphi_s^2) ds.$$

3. $(\varphi \bullet W)$ is clearly a martingale transform of the discrete martingale $\{W_{t_i} : i = 0, 1, 2, \dots, n\}$.4. If φ is deterministic, then $(\varphi \bullet W)_t$ is a linear combination of independent normal random variables (Brownian motion increments).

■

We now expand the class of integrands φ . For any stochastic process φ , we define

$$\|\varphi\|_W := \left(\mathbb{E} \left(\int_0^T \varphi_s^2 ds \right) \right)^{\frac{1}{2}}.$$

Let

$$\mathbb{L} := \{\varphi : \varphi \text{ is adapted and left continuous with right limits}\}$$

and

$$L^2(W) := \{\varphi : \|\varphi\|_W < \infty\}.$$

We will choose the space of integrands to be

$$\mathcal{I} := \mathbb{L} \cap L^2(W).$$

Notice that $\mathbb{S} \subseteq \mathcal{I}$ by construction. Also, if $\varphi \in \mathcal{I}$, then there exists a sequence $(\varphi^n) \subseteq \mathbb{S}$ such that

$$\|\varphi^n - \varphi\|_W \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Indeed, if $\varphi \in \mathcal{I}$ is also bounded, and we pick the partition $0 = t_0 < t_1 < \dots < t_n = T$ with mesh tending to zero, then

$$\varphi_t^n(\omega) := \sum_{i=1}^n \varphi_{t_{i-1}}(\omega) I_{(t_{i-1}, t_i]}(t), \quad n = 1, 2, \dots$$

is a sequence with the desired properties.

Definition 9.2.3. For any $\varphi \in \mathcal{I}$, we define the Ito integral of φ with respect to a Brownian motion W as the limit

$$\int_0^t \varphi_s dW_s := \lim_{n \rightarrow \infty} \int_0^t \varphi_s^n dW_s,$$

where (φ^n) is any sequence of simple process such that $\|\varphi - \varphi^n\|_W \rightarrow 0$ as $n \rightarrow \infty$.

Remark 9.2.1. The limit above is independent of the chosen sequence. ■

Remark 9.2.2. The Ito integral is linear. ■

The general Ito integral also has the same properties as the corresponding integral for elementary processes.

Theorem 9.2.3. If $\varphi \in \mathcal{I}$, then

1.

$$\mathbb{E} \left(\int_0^t \varphi_s dW_s \right) = 0 \quad \text{for all } 0 \leq t \leq T$$

2. (Ito isometry)

$$\mathbb{E} \left(\left(\int_0^t \varphi_s dW_s \right)^2 \right) = \mathbb{E} \left(\int_0^t \varphi_s^2 ds \right) = \int_0^t \mathbb{E}(\varphi_s^2) ds \quad \text{for all } 0 \leq t \leq T$$

3.

$$\left\{ \int_0^t \varphi_s dW_s : 0 \leq t \leq T \right\} \quad \text{is a martingale}$$

4. If φ is deterministic (i.e., $\varphi_t(\omega)$ does not depend on $\omega \in \Omega$), then

$$\int_0^t \varphi_s dW_s \sim N\left(0, \int_0^t \varphi_s^2 ds\right) \quad \text{for all } 0 \leq t \leq T.$$

Example 9.2.1. 1. The random variable

$$\int_0^1 t^2 dW_t$$

has a normal distribution with mean 0 and variance $\int_0^1 t^4 dt = 1/5$.

2. For $0 \leq t \leq T < \infty$, the process

$$I_t := \int_0^t W_s^2 dW_s$$

is a martingale. Furthermore,

$$\mathbb{E}(I_t) = 0 \text{ and } \mathbb{E}(I_t^2) = \mathbb{E}\left(\int_0^t W_s^4 ds\right) = \int_0^t \mathbb{E}(W_s^4) ds = \int_0^t 3s^2 ds = t^3.$$

9.3 Ito Diffusion Processes

An adapted stochastic process $X = \{X_t : 0 \leq t \leq T\}$ is an *Ito process* if there exists stochastic process $\mu = \{\mu_t : 0 \leq t \leq T\}$ and $\sigma = \{\sigma_t : 0 \leq t \leq T\}$ such that

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s \quad \text{for all } 0 \leq t \leq T, \quad (9.2)$$

where

$$\sigma \in \mathcal{I} \text{ and } \int_0^t |\mu_s| ds < \infty \quad \text{for all } 0 \leq t \leq T.$$

We often write (8.2) in differential notation as

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

and call it the *stochastic differential* of X . The process μ is called the *drift* of X and σ is called the *diffusion* coefficient of X . Note that if $\sigma \in \mathcal{I}$, then X is a martingale if and only if $\mu \equiv 0$; i.e, if and only if X is driftless.

Example 9.3.1. 1. A Brownian motion is an Ito process since

$$W_t = W_0 + \int_0^t 1 dW_s.$$

Thus, $dW_t = 0dt + 1dW_t$.

2. The process

$$X_t = t^3 + \int_0^t W_s dW_s = 0 + \int_0^t 3s^2 ds + \int_0^t W_s dW_s$$

is an Ito process. We have

$$dX_t = 3t^2 dt + W_t dW_t.$$

3. An Ito process can also be given implicitly. Define

$$X_t = X_0 + \int_0^t \mu X_s ds + \int_0^t \sigma X_s dW_s, \quad \mu \in \mathbb{R}, \quad \sigma \in (0, \infty).$$

Then X is an Ito process, called geometric Brownian motion. Its stochastic differential is

$$dX_t = \mu X_t dt + \sigma X_t dW_t.$$

We will later see how to ‘solve’ this stochastic differential equation to get X explicitly.

If X is an Ito process with differential

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

we will define the integral of a stochastic process φ with respect to X as

$$\int_0^t \varphi_s dX_s := \int_0^t \varphi_s \mu_s ds + \int_0^t \varphi_s \sigma_s dW_s,$$

provided the integrals on the right hand side exist.

One of the most important results in stochastic calculus is the following.

Theorem 9.3.1 (Ito’s Lemma). *Let $f : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ be a function of two variables $f = f(t, x)$ such that the partial derivatives*

$$\frac{\partial f}{\partial t}, \quad \frac{\partial f}{\partial x} \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2}$$

all exist and are continuous. Suppose that X is an Ito process with

$$dX_t = \mu_t dt + \sigma_t dW_t.$$

Then $Y_t = f(t, X_t)$ is also an Ito process, and

$$dY_t = df(t, X_t) = \left[\frac{\partial f}{\partial t} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} + \mu_t \frac{\partial f}{\partial x} \right] dt + \sigma_t \frac{\partial f}{\partial x} dW_t.$$

In integral notation, we have

$$Y_t = Y_0 + \int_0^t \left[\frac{\partial f}{\partial s} + \frac{1}{2} \sigma_s^2 \frac{\partial^2 f}{\partial x^2} + \mu_s \frac{\partial f}{\partial x} \right] ds + \int_0^t \sigma_s \frac{\partial f}{\partial x} dW_s.$$

Remark 9.3.1. An easier way to remember this formula is as follows. Let us introduce the following algebra of differentials:

$$(dt)^2 = 0, \quad dt dW_t = 0 \quad \text{and} \quad (dW_t)^2 = dt.$$

Then Ito’s lemma can be written as

$$dY_t = df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2.$$

Example 9.3.2. Let

$$dX_t = \mu dt + \sigma dW_t, \quad \mu \in \mathbb{R}, \quad \sigma \in (0, \infty).$$

We find the stochastic differential equation of $Y_t = 2t^2 e^{2X_t}$. Note that $f(t, x) = 2t^2 e^{2x}$ satisfies all the hypotheses of Ito’s lemma with

$$\frac{\partial f}{\partial t} = 4te^{2x}, \quad \frac{\partial f}{\partial x} = 4t^2 e^{2x} \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2} = 8t^2 e^{2x}.$$

Thus,

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2 = 4te^{2X_t} dt + 4t^2 e^{2X_t} dX_t + \frac{1}{2} 8t^2 e^{2X_t} (dX_t)^2 \\ &= (4te^{2X_t} + 4t^2 e^{2X_t} \mu + 4t^2 e^{2X_t} \sigma^2) dt + 4t^2 e^{2X_t} \sigma dW_t. \end{aligned}$$

Definition 9.3.1. An Ito diffusion process X is a solution to the following Stochastic Differential Equation (SDE)

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$$

for some real-valued functions μ and σ . That is, X is given implicitly by

$$X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s, \quad t \geq 0.$$

Example 9.3.3 (Geometric Brownian Motion). Consider the following SDE

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad X_0 = \text{given},$$

where μ and σ are constants, with $\sigma > 0$. We want to solve this equation by finding X explicitly in terms of Brownian motion W . We apply Ito's lemma to $Y_t = \ln X_t$ to get

$$dY_t = d \ln X_t = 0 dt + \frac{1}{X_t} dX_t - \frac{1}{2X_t^2} (dX_t)^2 = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t.$$

Hence

$$Y_t = Y_0 + \int_0^t \left(\mu - \frac{1}{2}\sigma^2 \right) ds + \int_0^t \sigma dW_s = Y_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t.$$

Therefore,

$$X_t = e^{Y_t} = \exp \left(\ln X_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \right) = X_0 \exp \left(\left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \right).$$

Notice that since

$$\ln X_t = Y_t = \ln X_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \sim N \left(\ln X_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t, \sigma^2 t \right),$$

X_t has a log-normal distribution. As an exercise, find the mean and variance of X_t .

Example 9.3.4 (Ornstein-Uhlenbeck process). Assume that X_0 is fixed and known, and let

$$dX_t = \alpha(\mu - X_t) dt + \sigma dW_t, \quad \alpha, \sigma > 0, \quad \mu \in \mathbb{R}.$$

We apply Ito's lemma to $Y_t = e^{\alpha t} X_t$:

$$dY_t = d(e^{\alpha t} X_t) = \alpha e^{\alpha t} X_t dt + e^{\alpha t} dX_t + 0 = \alpha \mu e^{\alpha t} dt + \sigma e^{\alpha t} dW_t,$$

which gives

$$Y_t = Y_0 + \int_0^t \alpha \mu e^{\alpha s} ds + \int_0^t \sigma e^{\alpha s} dW_s = Y_0 + \mu(e^{\alpha t} - 1) + \int_0^t \sigma e^{\alpha s} dW_s.$$

Therefore (note that $Y_0 = X_0$),

$$X_t = e^{-\alpha t} Y_t = X_0 e^{-\alpha t} + \mu(1 - e^{-\alpha t}) + e^{-\alpha t} \int_0^t \sigma e^{\alpha s} dW_s.$$

So, X_t has a normal distribution with

$$\mathbb{E}(X_t) = X_0 e^{-\alpha t} + \mu(1 - e^{-\alpha t})$$

and

$$\text{Var}(X_t) = \text{Var} \left(e^{-\alpha t} \int_0^t \sigma e^{\alpha s} dW_s \right) = e^{-2\alpha t} \int_0^t \sigma^2 e^{2\alpha s} ds = \frac{\sigma^2}{2\alpha} (1 - e^{-2\alpha t}).$$

Part III

Appendices

Appendix A

Naive Set Theory

Here we recall a few definitions and results from naive set theory.

A.1 Basics

Definition A.1.1. *A set is a collection of distinct objects.*

There are two methods we will use to describe a set:

1. A (complete or incomplete) list of the objects that belong to the set — e.g., $A = \{1, 2, 3\}$ or $B = \{5, 10, 15, 20, \dots\}$.
2. A condition which objects must satisfy in order to belong to the set — e.g.,

$$A = \{x : x \text{ is a colour}\} = \{\text{red, yellow}, \dots\}.$$

This representation will generally take the form:

$$A = \{x : \phi(x)\},$$

where $\phi(x)$ is a condition depending on the dummy variable x , which evaluates to “true” if x satisfies the condition and “false” otherwise. The set A then contains all those x ’s for which $\phi(x)$ evaluates to “true”.

We say that x is an *element* of a set A (written as $x \in A$) if x belongs to A . If A and B are sets, we say that A is a **subset** of B (written as $A \subseteq B$) if every element of A is also an element of B , i.e., $x \in A \implies x \in B$. We say that A and B are *equal* (written as $A = B$) if $A \subseteq B$ and $B \subseteq A$, i.e., if $x \in A \iff x \in B$. We will also assume that there exists a set S (or Ω), called the **universal set**, of all elements under consideration. If A and B are subsets of the universal set S , then we define

1. the *intersection* of A and B as

$$A \cap B := \{x \in S : x \in A \text{ and } x \in B\};$$

2. the *union* of A and B as

$$A \cup B := \{x \in S : x \in A \text{ or } x \in B\};$$

3. the *complement* of A in S as

$$A^c := \{x \in S : x \notin A\}.$$

There exists a set $\emptyset$ — called the *empty set* — that contains no elements. It is vacuously true that for any set A , $\emptyset \subseteq A$. We denote by $\mathcal{P}(A)$ or 2^A the set of all subsets of a set A , called the *power set* of A . If $A \cap B = \emptyset$, we say that A and B are *disjoint*. We also write $B \setminus A := \{x \in B : x \notin A\}$ for the *difference* between A and B . Here are some examples of commonly used sets:

- $\mathbb{N} := \{0, 1, 2, \dots\}$ — natural numbers
- $\mathbb{N}^+ := \{1, 2, 3, \dots\}$ — positive integers
- $\mathbb{Z} := \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ — integers
- $\mathbb{Q} := \left\{\frac{p}{q} : p, q \in \mathbb{Z} \text{ and } q \neq 0\right\} = \{\dots, -4, -\frac{5}{9}, 0, \frac{83}{5}, 120, \dots\}$ — rational numbers
- $\mathbb{R} := \{\frac{2}{3}, -20, \pi, e, \dots\}$ — real numbers.
- $\bar{\mathbb{R}} := \mathbb{R} \cup \{\infty, -\infty\}$ — extended real numbers.
- $\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}$, $i^2 = -1$ — complex numbers.

For every $A \subseteq \bar{\mathbb{R}}$, we define A^+ to be the set of all elements of A that are positive, i.e., $A^+ := \{x \in A : x > 0\}$. For instance, $\mathbb{Z}^+ := \{x \in \mathbb{Z} : x > 0\} = \mathbb{N}^+$.

Definition A.1.2. If I is a set, we call the collection of sets (a set of sets) $\mathcal{A} := \{A_\alpha : \alpha \in I\}$ a family of sets indexed by I . The set I is called the index set.

With this notation, we can write the collection $\{A_1, A_2, \dots, A_n\}$ as $\{A_\alpha : \alpha \in I\}$, where $I = \{1, 2, 3, \dots, n\}$. We can also use this notation to generalize the definitions of intersection and union to arbitrary families of sets as follows:

$$\bigcap_{\alpha \in I} A_\alpha = \cap \{A_\alpha : \alpha \in I\} := \{x \in S : x \in A_\alpha \text{ for every } \alpha \in I\}$$

and

$$\begin{aligned} \bigcup_{\alpha \in I} A_\alpha &= \cup \{A_\alpha : \alpha \in I\} := \{x \in S : x \in A_\alpha \text{ for some } \alpha \in I\} \\ &= \{x \in S : \exists \alpha \in I \text{ such that } x \in A_\alpha\}. \end{aligned}$$

If $I = \mathbb{N}^+$, we will write

$$\bigcup_{\alpha \in I} A_\alpha = \bigcup_{i=1}^{\infty} A_i.$$

Similar notation applies to intersections.

Exercise A.1.1. Let $\{A_\alpha : \alpha \in I\}$ be a collection of sets and A be a set. Show the following:

1.

$$\left(\bigcup_{\alpha \in I} A_\alpha \right)^c = \bigcap_{\alpha \in I} A_\alpha^c$$

2.

$$\left(\bigcap_{\alpha \in I} A_\alpha \right)^c = \bigcup_{\alpha \in I} A_\alpha^c$$

3.

$$A \cap \bigcup_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} (A \cap A_\alpha)$$

4.

$$A \cup \bigcap_{\alpha \in I} A_\alpha = \bigcap_{\alpha \in I} (A \cup A_\alpha)$$

■

Definition A.1.3. A collection of sets $\{A_\alpha : \alpha \in I\}$ is said to be pairwise disjoint if $A_i \cap A_j = \emptyset$ for $i, j \in I$ with $i \neq j$.

Note that if $\bigcap_{\alpha \in I} A_\alpha = \emptyset$, it does not follow that the collection $\{A_\alpha : \alpha \in I\}$ is pairwise disjoint.

We can extend the arithmetic operations and order relation of $\mathbb{R}$ to the extended real numbers $\bar{\mathbb{R}}$ as follows. If $a \in \mathbb{R}$, then

$$a \pm \infty = \pm\infty \pm a = \pm\infty, \quad \infty + \infty = \infty, \quad -\infty - \infty = -\infty \text{ and } \frac{a}{\pm\infty} = 0.$$

We also have $a \times \pm\infty = \pm\infty$ if $a > 0$ or $a = \infty$ and $a \times \pm\infty = \mp\infty$ if $a < 0$ or $a = -\infty$. We will make the convention that $0 \times \infty = 0$. We do not define $\infty - \infty$. Finally, $-\infty < a < \infty$ for every $a \in \mathbb{R}$. We can then define the following intervals in $\bar{\mathbb{R}}$ ($a, b \in \bar{\mathbb{R}}$):

$$(a, b) := \{x \in \bar{\mathbb{R}} : a < x < b\}, \quad [a, b] := \{x \in \bar{\mathbb{R}} : a \leq x \leq b\}, \quad (a, b] := \{x \in \bar{\mathbb{R}} : a < x \leq b\}$$

and

$$[a, b) := \{x \in \bar{\mathbb{R}} : a \leq x < b\}.$$

Remark A.1.1. Unfortunately, the usual cancellation in $\mathbb{R}$ do not extend to $\bar{\mathbb{R}}$. That is, if $x, y, z \in \bar{\mathbb{R}}$ are such that $x + z = y + z$, it does not follow that $x = y$ if $z = \pm\infty$. A similar statement applies to multiplication and division.

Exercise A.1.2. Show the following

1.

$$\bigcup_{n=1}^{\infty} \left(0, 2 + \frac{1}{n}\right] = (0, 3]$$

2.

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, 1 + \frac{1}{n}\right] = [0, 1]$$

3.

$$\bigcup_{n=1}^{\infty} \left(\frac{(-1)^n}{n^3}, n^2\right] = (-1, \infty).$$

■

Definition A.1.4. Let A and B be sets. The Cartesian product of A and B is the set

$$A \times B := \{(x, y) : x \in A \text{ and } y \in B\}.$$

In general, if $\{A_1, \dots, A_n\}$ is a family of sets, we define their Cartesian product as

$$\prod_{i=1}^n A_i := A_1 \times \dots \times A_n := \{(x_1, \dots, x_n) : x_1 \in A_1, \dots, x_n \in A_n\}$$

Example A.1.1. Let $A = \{1, 2\}$, $B = \{a, b, c\}$. Then

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

■

A.2 Relations and Functions

A **relation** R between two sets A and B is a subset of $A \times B$. A relation associates some elements of A to some elements of B . If R is a relation between A and B , we say that $x \in A$ is *related* to $y \in B$ (or x and y are related) if the pair (x, y) belongs to R ; i.e., if $(x, y) \in R$. If $A = B$, we call R a relation on A (or B). We will sometimes write $x \sim y$ to say that x is related to y (i.e., $(x, y) \in R$).

Example A.2.1. If we take $A = \{1, 2\}$, $B = \{a, b, c\}$ and $R = \{(1, a), (2, b)\}$, then $1 \sim a$, while 2 is not related to c , for instance. ■

Definition A.2.1. Let A be a set and R be a relation on A (i.e. $R \subseteq A \times A =: A^2$). We say that R is

1. *reflexive* $\iff$ for every $x \in A$, x is related to itself (i.e. $x \sim x$ for every $x \in A$)
2. *symmetric* $\iff \forall x, y \in A, [x \sim y \implies y \sim x]$
3. *transitive* $\iff \forall x, y, z \in A, [x \sim y \text{ and } y \sim z \implies x \sim z]$
4. *an equivalence relation* if R is reflexive, symmetric and transitive.

Example A.2.2. Consider the relation R on the set of integers $\mathbb{Z}$ defined by

$$x \sim y \iff x - y \text{ is even.}$$

We show that R is an equivalence relation.

1. Reflexivity follows from the fact that for every $x \in \mathbb{R}$, $x - x = 0$, which is an even number, so x is related to itself.
2. For symmetry, assume that $x - y$ is even (i.e., x is related to y), then $y - x = -(x - y)$ is also even (multiplying an even number by an integer, in this case -1). So R is symmetric.
3. Finally, if $x - y$ is even and $y - z$ is even, then $x - z = (x - y) + (y - z)$ is even (sum of even numbers). Thus R is transitive.

Therefore R is an equivalence relation. ■

Definition A.2.2. Let R be an equivalence relation on a set A and $x \in A$. The *equivalence class* of x , denoted by $[x]$, is defined as the set of all elements of A that are related to x ; i.e.,

$$[x] := \{y \in A : x \sim y\} = \{y \in A : y \sim x\}.$$

In the previous example for instance, the equivalence class of 3 is all odd integers.

Lemma A.2.1. Let R be an equivalence relation on a set A and $x, y \in A$. Then $x \sim y \iff [x] = [y]$.

Proof. Assume that $x \sim y$ and let $a \in [x]$. Then $y \sim x$ and $x \sim a \implies y \sim a$. Hence $a \in [y]$. Similarly one shows that $a \in [y] \implies a \in [x]$. Thus $[x] = [y]$.

Now assume that $[x] = [y]$. By reflexivity, $x \in [x] = [y]$, hence x is also related to y . ■

Proposition A.2.1. Let R be an equivalence relation on a set A . Then the equivalence classes of R are non-empty, disjoint, and their union is equal to A . That is, the equivalence classes of R form a partition of A .

Proof. Since R is reflexive, for any $x \in A$, $x \in [x] \neq \emptyset$.

Now let $x, y \in A$. We show that exactly one of the following two alternatives is true:

1. $[x] = [y]$; or
2. $[x] \cap [y] = \emptyset$.

If $x \sim y$, then (1) holds by the Lemma above. If $x \not\sim y$, then (1) does not hold by the Lemma above, and we now show that $[x] \cap [y] = \emptyset$ in this case. Suppose there exists $z \in [x] \cap [y]$. Then $x \sim z$ and $z \sim y$. Hence by transitivity, $x \sim y$, which is a contradiction. Thus $[x] \cap [y] = \emptyset$. So, the equivalence classes are pairwise disjoint.

We finally show that

$$\bigcup_{x \in A} [x] = A.$$

First, it is clear that $\bigcup_{x \in A} [x] \subseteq A$. So let $y \in A$. By reflexivity, $y \in [y] \subseteq \bigcup_{x \in A} [x]$. ■

Example A.2.3. Let $A = [0, 2\pi)$ and define the relation R on A as

$$\text{for every } x, y \in A, \quad x \sim y \iff \sin x = \sin y.$$

Then R is an equivalence relation.

1. Clearly, $\sin x = \sin x$ for every $x \in A$, hence $x \sim x$, implying that R is reflexive.
2. If $x \sim y$, then $\sin x = \sin y \implies \sin y = \sin x \implies y \sim x \implies R$ is symmetric.
3. If $x \sim y$ and $y \sim z$, then $\sin x = \sin y = \sin z \implies x \sim z \implies R$ is transitive.

Now we find the equivalence classes.

$$\begin{aligned} [0] &= \{x \in [0, 2\pi) : \sin x = \sin 0 = 0\} = \{0, \pi\} \\ [0.5] &= \{x \in [0, 2\pi) : \sin x = \sin 0.5\} = \{0.5, \pi - 0.5\} \\ [4] &= \{x \in [0, 2\pi) : \sin x = \sin 4\} = \{4, 3\pi - 4\}. \end{aligned}$$

In general,

$$[x] = \begin{cases} \{x, \pi - x\} & \text{if } 0 \leq x \leq \pi \\ \{x, 3\pi - x\} & \text{if } \pi < x < 2\pi. \end{cases}$$

Convince yourself that these equivalence classes form a partition of $[0, 2\pi)$. What we have done is to identify elements of $[0, 2\pi)$ that have the same sine as “equal” (or “the same”). ■

Functions are special kinds of relations between sets. A *function* f from a set X to a set Y (written as $f : X \rightarrow Y$) is a relation between X and Y that satisfies the following additional condition:

for every $x \in X$ there exists a unique $y \in Y$ such that x and y are related

$$\text{i.e., } \forall x \in X \exists! y \in Y \text{ such that } (x, y) \in f. \quad (\text{A.1})$$

We write $y = f(x)$ if (A.1) holds and refer to X and Y (respectively) as the *domain* and *codomain* of f . In simple terms, a function is a relation that passes the “vertical line test”—it is *left total* and *right unique*. The following definitions are very important for the course.

Definition A.2.3. Let X and Y be sets and $f : X \rightarrow Y$ be a function between them. For $A \subseteq X$ and $B \subseteq Y$, we define

1. The direct image of A under f as

$$f(A) := \{f(x) : x \in A\} \subseteq Y.$$

2. The pre-image of B under f as

$$f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X.$$

$f(X)$ is called the range of f .

Example A.2.4. Take $X = \{1, 2, 3\}$, $Y = \{a, b, c\}$, $f : X \rightarrow Y$ be defined by $f(1) = a$, $f(2) = f(3) = b$, $A = \{1, 2\}$ and $B = \{b, c\}$. Then

$$f(A) = \{f(1), f(2)\} = \{a, b\} \text{ and } f^{-1}(B) = \{x \in X : f(x) \in \{b, c\}\} = \{2, 3\}.$$

The range of f is $f(X) = \{a, b\}$. ■

Note that in general, the range is not equal to the codomain.

Example A.2.5. Take $X = Y = \mathbb{R}$ and $f : X \rightarrow Y$ be defined by $f(x) = x^2$ for every $x \in \mathbb{R}$. Then $f((-5, 5)) = \{f(x) : x \in (-5, 5)\} = \{x^2 : x \in (-5, 5)\} = [0, 25)$, while

$$f^{-1}((-10, 20)) = \{x \in \mathbb{R} : f(x) \in (-10, 20)\} = \{x \in \mathbb{R} : x^2 \in (-10, 20)\} = (-\sqrt{20}, \sqrt{20}).$$

The range of f is $f(\mathbb{R}) = [0, \infty)$. ■

The most important result in this section is the following:

Proposition A.2.2. Let X and Y be sets and $f : X \rightarrow Y$ be a function. If $\{A_\alpha : \alpha \in I\}$ is a collection of subsets of X and $\{B_\alpha : \alpha \in J\}$ is a collection of subsets of Y , then

1.

$$f\left(\bigcup_{\alpha \in I} A_\alpha\right) = \bigcup_{\alpha \in I} f(A_\alpha)$$

2.

$$f\left(\bigcap_{\alpha \in I} A_\alpha\right) \subseteq \bigcap_{\alpha \in I} f(A_\alpha)$$

3.

$$f^{-1}\left(\bigcup_{\alpha \in J} B_\alpha\right) = \bigcup_{\alpha \in J} f^{-1}(B_\alpha)$$

4.

$$f^{-1}\left(\bigcap_{\alpha \in J} B_\alpha\right) = \bigcap_{\alpha \in J} f^{-1}(B_\alpha)$$

5.

$$f^{-1}(B_{\alpha_0}^c) = (f^{-1}(B_{\alpha_0}))^c, \quad \alpha_0 \in J.$$

Proof. We only prove 3 and leave the rest as an exercise. We need to show that

$$f^{-1}\left(\bigcup_{\alpha \in J} B_\alpha\right) = \bigcup_{\alpha \in J} f^{-1}(B_\alpha).$$

To this end, let $x \in f^{-1}(\bigcup_{\alpha \in J} B_\alpha)$. This means that $f(x) \in \bigcup_{\alpha \in J} B_\alpha \iff \exists \alpha \in J$ such that $f(x) \in B_\alpha \iff \exists \alpha \in J$ such that $x \in f^{-1}(B_\alpha) \iff x \in \bigcup_{\alpha \in J} f^{-1}(B_\alpha)$. ■

Definition A.2.4. Let X and Y be sets and $f : X \rightarrow Y$ be a function between them. We say that f is

1. **One-to-one** or *injective* $\iff \forall x_1, x_2 \in X [f(x_1) = f(x_2) \implies x_1 = x_2]$ (or $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$)
2. **Onto** or *surjective* $\iff$ for every $y \in Y$, there exists $x \in X$ such that $y = f(x)$. That is, if $f(X) = Y$ (range = codomain).
3. **Bijective** $\iff f$ is both injective and surjective.

If $f : X \rightarrow Y$ is bijective, then there exists a unique bijection $f^{-1} : Y \rightarrow X$ such that $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$ for every $x \in X$ and every $y \in Y$. The function f^{-1} is called the *inverse* of f , and in this case, f is said to be *invertible*.

Example A.2.6. Define $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow [0, \infty)$ and $h : [0, \infty) \rightarrow [0, \infty)$ by $f(x) = g(x) = h(x) = x^2$ for every x . Convince yourself that f is neither injective nor surjective, g is surjective but not injective and h is bijective. The inverse of h is the function $h^{-1} : [0, \infty) \rightarrow [0, \infty)$ defined by $h^{-1}(x) = \sqrt{x}$ for every $x \in [0, \infty)$. ■

We end this section with an axiom that has had enormous influence in many branches of mathematics, and specifically to measure theory. Its statement makes it seem innocent, yet its consequences have led many mathematicians to question its validity.

Axiom of Choice: Let $\{A_\alpha : \alpha \in I\}$ be a non-empty collection of sets. There exists a function $f : I \rightarrow \bigcup_{\alpha \in I} A_\alpha$ such that $f(\alpha) \in A_\alpha$ for every $\alpha \in I$.

The function f advertised above is called a *choice function*. In simple terms, the axiom of choice says that for any non-empty collection of sets, one can pick an element from each set.

A.3 Cardinality

Here we discuss what it means for two sets to have the same ‘number of elements’.

Definition A.3.1. Let X and Y be sets. We say that X and Y have the same cardinality (or X and Y are *bijective*) if there exists a bijection $f : X \rightarrow Y$. We write $|X| = |Y|$ in this case.

Unfortunately, this definition does not tell us what the cardinality of a set is — it only lets us compare sets and decide which ones are ‘equal’. One can define an equivalence relation (on the class of all sets — see references), where two sets X and Y are related if and only if $|X| = |Y|$. From this relation, we can then define the cardinality of a set X as its equivalence class under this relation. We will not pursue this any further. It will be useful, however, to think of $|X|$ as “the number of elements of X ” when X is finite (see below).

For each $n \in \mathbb{N}^+$ define the set (n) as $(n) := \{1, 2, 3, \dots, n\}$. A set X is *finite* if $X = \emptyset$ or $|X| = |(n)|$ for some $n \in \mathbb{N}^+$. Otherwise, X is *infinite*.

Theorem A.3.1. Let X and Y be finite sets with the same cardinality, and $f : X \rightarrow Y$ be a function between them. Then the following are equivalent:

1. f is injective
2. f is surjective
3. f is bijective

Proof. Exercise. ■

Thus, for finite sets, having the same cardinality means all injective functions are necessarily surjective, and hence bijective. Now let us see what goes wrong when dealing with infinite sets.

Example A.3.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \arctan(x)$. Then f is injective but not surjective. On the other hand, $g : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ defined by $g(x) = f(x) = \arctan(x)$ is bijective (check this!). This means that $\mathbb{R}$ and $(-\frac{\pi}{2}, \frac{\pi}{2})$ have the same cardinality!

The previous example seems a bit surprising: how can a set have the same cardinality as one of its proper subsets¹? We give a name to such weirdos.

Definition A.3.2. A set is *Dedekind-infinite* if it is bijective to one of its proper subsets.

Using the Axiom of Choice (or some weaker version of it), it can be shown that a set is Dedekind-infinite if and only if it is infinite.

Example A.3.2. Let $E := \{0, 2, 4, 6, \dots\} = \{\text{non-negative even integers}\}$ and $O := \{1, 3, 5, 7, \dots\} = \{\text{positive odd integers}\}$. It is clear that $f : \mathbb{N} \rightarrow E$ and $g : \mathbb{N} \rightarrow O$ defined by $f(n) = 2n$ and $g(n) = 2n - 1$ for every $n \in \mathbb{N}$ are bijections. Thus, $|\mathbb{N}| = |E| = |O|$.

Definition A.3.3. A set X is *countably infinite* if $|X| = |\mathbb{N}|$; X is *countable* if it is finite or countably infinite.

The previous example shows that the non-negative even numbers and the non-negative odd numbers are both countably infinite, and hence countable. The cardinality of the set of natural numbers (and that of all countably infinite sets) is denoted by $\aleph_0$ (*aleph null*). We use countable sets a lot in this course, so the following results are very important.

Proposition A.3.1. The set of all integers $\mathbb{Z}$ is countable (countably infinite).

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even} \\ -\frac{(n+1)}{2} & \text{otherwise.} \end{cases}$$

Then f is a bijection. ■

Here is a trivial but important observation: when a set X is countably infinite, the existence of a bijection $f : \mathbb{N} \rightarrow X$ means that each element of X is associated with a unique natural number n ($n \mapsto x_n$). That is, we can write the elements of X as follows

$$X = \{x_0, x_1, x_2, \dots, x_n, \dots\} = \{x_n : n \in \mathbb{N}\}.$$

So, every countable set is simply a sequence of distinct elements. We can do the same for a collection/family of sets. By a countable collection of sets we mean a collection of sets indexed by a countable set (usually the natural numbers). That is, a collection $\{A_\alpha : \alpha \in I\}$ is countable if I is countable. This means that we can list the members of this collection as follows

$$\{A_0, A_1, A_2, A_3, \dots\}.$$

Theorem A.3.2. Let $\{A_n : n \in I\}$ be a countable collection of countable sets. Then

$$A = \bigcup_{n \in I} A_n$$

is also countable.

¹A set A is a proper subset of a set X if $A \subseteq X$ and there exists $x \in X$ such that $x \notin A$.

Proof. Exercise. ■

You didn't see that coming, did you? Well, how about this:

Theorem A.3.3. *The set of rational numbers $\mathbb{Q}$ is countable.*

Proof. For each $n \in \mathbb{Z} \setminus \{0\}$, define $A_n := \{\dots, \frac{-2}{n}, \frac{-1}{n}, 0, \frac{1}{n}, \frac{2}{n}, \dots\} = \{\frac{x}{n} : x \in \mathbb{Z}\}$ to be the set of all rational numbers whose denominator is n . Then

$$\mathbb{Q} = \bigcup_{n \in \mathbb{Z} \setminus \{0\}} A_n.$$

Since each A_n is countable, it follows that $\mathbb{Q}$ is also countable by the previous result. ■

Now you might asking yourself this question: “Are all infinite sets countable?”.

Theorem A.3.4. *The set of real numbers $\mathbb{R}$ is uncountable (not countable).*

Proof. We need to show that there is no bijection between $\mathbb{N}$ and $\mathbb{R}$. We will actually prove a stronger statement that there is no bijection between $\mathbb{N}$ and $[0, 1]$. Assume that $[0, 1]$ is countable. This simply means that $[0, 1] = \{a_0, a_1, a_2, \dots\} = \{a_n : n \in \mathbb{N}\}$. Now each a_n has a unique² decimal expansion $a_n = 0.a_n^1 a_n^2 a_n^3 \dots a_n^m \dots$. For each $n \in \mathbb{N}$, we pick $b_n \in \{0, 1, 2, \dots, 9\}$ such that $b_n \neq a_n^n$. Convince yourself that the number $b := 0.b_1 b_2 b_3 \dots$ is not in the list, even though clearly $b \in [0, 1]$. ■

Corollary A.3.1. *Every non-empty interval in $\mathbb{R}$ is uncountable.*

Proof. Take, for instance, the closed interval $[a, b]$ and define the bijection $f : [a, b] \rightarrow [0, 1]$ by $f(x) = \frac{x-a}{b-a}$. The rest is left as an exercise. ■

It gets worse:

Corollary A.3.2. *All non-empty intervals in $\mathbb{R}$ have the same cardinality as $\mathbb{R}$.*

Proof. Exercise ■

The cardinality of the reals is called the *Cardinality of the Continuum* and is denoted by $\mathfrak{c}$.

²Almost unique, as $1 = 0.999\dots = 1.000\dots$. But there is only one expansion that does not end with zeros